Combinatorial and topological games

Baire spaces and applications in classical mathematics

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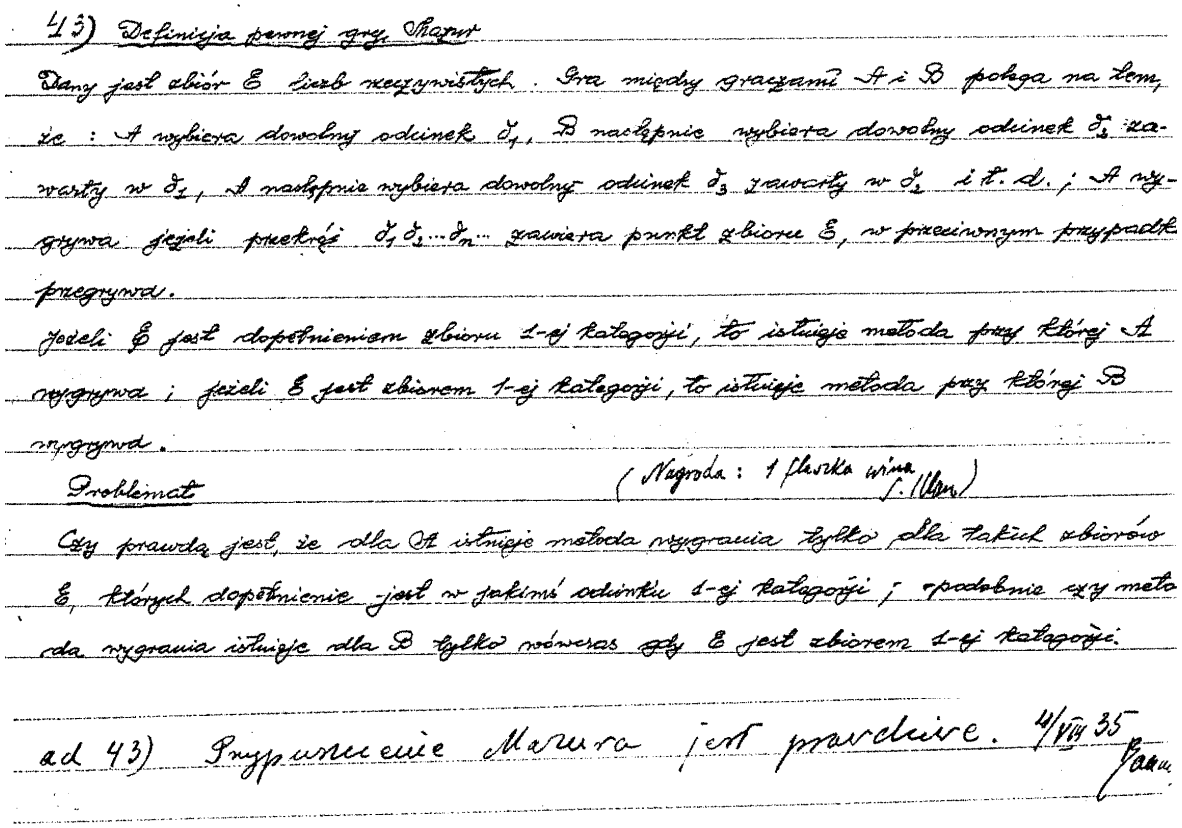
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Introduction

In 1935 Stanisław Mazur entered a problem in the Scottish Book (see Figure 1). The book was a notebook kept at the Scottish Café, in the beautiful city of Lwów (see Figure 2), and the mathematicians of Lwów did a good part of their thinking at the café's tables. The problem concerns a game in which two players take turns choosing intervals of the real line, each inside the previous one. Mazur asked when the first player can force the intersection of the intervals to meet a given set, and when the second player can keep the intersection disjoint from it. Banach found the answer but never published a proof. The first published proof is Oxtoby's, in a paper of 1957 whose title calls the game by the name it has kept: the Banach–Mazur game.


43) Definicja pewnej gry. Mazur

Dany jest zbiór E liczb rzeczywistych. Gra między graczami A i B polega na tem, że: A wybiera dowolny odcinek δ_1, B następnie wybiera dowolny odcinek δ_2 zawarty w δ_1, A następnie wybiera dowolny odcinek δ_3 zawarty w δ_2 i t. d.; A wygrywa jeżeli przekrój $\delta_1 \delta_2 \dots \delta_n \dots$ zawiera punkt zbioru E, w przeciwnym przypadku przegrywa.

Jeżeli E jest dopełnieniem zbioru 1-ej kategorji, to istnieje metoda przy której A wygrywa; jeżeli E jest zbiorem 1-ej kategorji, to istnieje metoda przy której B wygrywa.

Problemat (Nagroda: 1 flaszka wina S. Ulam)

Czy prawdą jest, że dla A istnieje metoda wygrania tylko dla takich zbiorów E, których dopełnienie jest w jakimś odcinku 1-ej kategorji; podobnie czy metoda wygrania istnieje dla B tylko wówczas gdy E jest zbiorem 1-ej kategorji.

ad 43) Przypuszczenie Mazura jest prawdziwe. 4/VIII 35 Banach

Figure 1. Mazur's Problem 43 in the Scottish Book, with the prize he offered and his signature: "Nagroda: 1 flaszka wina" (a bottle of wine). Below, from the opposite page: "ad 43) Przypuszczenie Mazura jest prawdziwe. 4/VIII 35" ("Mazur's conjecture is true," 4 August 1935), signed by Banach. Courtesy of the Banach family, via the Stefan Banach Portal, Adam Mickiewicz University, Poznań (`kielich.amu.edu.pl/Stefan_Banach`).

A game is, at first sight, a curious way of asking a question about a set. Its interest lies in the answer. In Part I we show that the second player can keep the play inside a set exactly when the set is large in the sense of Baire category (see Theorem 3.5). Category and the game are thus one notion under two modes of presentation: the one given by countable unions of nowhere dense sets, the other by what a player can enforce. The second presentation proved

the more fertile. Over the last sixty years the rules have been changed in many ways: the sizes of the moves have been prescribed, the players have been limited in how much they may compute, and they have been allowed to play vectors, points or covers instead of intervals. Each change has found its problem in some unexpected corner of mathematics. We first develop the theory of the game and then tell twelve stories in which a game, often one of these variants, settles a problem.

Part I is devoted to the Banach–Mazur game, its strengthening, the strong Choquet game, and what the two games reveal about Baire spaces and complete metrizability. The part is self-contained and can be read as a short course on its own.

Part II is about applications of the Banach–Mazur game itself and its close relatives, the games of Choquet type, in which the two players take turns shrinking a set:

Number theory: Schmidt's games and badly approximable numbers (Schmidt, 1966).

Dynamical systems and Lie theory: Bounded orbits of flows on spaces of lattices, and Margulis's conjecture, proved by Kleinbock and Margulis (1996).

Theoretical computer science: Baire category inside complexity classes (Mehlhorn, 1973; Lutz, 1990).

Topological groups: When separate continuity implies joint continuity (Ellis, 1957; Namioka, 1974; Kenderov, Kortezov and Moors, 2001).

Infinite-dimensional Ramsey theory: The Ellentuck topology (Ellentuck, 1974) and the Kastanas game (Kastanas, 1983).

Functional analysis: Asplund spaces (Namioka–Phelps and Stegall, 1975; the game of Kenderov and Moors, 1999) and Gowers's dichotomy for Banach spaces (Gowers, 1996).

Set theory: Nonprincipal ultrafilters on ω, the Baire property and the axiom of choice (Sierpiński, 1938; Jalali-Naini, 1976; Talagrand, 1980), and Galvin's games for selective ultrafilters and P-points.

In Part III we turn to other famous topological games, in which the players do not simply shrink a set:

Set-theoretic topology: The Menger game and the refutation of Menger's conjecture (Fremlin and Miller, 1988).

Descriptive set theory: Borel determinacy (Martin, 1975) and the Wadge hierarchy (Wadge, 1972).

Three beautiful applications of games that are not topological at all make up Part IV:

Probabilistic combinatorics: The Erdős–Selfridge theorem and derandomization (Erdős and Selfridge, 1973).

Fixed point theory: Hex and the Brouwer fixed point theorem (Gale, 1979).

Model theory: Ehrenfeucht–Fraïssé games (Fraïssé, 1954; Ehrenfeucht, 1961).

Part V, an epilogue, is the story of the Scottish Book: how it survived the war, how its problems traveled to Los Alamos, Denton and Toronto, and how the games it started came back to settle problems of Banach himself.

After Part I, most sections of Parts II–IV stand on their own; at the opening of Part II we indicate which of its sections lean on which. In the last section of Part II the Banach–Mazur game confronts the axiom of choice directly. We show there that a nonprincipal ultrafilter on ω defines a game that nobody wins; consequently, if every game of this kind is determined, there is no nonprincipal ultrafilter on ω (see Theorem 13.1).

Since every game in Part II is the original one with a few rules changed, we record the original here, in the form in which we will keep modifying it. Mazur's question was about a set E at which the first player aims; we write $S = \mathbb{R} \setminus E$ for the set at which the second player aims, and give the payoff to her (see the conventions below). In Part I we prove the theorem in the box for an arbitrary topological space in place of the real line (see Theorem 3.5).

The Banach–Mazur game. Fix a set $S \subseteq \mathbb{R}$. Player I chooses a closed interval U_0 of positive length; Player II chooses a closed interval $V_0 \subseteq U_0$ of positive length; Player I chooses $U_1 \subseteq V_0$; and so on, forever. Player II wins if every point of $\bigcap_k V_k$ belongs to S. Theorem (Banach; Oxtoby, 1957): Player II has a winning strategy if and only if the complement $\mathbb{R} \setminus S$ is meager, that is, if and only if the set S is "large" in the sense of Baire category.

Behind all the variations of the rules, two features of games recur, and between them they account for much of what follows. The first is that a winning strategy is built one move at a time: a global question, whether a set is large, becomes a local one, what to play next. The second is that strategies can be *combined*: a player can pursue several goals at once by taking turns between them, one round for this goal, the next round for that one. This second feature explains why sets defined by games survive countable intersections. Comeager sets and sets of full measure survive them too, but largeness in the sense of dimension does not: two sets of full Hausdorff dimension can be disjoint, as the badly approximable numbers and their complement are. The winning sets of Schmidt's game have both virtues: they have full dimension, and they are closed under countable intersections. We show both facts in Section 7, where we also tell the history of the problem.

Conventions. We follow the usage of Part I throughout: Player I moves first; in games of Banach–Mazur type, Player II answers inside Player I's move; and the payoff set belongs to Player II: she is the one trying to produce an object with a given property, and he is the one trying to stop her. Where an author used other names for the players (Spoiler and Duplicator, Maker and Breaker, ONE and TWO, β and α, Alice and Bob), we say which of them is our Player I.

The sections of Parts II–IV share roughly the same plan: the problem, its history, the game, the main result, and how the game solves the problem.

Part I. Topological games and Baire spaces

A set of real numbers may be large in more than one sense: in measure, in dimension, or in category. Of these, the third is the one most naturally described by a game. In this part we study *topological games*: infinite games in which two players alternately choose open sets (in one of them Player I also chooses points), and in which a topological property of the underlying space is characterized by which player has a winning strategy.

Roadmap

Two theorems carry this part, and each of them turns a property of a space into a fact about a game. The first is that a space is a Baire space exactly when the first player has no winning strategy in the Banach–Mazur game (see Theorem 3.3): that the space is Baire and that Player I is powerless are one fact, stated once in the language of topology and once in the language of play. The second is that a metrizable space is completely



(a) (b)

FIGURE 2. The Scottish Café. (a) Plac Akademicki in Lwów, 1925; the café occupied the ground floor of the turreted corner building on the right, today Prospekt Shevchenka 27. Photograph by Marek Münz; National Library of Poland (Polona), public domain. (b) The interior of the café in the 1930s; photographer unknown, public domain.

metrizable exactly when the second player has a winning strategy in the strong Choquet game (see Theorem 4.4).

In Section 1 we recall category and Baire spaces, and in Section 2 we say what a game is. The first theorem is proved in Section 3. In Section 4 the moves of Player I acquire points; there we prove the easy half of the second theorem and derive from it the Baire category theorem (see Corollary 4.5), and we extend that half to Čech-complete spaces (see Subsection 4.1). The hard half, which we prove in Section 5, is that a winning strategy for Player II produces a complete metric. The proof uses three classical tools that are worth knowing for their own sake: Alexandrov's theorem on G_δ sets, Stone's theorem on locally finite refinements, and Kőnig's lemma. We close with a single chain of implications that gathers all these results (see Subsection 5.3). A reader who knows Baire category may begin with Section 2. If the part has a moral, it is that a strategy can stand in for a construction: the complete metric of Section 5 is read off from what Player II knows how to do.

Prerequisites. We assume a first course in metric spaces and the basic vocabulary of point-set topology: open and closed sets, closure and interior, dense sets, continuity, and Cauchy sequences and completeness. We also use the following notions:

(1) A topological space X is *metrizable* if there is a metric on X whose open sets are exactly the open sets of X; such a metric is called *compatible*. The space is *completely metrizable* if it has a compatible metric that is complete. Completeness is a property of the metric, whereas complete metrizability is a property of the topology: the irrational numbers are not complete in the usual metric of $\mathbb{R}$, yet they carry a complete compatible metric (see Section 5).
(2) A G_δ *set* is a countable intersection of open sets.

(3) Every metric space (X, d) is a dense subspace of a complete metric space, its *completion*, which is unique up to an isometry that fixes X; for instance, $\mathbb{R}$ is the completion of $\mathbb{Q}$.
(4) *Zorn's lemma*: if every chain in a non-empty partially ordered set has an upper bound, then the set has a maximal element. We use it to find maximal families of pairwise disjoint open sets.

1. Baire category

Baire category measures the size of a set by topology alone: a set is small when it is a countable union of sets that are thin everywhere. In this section we recall the notions of category, and we show that a space is a Baire space exactly when none of its non-empty open subsets is small in this sense, that is, of the first category (see Theorem 1.3 below).

Definition 1.1. Let X be a topological space, and let $E \subseteq X$ be given. We denote the closure of E by $\overline{E}$, and its interior by E°.

(1) The set E is *nowhere dense* in X if $(\overline{E})^\circ = \emptyset$, that is, if its closure has empty interior.
(2) The set E is of the *first category* in X (or *meager*) if it is the union of a countable family of nowhere dense sets. A set that is not of the first category is of the *second category* (or *non-meager*).
(3) The set E is *comeager* (or *residual*) in X if its complement $X \setminus E$ is of the first category.
(4) The space X is called a *Baire space* if the intersection of every countable family of dense open subsets of X is dense in X.
(5) The set E has the *Baire property* if there is an open set O such that the symmetric difference $E \triangle O$ is of the first category; that is, if E is an open set up to an error of the first category.

Remark 1.2. The notions in Definition 1.1 classify sets by their topological "size"; they form a qualitative analogue of measure theory.

(1) A nowhere dense set is thin everywhere. Equivalently (see Exercise 1 at the end of this section), a set E is nowhere dense exactly when every non-empty open set contains a non-empty open set disjoint from E. Every finite subset of $\mathbb{R}$ is nowhere dense, and so is the Cantor ternary set (see Figure 3).
(2) A set of the first category is topologically negligible, and a countable union of such sets is again of the first category, just as a countable union of Lebesgue-null sets is null. The rationals $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$ form a set of the first category in $\mathbb{R}$. The analogy has limits: the line $\mathbb{R}$ is the union of a set of the first category and a set of Lebesgue measure zero [Oxt80, Chapter 1], so a set that is small in one sense may be large in the other.
(3) A comeager set contains "most" points of the space. A property is said to hold of the *generic* point of X if the set of points that have it is comeager. In $\mathbb{R}$, the set of irrational numbers is comeager, so the generic real number is irrational.
(4) In Theorem 1.3 below we show that X is a Baire space exactly when no non-empty open subset of X is negligible. This *local* condition is stronger than asking that X itself be of the second category in X. Consider the disjoint union $X = (\mathbb{Q} \times \{0\}) \cup (\mathbb{R} \times \{1\}) \subseteq \mathbb{R}^2$ (see Figure 4). Its open subset $\mathbb{R} \times \{1\}$ is of the second category in X by the Baire category theorem (see Corollary 4.5), and therefore so is X; yet the space X is not a Baire space, since its open subset $\mathbb{Q} \times \{0\}$ is of the first category.

(5) Every open set O has the Baire property, because $O \triangle O = \emptyset$. Every closed set F has it as well: the set $F \triangle F^\circ = F \setminus F^\circ$ is closed and has empty interior, hence it is nowhere dense. The sets with the Baire property form a σ-algebra (see [Oxt80]), and therefore every Borel set has the Baire property.



FIGURE 3. The first stages of the Cantor ternary set $C = \bigcap_n C_n$: the set C_{n+1} is obtained from C_n by removing the open middle third of each of its intervals. The set C is closed and uncountable, yet every open interval contains a smaller open interval that misses C; it is therefore nowhere dense.



FIGURE 4. The space $X = (\mathbb{Q} \times \{0\}) \cup (\mathbb{R} \times \{1\})$ of Remark 1.2(4). Each line is an open subset of X. The upper line makes X of the second category in itself; the lower line, a countable union of points, is a non-empty open set of the first category, which keeps X from being a Baire space.

Theorem 1.3. *A topological space X is a Baire space if and only if no non-empty open subset of X is of the first category.*

Proof. The two conditions are exchanged by passing to complements: a closed set is nowhere dense exactly when its complement is a dense open set.

Suppose first that X is a Baire space, and let $G \subseteq X$ be open and non-empty. If $G = \bigcup_{n=1}^\infty E_n$ with each E_n nowhere dense, then each set $V_n = X \setminus \overline{E_n}$ is open and dense in X. Since X is a Baire space, the intersection $\bigcap_{n=1}^\infty V_n$ is dense in X, and it therefore meets the non-empty open set G. However, we have

$$G \cap \bigcap_{n=1}^{\infty} V_n = G \setminus \bigcup_{n=1}^{\infty} \overline{E_n} = \emptyset,$$

a contradiction, which shows that G is of the second category.

Conversely, suppose that no non-empty open subset of X is of the first category. Let $(V_n)_{n\geq 1}$ be a sequence of dense open subsets of X, and let $O \subseteq X$ be open and non-empty; we show that O meets $\bigcap_{n=1}^\infty V_n$. For each n, the set $E_n = X \setminus V_n$ is closed, and it has empty interior because V_n is dense; thus each E_n is nowhere dense. If O missed $\bigcap_{n=1}^\infty V_n$, it would be contained in $\bigcup_{n=1}^\infty E_n$, and would therefore be a non-empty open set of the first category, contrary to our hypothesis. Hence $O \cap \bigcap_{n=1}^\infty V_n \neq \emptyset$. Since O was arbitrary, the intersection $\bigcap_{n=1}^\infty V_n$ is dense, which proves that X is a Baire space. ☺

Exercises.

(1) Let X be a topological space, and let $E \subseteq X$ be given. Show that the definition of a nowhere dense set in Definition 1.1(1) is equivalent to the description in Remark 1.2(1), and that the closure of a nowhere dense set is nowhere dense.
(2) Show that the union of two nowhere dense subsets of a topological space is nowhere dense. Show by an example in $\mathbb{R}$ that the union of countably many nowhere dense sets need not be nowhere dense.
(3) Let X be a topological space. Show that X is a Baire space if and only if every comeager subset of X is dense in X. Deduce that in a non-empty Baire space no comeager set is of the first category.
(4) Let X be a Baire space, and let $U \subseteq X$ be a non-empty open set. Show that U, with the subspace topology, is a Baire space. (Hint: show that a subset of U that is nowhere dense in U is nowhere dense in X, and use Theorem 1.3.)
(5) $(*)$ Let X be a Baire space, and let $D \subseteq X$ be a dense G_δ set. Show that D, with the subspace topology, is a Baire space. (Hint: a dense open subset of D has the form $W \cap D$ with W open and dense in X.)

2. Infinite games

In this section we fix the formal framework for the games of these notes: trees of positions, plays, payoff sets, and strategies.

The definitions have four tasks. They must make precise what a player knows when moving, namely the finite sequence of moves made so far, and what a player may do, namely choose one of the moves allowed in that position. They must fix what counts as winning: the membership of the infinite play in a given set. Finally, they must capture what it means to have a method of play: a rule that assigns a move to every position at which the player is to move. Trees of positions, plays, payoff sets and strategies are these four requirements, made exact.

Definition 2.1. Let A be a non-empty set. We denote by $A^{<\omega}$ the collection of all finite sequences $s = (s_0, \dots, s_{n-1})$ of elements of A. The length n of s is denoted by $|s|$; the empty sequence $\langle\rangle$ has length 0. If $s, t \in A^{<\omega}$, $|t| \leq |s|$, and $t_i = s_i$ for $0 \leq i < |t|$, we write $t \subseteq s$ and say that s *extends* t; when sequences are regarded as functions defined on initial segments of ω, the relation is literally inclusion. For $s \in A^{<\omega}$ and $a \in A$, we write $s^\frown a = (s_0, \dots, s_{|s|-1}, a)$ for the extension of s by a.

A non-empty subset $T \subseteq A^{<\omega}$ is called a *tree* if $t \in T$ whenever $s \in T$ and $t \subseteq s$. For each $s \in T$, we write

$$\mathrm{Succ}(s) = \{a \in A : s^\frown a \in T\}.$$

We assume throughout that $\mathrm{Succ}(s) \neq \emptyset$ for every $s \in T$. An infinite sequence $x = (a_n)_{n=0}^\infty \in A^\omega$ is a *play* in T if $(a_0, \dots, a_{n-1}) \in T$ for all $n \geq 1$. The set of all such plays is denoted by $[T]$.

Definition 2.2. An *infinite game* is a pair (T, W), where $T \subseteq A^{<\omega}$ is a tree and $W \subseteq [T]$. Two players, I and II, alternately choose elements $a_n \in A$, subject to the rule that $(a_0, \dots, a_n) \in T$: Player I chooses a_{2n}, and Player II chooses a_{2n+1} $(n = 0, 1, 2, \dots)$. Player II *wins* the resulting play $x = (a_n) \in [T]$ if $x \in W$; otherwise, Player I wins. The elements of T are called *positions*.

A *strategy* for Player I is a function σ, defined on the positions of even length, such that $\sigma(s) \in \mathrm{Succ}(s)$ for every such position s. A play or position is *consistent* with σ if

$a_{2n} = \sigma(a_0, \dots, a_{2n-1})$ whenever a_{2n} is defined. We say that σ is a *winning strategy* for Player I if every play consistent with σ lies in $[T] \setminus W$.

A strategy τ for Player II and the notion of a winning strategy for Player II are defined analogously.

Remarks 2.3. (1) In these notes W is the set of plays won by Player II. Many texts, among them Kechris's book [Kec95], use the opposite convention, in which the payoff set is Player I's. Ours suits the games below, in which Player II is the player who tries to build something: a point common to all the moves, or an intersection that lies inside a given set.

(2) The two players need not choose from the same kind of object: if Player I chooses from a set A_{I} and Player II from a set A_{II}, we take $A = A_{\mathrm{I}} \cup A_{\mathrm{II}}$ and let the tree T record which moves are legal.

(3) At most one player has a winning strategy: if σ and τ were winning strategies for Players I and II, the unique play consistent with both (see Exercise 2 at the end of this section) would lie both outside and inside W.

(4) A game is called *determined* if one of the two players has a winning strategy. Not every game is determined. There are Baire spaces X in which neither player has a winning strategy in the Banach–Mazur game $\mathrm{BM}(X)$ of Section 3 (see the remark after Theorem 3.3), and the axiom of choice produces undetermined games in which the players choose finite binary strings (see Section 13). On the other side stands Martin's theorem (see Section 15): every game in which the players choose natural numbers, and whose payoff set is a Borel subset of ω^ω, is determined.

(5) A strategy sees the whole position, not only the last move: the move that σ prescribes at a position $(a_0, \dots, a_{2n-1})$ may depend on everything played so far (see Figure 5). This memory can be indispensable (see Exercise 5 at the end of this section).

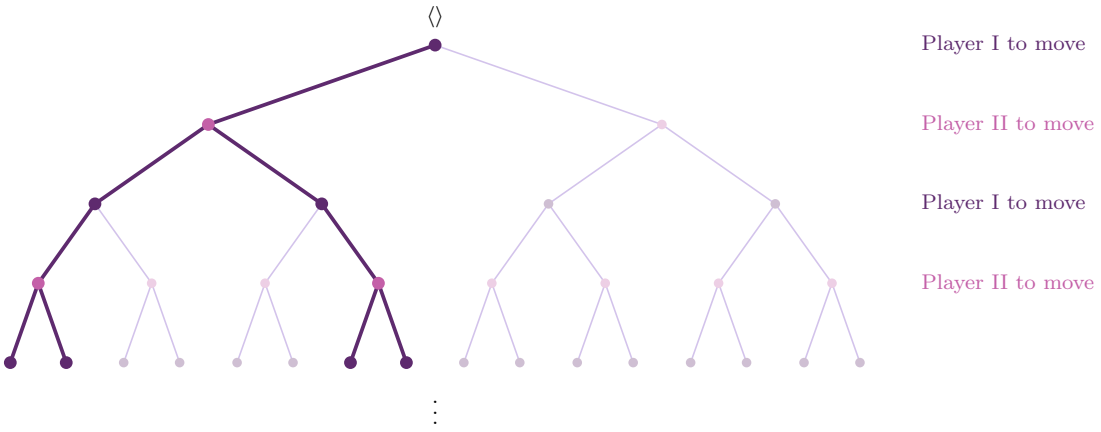


FIGURE 5. A game tree, drawn with two moves available at each position. The levels alternate between Player I (plum) and Player II (pink). A strategy σ for Player I picks one move at each of his positions. The positions consistent with σ (dark) form a subtree that branches only where Player II is to move, and its infinite branches are the plays consistent with σ. The strategy is winning if all of these plays lie outside W.

Exercises.

(1) Let $A = \{0, 1\}$ and $T = A^{<\omega}$. For each of the following payoff sets $W \subseteq [T]$, decide which player has a winning strategy in (T, W) and describe one: (a) the plays containing at least one 1; (b) the plays with $a_n = 0$ for every n; (c) the plays with $a_1 = a_0$; (d) the plays in which Player I plays 1 infinitely often.
(2) Let (T, W) be an infinite game, and let σ and τ be strategies for Player I and Player II, respectively. Show that there is exactly one play consistent with both σ and τ, a fact used in Remarks 2.3(3).
(3) Let T be a tree, and let $W \subseteq W' \subseteq [T]$ be given. Show that a winning strategy for Player II in (T, W) is winning in (T, W'), and that a winning strategy for Player I in (T, W') is winning in (T, W).
(4) Let (T, W) be an infinite game, and suppose that there is an integer N such that whether a play lies in W depends only on its first N moves. Show that the game is determined. (Hint: decide who wins from each position of length N, and work backwards to $\langle\rangle$.)
(5) $(*)$ Let $A = \{0, 1\}$, $T = A^{<\omega}$, and let W be the set of plays with $a_{2n+1} = a_0$ for every n. Show that Player II has a winning strategy, but that no strategy for her whose move depends only on the last move of Player I is winning.

3. The Banach–Mazur game

In the Banach–Mazur game, two players take turns shrinking an open set, and the second player wants the sets to keep a point in common. In this section we define the game and show that a space is a Baire space exactly when the first player has no winning strategy (see Theorem 3.3 below). We then let the players aim at a given set, and prove the theorem of Banach and Oxtoby: the second player can keep the play inside a set exactly when its complement is of the first category (see Theorem 3.5 below).

Definition 3.1. Let X be a non-empty topological space, and let $\mathcal{O}^+(X)$ be the collection of all non-empty open subsets of X. The *Banach–Mazur game* on X, denoted by $\mathrm{BM}(X)$, is the game (T, W) where:

(1) The set of moves is $A = \mathcal{O}^+(X)$.
(2) The tree T consists of all finite sequences $(U_0, V_0, U_1, V_1, \dots)$ of sets in $\mathcal{O}^+(X)$ with $U_0 \supseteq V_0 \supseteq U_1 \supseteq V_1 \supseteq \cdots$.
(3) The payoff set is $W = \{(U_0, V_0, U_1, \dots) \in [T] : \bigcap_{n=0}^{\infty} U_n \neq \emptyset\}$.

Thus Player I plays the sets U_n, Player II plays the sets V_n, and Player II wins if the sets played have a point in common.

Example 3.2. (1) In $\mathrm{BM}(\mathbb{Q})$, Player I has a winning strategy. Fix an enumeration $q_0, q_1, \dots$ of $\mathbb{Q}$, and let Player I play $U_0 = \mathbb{Q} \setminus \{q_0\}$ and $U_n = V_{n-1} \setminus \{q_n\}$ for $n \geq 1$. These sets are open because the points of $\mathbb{Q}$ are closed, and non-empty because $\mathbb{Q}$ has no isolated points; since every q_n is missing from U_n, the intersection $\bigcap_n U_n$ is empty (see Figure 6).

(2) In $\mathrm{BM}(\mathbb{R})$, Player II has a winning strategy: in answer to U_n, she plays a bounded open interval V_n with $\overline{V_n} \subseteq U_n$. The compact sets $\overline{V_n}$ are nested and non-empty, hence they have a common point, and that point lies in every U_n because $\overline{V_n} \subseteq U_n$.

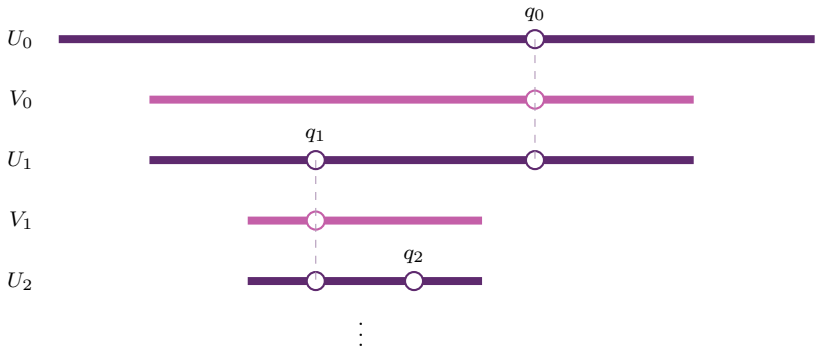


FIGURE 6. Player I wins BM($\mathbb{Q}$) (see Example 3.2(1)). The moves of Player I are drawn in plum and those of Player II in pink, as intervals of rationals. At step n Player I removes the rational q_n (a hole), and a hole, once made, persists in every later set. Every rational is removed at some step, so the sets have no point in common.

Theorem 3.3. *Let X be a non-empty topological space.*

(1) *Player I has a winning strategy in* BM(X) *if and only if X contains a non-empty open set of the first category.*
(2) *The space X is a Baire space if and only if Player I has no winning strategy in* BM(X).

Proof. By Theorem 1.3, part (2) is an immediate consequence of part (1), which we now prove. The idea is that a nowhere dense set can be avoided in a single move. If a non-empty open set is a countable union of nowhere dense sets, Player I opens with it and avoids one of the pieces at each later move; conversely, the plays of a winning strategy for Player I, laid side by side, show that his first move is of the first category.

Suppose first that $G \subseteq X$ is open, non-empty, and of the first category. Then there are nowhere dense sets $E_0, E_1, \ldots$ with $G = \bigcup_{n=0}^{\infty} E_n$. Player I plays $U_0 = G$. If Player II responds with $V_0 \subseteq U_0$, the set $V_0 \setminus \overline{E_0}$ is open, and it is non-empty because $\overline{E_0}$ has empty interior. Player I may therefore choose a non-empty open set $U_1 \subseteq V_0 \setminus \overline{E_0}$. In general, given V_{n-1}, Player I chooses an open set

$$U_n \subseteq V_{n-1} \setminus \overline{E_{n-1}}, \qquad U_n \neq \emptyset.$$

This defines a strategy σ for Player I. Every play consistent with σ satisfies $U_n \cap E_{n-1} = \emptyset$ for all $n \geq 1$, whence

$$\bigcap_{n=0}^{\infty} U_n \subseteq G \setminus \bigcup_{n=0}^{\infty} E_n = \emptyset.$$

Thus the strategy σ is winning for Player I.

Conversely, suppose that Player I has a winning strategy σ, and put $U_0 = \sigma(\langle\rangle)$. We show that the non-empty open set U_0 is of the first category. For a position p of odd length $2n+1$ consistent with σ, we shall write U^p for the last entry of p, the set U_n played by Player I.

We build, for each $n \geq 0$, a family $\mathcal{P}_n$ of positions of length $2n+1$ consistent with σ, such that:

(a) the sets U^p, $p \in \mathcal{P}_n$, are pairwise disjoint;

(b) their union $D_n = \bigcup_{p\in\mathcal{P}_n} U^p$ is dense in U_0;
(c) every $p \in \mathcal{P}_{n+1}$ extends some $q \in \mathcal{P}_n$.

Let $\mathcal{P}_0 = \{(U_0)\}$. Suppose that $\mathcal{P}_n$ has been built, and let $q \in \mathcal{P}_n$ be given. Each non-empty open $V \subseteq U^q$ is a legal move for Player II at q, and leads to the position $q^\frown V^\frown \sigma(q^\frown V)$ of length $2n+3$. By Zorn's lemma (the union of a chain of such families keeps the property), we may choose a family $\mathcal{P}(q)$ of such positions, maximal with respect to the property that their last entries are pairwise disjoint. The union of these last entries is dense in U^q: otherwise some non-empty open $O \subseteq U^q$ would miss all of them, and the position $q^\frown O^\frown \sigma(q^\frown O)$, whose last entry lies in O, could be added to $\mathcal{P}(q)$, contradicting maximality. Let $\mathcal{P}_{n+1}$ be the union $\bigcup_{q\in\mathcal{P}_n} \mathcal{P}(q)$. Property (c) holds by construction. Property (a) holds because last entries of positions in the same $\mathcal{P}(q)$ are disjoint, and those in $\mathcal{P}(q)$ and $\mathcal{P}(q')$ with $q \neq q'$ lie in the disjoint sets U^q and $U^{q'}$. For (b), we argue by induction on n; the case $n = 0$ is trivial, since $D_0 = U_0$. By the inductive hypothesis, every non-empty open set $O \subseteq U_0$ meets D_n, hence meets U^q for some $q \in \mathcal{P}_n$. The union of the last entries of the positions in $\mathcal{P}(q)$ is dense in U^q, and it therefore meets the non-empty open set $O \cap U^q$; as this union is contained in D_{n+1}, the set O meets D_{n+1}. Figure 7 shows the first two levels.

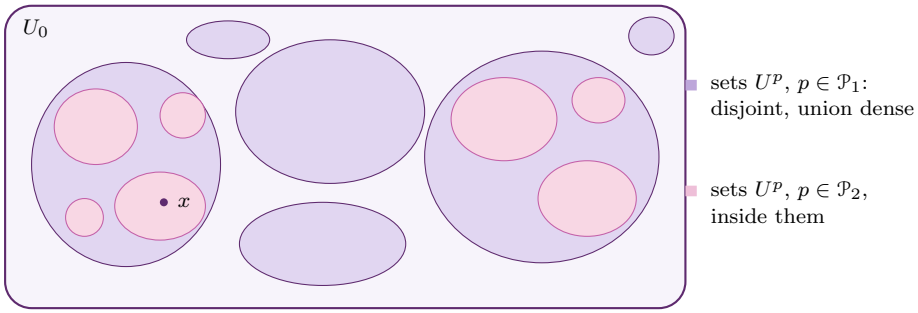


FIGURE 7. The proof of Theorem 3.3: the first two levels of the construction. Inside U_0, the answers of σ at the first level form a maximal family of pairwise disjoint open sets, whose union D_1 is dense in U_0; inside each of them, the second level does the same. A point x lying in every D_n picks out exactly one set at each level, and these sets are the moves of Player I in a single play consistent with σ.

Each D_n is open and dense in U_0; hence the set $U_0 \setminus D_n$ is nowhere dense in X. Suppose, toward a contradiction, that U_0 is of the second category. Then the first-category set $\bigcup_n (U_0 \setminus D_n)$ does not contain U_0, and there is a point $x \in \bigcap_n D_n$. By (a), for each n there is exactly one $p_n \in \mathcal{P}_n$ with $x \in U^{p_n}$. Property (c) provides a position $q \in \mathcal{P}_n$ extended by p_{n+1}; since the moves along a play are nested, we have $x \in U^{p_{n+1}} \subseteq U^q$, and property (a) forces $q = p_n$. Thus the positions $p_0 \subseteq p_1 \subseteq \cdots$ fit together into a single play consistent with σ, and the point x lies in every set U^{p_n} that Player I plays in it. This contradicts the assumption that σ is winning. Hence the opening move U_0 is a non-empty open set of the first category, and the proof is complete.

Remark 3.4. When X is a Baire space, part (2) rules out a winning strategy for Player I in $\mathrm{BM}(X)$, but Player II need not have one either: there are Baire spaces in which neither player has a winning strategy (see [Tel87]). This happens already for subspaces of the real line: if $B \subseteq \mathbb{R}$ meets every uncountable closed subset of $\mathbb{R}$, and so does its complement (Bernstein

constructed such sets in 1908), then the subspace B is a Baire space on which neither player of $\mathrm{BM}(B)$ has a winning strategy.

In Mazur's original game, and in most applications in Part II, the players aim at a given set. Let X be a non-empty topological space, and let $S \subseteq X$ be given. The game $\mathrm{BM}(X, S)$ has the same moves as $\mathrm{BM}(X)$, but Player II wins the play $(U_0, V_0, U_1, \dots)$ if

$$\bigcap_{n=0}^{\infty} U_n \subseteq S,$$

that is, if no point outside S survives. Since the moves are nested, the intersection may be taken over the moves of either player: $\bigcap_n U_n = \bigcap_n V_n$. Unlike in $\mathrm{BM}(X)$, Player II also wins when the intersection is empty; Player I wins only if he produces a point outside S.

Theorem 3.5 (Banach; Oxtoby). *Let X be a non-empty topological space, and let $S \subseteq X$ be given.*

(1) *Player II has a winning strategy in $\mathrm{BM}(X, S)$ if and only if $X \setminus S$ is of the first category in X.*

(2) *If X is a complete metric space, then Player I has a winning strategy in $\mathrm{BM}(X, S)$ if and only if there is a non-empty open set U such that $U \cap S$ is of the first category in X.*

Proof. Both parts rest on the two ideas of the proof of Theorem 3.3: a nowhere dense set can be avoided in a single move, and a winning strategy, unfolded into maximal families of positions, exhibits a set of the first category. (1) Suppose first that $X \setminus S = \bigcup_n E_n$ with each E_n nowhere dense. In answer to U_n, Player II plays a non-empty open set $V_n \subseteq U_n \setminus \overline{E_n}$. A point of $\bigcap_n V_n$ lies outside every E_n, hence in S, so this strategy is winning.

Conversely, let τ be a winning strategy for Player II. We repeat the construction in the proof of Theorem 3.3, now with the moves of Player II. For a position p of non-zero even length consistent with τ, we write V^p for its last entry. As in that proof, Zorn's lemma provides families $\mathcal{P}_n$ of positions of length $2n + 2$ consistent with τ, such that the sets V^p, $p \in \mathcal{P}_n$, are pairwise disjoint, their union D_n is dense in X, and every $p \in \mathcal{P}_{n+1}$ extends some $q \in \mathcal{P}_n$. For $\mathcal{P}_0$ we take a maximal family of positions $(U, \tau(U))$ whose last entries are pairwise disjoint; the union of these last entries is dense, because for every non-empty open set O the answer $\tau(O)$ lies in O. At a position $q \in \mathcal{P}_n$, every non-empty open $U \subseteq V^q$ is a legal move for Player I, and the answer of τ lies in U, so the maximality argument goes through unchanged. Now let $x \in \bigcap_n D_n$ be given. Exactly as before, the positions $p_n \in \mathcal{P}_n$ with $x \in V^{p_n}$ fit together into a single play consistent with τ, and every set played in it contains x. Since τ is winning, the point x lies in S. Thus the intersection $\bigcap_n D_n$ is contained in S, whence $X \setminus S \subseteq \bigcup_n (X \setminus D_n)$. Each $X \setminus D_n$ is closed with empty interior, hence nowhere dense, and therefore the complement $X \setminus S$ is of the first category.

(2) Let d be the metric of X, which is complete by hypothesis. Suppose first that U is a non-empty open set and that $U \cap S = \bigcup_n E_n$ with each E_n nowhere dense. Player I plays $U_0 = U$ and, in answer to V_{n-1}, a non-empty open set U_n with $\overline{U_n} \subseteq V_{n-1} \setminus \overline{E_{n-1}}$ and $\operatorname{diam} U_n < 2^{-n}$. The closed sets $\overline{U_n}$ are nested and their diameters tend to 0, so by completeness they have a point x in common. The point x lies in every U_n, because $x \in \overline{U_{n+1}} \subseteq V_n \subseteq U_n$; in particular, it lies in $U = U_0$. It lies in no E_n, because $\overline{U_{n+1}}$ misses $\overline{E_n}$. Hence Player I has produced a point of $U \setminus S$, namely x, and he has won.

Conversely, let σ be a winning strategy for Player I, and put $U_0 = \sigma(\langle\rangle)$. We show that $U_0 \cap S$ is of the first category. Once more we follow the construction in the proof of Theorem 3.3, with one change: at level n, the families $\mathcal{P}(q)$ use only moves V of Player II with $\operatorname{diam} V < 2^{-n}$. Every non-empty open set contains such a set V (a small open ball), and the maximality argument shows, as before, that the unions D_n are dense in U_0. Every point $x \in \bigcap_n D_n$ again picks out a play consistent with σ whose intersection contains x. The diameters of the moves tend to 0, so this intersection is exactly $\{x\}$. Since σ is winning, the intersection contains a point outside S, and that point can only be x. Thus the intersection $\bigcap_n D_n$ lies in $U_0 \setminus S$, and the set $U_0 \cap S$ is contained in $\bigcup_n (U_0 \setminus D_n)$, which is of the first category because each $U_0 \setminus D_n$ is nowhere dense. Hence Player I's first move U_0 is the required non-empty open set. ☺

Remark 3.6. By part (1), Player II wins $\mathrm{BM}(\mathbb{R}, S)$ when S is the set of irrational numbers: she removes the rationals one at a time, just as Player I did in Example 3.2(1). Part (1) needs no hypothesis on the space. In part (2), the "only if" direction uses the metric but not its completeness, which is needed only in the "if" direction, where Player I must produce a point. Without completeness that direction fails: in $X = \mathbb{Q}$ with $S = \emptyset$, the set $U \cap S$ is empty for every U, yet Player II wins by part (1), because $\mathbb{Q}$ is of the first category in itself.

Exercises.

(1) Let C be the Cantor ternary set. For each of the sets S below, decide which player has a winning strategy in $\mathrm{BM}(\mathbb{R}, S)$, and describe one explicitly: (a) the irrational numbers; (b) the rational numbers; (c) the set C; (d) the set $\mathbb{R} \setminus C$. Check your answers against Theorem 3.5.

(2) Let X be a non-empty compact Hausdorff space. Show that Player II has a winning strategy in $\mathrm{BM}(X)$. (Hint: such a space is regular, so every point of a non-empty open set U has an open neighborhood whose closure lies in U.)

(3) Let X be a topological space, and let $S_0, S_1, \dots \subseteq X$ be given. Suppose that Player II has a winning strategy τ_k in $\mathrm{BM}(X, S_k)$ for every k. Show directly that she has a winning strategy in $\mathrm{BM}(X, \bigcap_k S_k)$. (Hint: split the rounds into pairwise disjoint infinite sets R_k, and let her follow τ_k in the rounds that belong to R_k.) Then give a second proof using Theorem 3.5.

(4) Let X be a Baire space, and let $U \subseteq X$ be a non-empty open set. Use Theorem 3.3 to show that U is a Baire space, by turning a winning strategy for Player I in $\mathrm{BM}(U)$ into one in $\mathrm{BM}(X)$.

(5) $(*)$ Let X be a complete metric space, and let $S \subseteq X$ have the Baire property. Show that $\mathrm{BM}(X, S)$ is determined. (Hint: if $U \cap S$ is of the first category for no non-empty open set U, show that every open set O with $S \triangle O$ of the first category is dense.)

4. The strong Choquet game

The Banach–Mazur game detects category, but not completeness: a metrizable space X in which Player II wins $\mathrm{BM}(X)$ need not be completely metrizable (see Remark 4.6 below). In this section we let Player I name a point with each of his moves, and we show that the resulting game characterizes complete metrizability among metrizable spaces (see Theorem 4.4 below). Here we prove the easy half of the theorem and derive from it the Baire category theorem (see Corollary 4.5 below); the hard half is proved in Section 5. At the end of the section we leave metric spaces and turn to Čech-completeness, a form of completeness that makes

sense in every Tychonoff space: in Subsection 4.1 we show that every Čech-complete space is strongly α-favorable, and that the converse fails.

Definition 4.1. Let X be a non-empty topological space. The *strong Choquet game* on X, denoted by $\mathrm{Ch}(X)$, is played as follows. Player I chooses a pair (x_0, U_0) with $U_0 \in \mathcal{O}^+(X)$ and $x_0 \in U_0$. Player II responds with $V_0 \in \mathcal{O}^+(X)$ such that $x_0 \in V_0 \subseteq U_0$. At step $n \geq 1$, Player I chooses (x_n, U_n) such that $x_n \in U_n \subseteq V_{n-1}$, and Player II responds with $V_n \in \mathcal{O}^+(X)$ such that $x_n \in V_n \subseteq U_n$. Player II wins the play if $\bigcap_{n=0}^{\infty} U_n \neq \emptyset$.

In the format of Definition 2.2, Player I chooses from $X \times \mathcal{O}^+(X)$ and Player II from $\mathcal{O}^+(X)$. If the points x_n are omitted, what remains is exactly the Banach–Mazur game $\mathrm{BM}(X)$, which in this context is also called the *Choquet game*. Choquet called the two players β (the first) and α (the second). Following him, we call a topological space X *α-favorable* if Player II has a winning strategy in $\mathrm{BM}(X)$, and *strongly α-favorable* if she has a winning strategy in $\mathrm{Ch}(X)$; Kechris [Kec95] calls such spaces *Choquet* and *strong Choquet* spaces.

Lemma 4.2. *Every strongly α-favorable space is α-favorable.*

Proof. The idea is that Player II can supply the missing points herself. Let X be a strongly α-favorable space, and let τ be a winning strategy for Player II in $\mathrm{Ch}(X)$. In $\mathrm{BM}(X)$, when Player I plays U_n, Player II picks a point $x_n \in U_n$ and answers with the set that τ assigns to the position $(x_0, U_0), V_0, \ldots, V_{n-1}, (x_n, U_n)$ of $\mathrm{Ch}(X)$. Since $x_n \in U_n \subseteq V_{n-1}$, the moves (x_n, U_n) are legal in $\mathrm{Ch}(X)$. Every play of $\mathrm{BM}(X)$ that follows this rule therefore arises, by omitting the points, from a play of $\mathrm{Ch}(X)$ consistent with τ, and the sets U_n have a common point because τ is winning. This rule is therefore a winning strategy for Player II in $\mathrm{BM}(X)$. ☺

Example 4.3. Player I wins $\mathrm{Ch}(\mathbb{Q})$: he follows the strategy of Example 3.2(1), adding any point of each set he plays.

Theorem 4.4 (G. Choquet). *A metrizable space is completely metrizable if and only if it is strongly α-favorable.*

Proof. Let X be a metrizable space, and suppose first that X is completely metrizable, with a compatible complete metric d. We define a strategy τ for Player II whose answers are balls with radii tending to 0; the points of Player I then form a Cauchy sequence, and its limit is a common point of the play. When Player I plays (x_n, U_n), Player II chooses r_n with $0 < r_n < 2^{-n}$ such that the closed ball $B[x_n, r_n] = \{y \in X : d(x_n, y) \leq r_n\}$ is contained in U_n, and plays $V_n = B(x_n, r_n)$.

Let $(x_0, U_0), V_0, (x_1, U_1), V_1, \ldots$ be a play consistent with τ. For $m > n$ we have $x_m \in U_m \subseteq V_n = B(x_n, r_n)$, whence $d(x_m, x_n) < 2^{-n}$. Thus the points x_n form a Cauchy sequence, which converges to some $x \in X$ because d is complete. For each n, the points x_m with $m \geq n$ all lie in the closed ball $B[x_n, r_n]$; so does their limit, and that ball is contained in U_n. Consequently, the limit x lies in every U_n, and the strategy τ is winning for Player II (see Figure 8).

The converse, that a strongly α-favorable metrizable space is completely metrizable, is proved in Section 5. ☺

Corollary 4.5 (Baire category theorem). *Every completely metrizable space is a Baire space.*

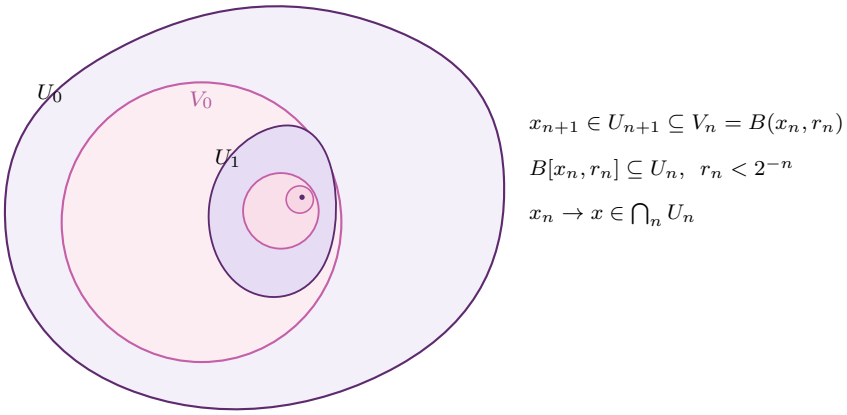


FIGURE 8. The easy half of Theorem 4.4. Player II answers each point x_n with a ball about it whose closure lies inside the set just played and whose radius is less than 2^{-n}. The points x_n form a Cauchy sequence, and its limit lies in every closed ball, hence in every U_n.

Proof. Let X be completely metrizable. By Theorem 4.4 and Lemma 4.2, Player II has a winning strategy in $\mathrm{BM}(X)$. Since at most one player has a winning strategy (see Remarks 2.3(3)), Player I has none, and by Theorem 3.3 the space X is a Baire space. ☺

Remark 4.6. The points x_n cannot be dropped from Theorem 4.4: an α-favorable metrizable space need not be completely metrizable. Let $X = \{(a, b) \in \mathbb{R}^2 : b > 0\} \cup (\mathbb{Q} \times \{0\})$. The upper half plane $H = \{(a, b) \in \mathbb{R}^2 : b > 0\}$ is open and dense in X, and Player II wins $\mathrm{BM}(X)$ by always answering with an open disk whose closure lies in $U_n \cap H$ and whose radius is less than 2^{-n}: these closed disks are nested and compact, and their common point lies in every U_n. The space X, however, is not completely metrizable. Indeed, a closed subspace of a completely metrizable space is completely metrizable (a Cauchy sequence in the subspace converges in the space, and its limit lies in the subspace), and is therefore a Baire space by Corollary 4.5; yet the closed subspace $\mathbb{Q} \times \{0\}$ of X is not a Baire space. By the hard half of Theorem 4.4 (proved in Section 5), it follows that Player II has no winning strategy in $\mathrm{Ch}(X)$. A direct argument, independent of Section 5, leads to the same conclusion: in $\mathrm{Ch}(X)$ Player I can choose his points on the axis, and Player II must answer with sets that contain them; Figure 9 shows how Player I then wins.

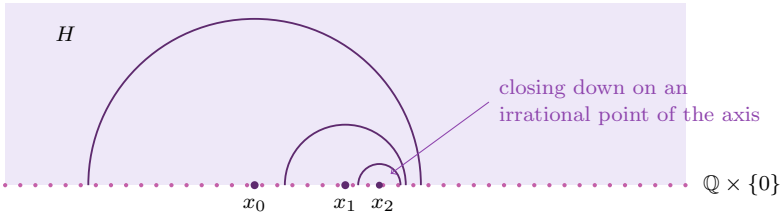


FIGURE 9. Player I wins $\mathrm{Ch}(X)$ for the space X of Remark 4.6. He plays rational points of the axis, with small balls of X about them, inside the previous answer of Player II, and avoids one more rational each time. The balls close down on a point of the axis that is not rational, hence not in X.

4.1. **Čech-complete spaces.** Because Player I names points, the strong Choquet game detects complete metrizability among metrizable spaces, which the Banach–Mazur game cannot do (see Remark 4.6). Completeness, however, is a metric notion; it has a purely topological counterpart, and one half of Theorem 4.4 extends to it. Recall that a *compactification* of a Tychonoff space X is a compact Hausdorff space that contains X as a dense subspace; the Stone–Čech compactification βX is the largest one.

Definition 4.7 (E. Čech, 1937). A Tychonoff space X is *Čech-complete* if it is a G_δ subset of its Stone–Čech compactification βX.

A Tychonoff space is Čech-complete if and only if it is a G_δ subset of some compactification, and it is then a G_δ subset of every compactification (see Engelking's book [Eng89, Section 3.9]). In particular, every compact Hausdorff space is Čech-complete, and so is every locally compact Hausdorff space, since such a space is open in its one-point compactification. For metrizable spaces nothing new arises: a metrizable space is Čech-complete exactly when it is completely metrizable (Engelking proves this theorem of Čech in his book [Eng89, Theorem 4.3.26]). By Theorem 4.4, Čech-completeness and strong α-favorability therefore coincide on metrizable spaces. Outside metrizable spaces only one implication survives.

Proposition 4.8. *Every Čech-complete space is strongly α-favorable.*

Proof. The idea is to play the game in βX, where compactness supplies a common point of the play, while the G_δ condition keeps that point in X. Let X be Čech-complete, and let $G_0, G_1, \dots$ be open subsets of βX with $X = \bigcap_n G_n$; closures below are taken in βX. Along with each answer V_n, Player II records an open set $W_n \subseteq \beta X$ with $V_n = W_n \cap X$, and she chooses the two sets as follows. When Player I plays (x_n, U_n), she takes an open set $\widetilde{U}_n \subseteq \beta X$ with $\widetilde{U}_n \cap X = U_n$; for $n \geq 1$ she may assume that $\widetilde{U}_n \subseteq W_{n-1}$, because $U_n \subseteq V_{n-1} \subseteq W_{n-1}$, and the set $\widetilde{U}_n \cap W_{n-1}$ therefore has the same trace U_n on X. The point x_n lies in the open set $\widetilde{U}_n \cap G_n$, and the compact Hausdorff space βX is regular; hence there is an open set W_n with $x_n \in W_n$ and $\overline{W_n} \subseteq \widetilde{U}_n \cap G_n$. Player II answers with $V_n = W_n \cap X$, a legal move because $x_n \in V_n \subseteq \widetilde{U}_n \cap X = U_n$.

In every play consistent with this strategy, the inclusions $\overline{W_{n+1}} \subseteq \widetilde{U}_{n+1} \subseteq W_n$ hold for every n, and the sets $\overline{W_n}$ thus form a decreasing sequence of compact sets, each containing the point x_n. By compactness, their intersection contains a point y (see Figure 10). Since each $\overline{W_n}$ lies in $\widetilde{U}_n \cap G_n$, the point y lies in $\bigcap_n G_n = X$; consequently it lies in $\widetilde{U}_n \cap X = U_n$ for every n, the play has the common point y, and the strategy is winning. ☺

As in the proof of Corollary 4.5, it follows that every Čech-complete space is a Baire space (see Figure 11). The converse of Proposition 4.8 fails, even among paracompact spaces: the Michael line, obtained from the real line by isolating the irrational points, is strongly α-favorable without being Čech-complete.

Example 4.9 (The Michael line). Let M be the real line with the topology in which a set is open if it is the union of a set open in the usual topology and a set of irrational numbers. Every irrational number is then an isolated point, while every rational number keeps its usual neighborhoods (see Figure 12). The space M is Hausdorff and paracompact (see Engelking's book [Eng89, Example 5.1.32]), hence normal, and in particular Tychonoff.

(1) *The space M is strongly α-favorable.* When Player I plays (x_n, U_n) with x_n irrational, Player II answers with $V_n = \{x_n\}$; every later move equals $\{x_n\}$, and the point x_n lies in every set of the play. When x_n is rational, the set U_n contains a usual open interval about x_n, and Player II answers with an interval $V_n = (x_n - r_n, x_n + r_n)$, where $r_n > 0$ is so small that the closed interval $[x_n - r_n, x_n + r_n]$ lies in U_n. If

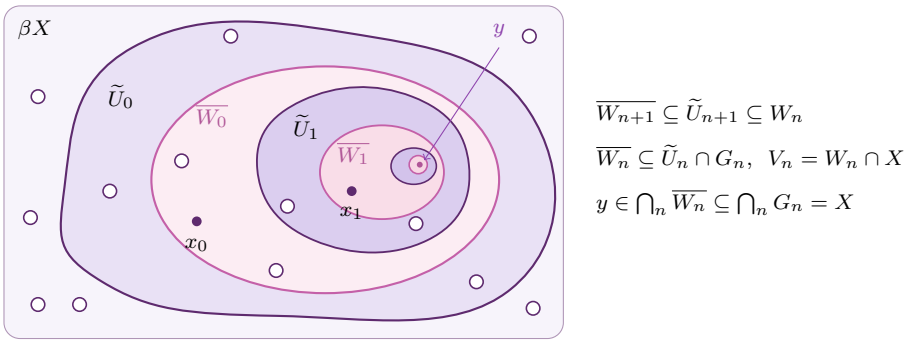


FIGURE 10. The proof of Proposition 4.8, drawn in βX. The white dots are points of βX that are not in X. The open sets $\widetilde{U}_n$ of βX that lift the moves of Player I (plum) alternate with the closures of the sets W_n of Player II (pink), each inside the previous one. Each closure $\overline{W_n}$ lies in G_n and therefore misses the points outside G_n; since every point of βX outside X lies outside some G_n, the white dots are left behind one after another. The nested compact sets $\overline{W_n}$ meet in a point y, which lies in every G_n, hence in X.

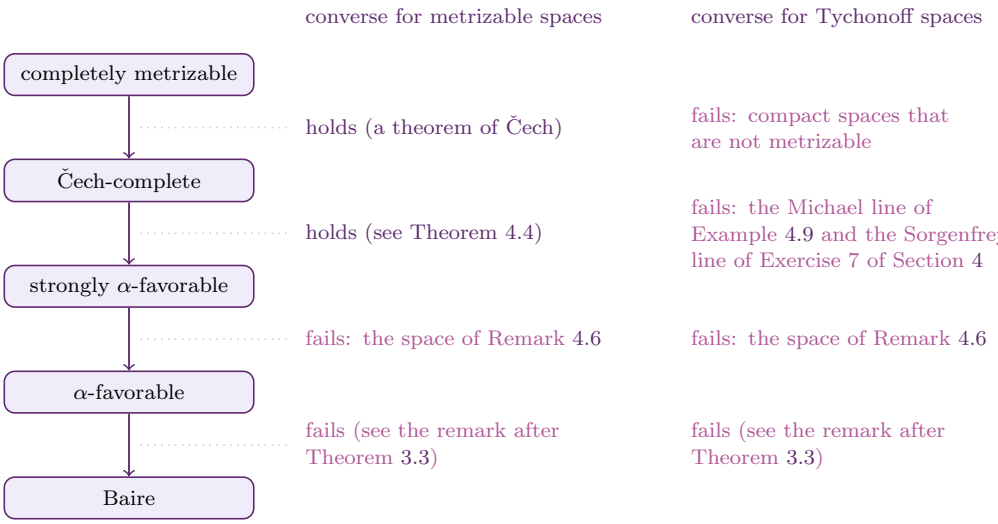


FIGURE 11. Completeness properties of a Tychonoff space, from the strongest to the weakest. The arrows come from Proposition 4.8, Lemma 4.2 and Theorem 3.3, together with the fact that complete metrizability implies Čech-completeness. The two columns on the right tell whether the converse of each arrow holds among metrizable spaces and among all Tychonoff spaces, and name a space that refutes it when it fails.

Player I plays only rational points, these closed intervals are nested, since each lies in the previous answer; being compact, they have a common point, and that point lies in every U_n.

(2) *The space M is not Čech-complete.* The set $\mathbb{Q}$ is closed in M, because its complement consists of isolated points, and its subspace topology is the usual one. If M were Čech-complete, then its closed subspace $\mathbb{Q}$ would be Čech-complete as well, since closed

subspaces of Čech-complete spaces are Čech-complete (see Engelking's book [Eng89, Theorem 3.9.6]), and hence a Baire space; yet $\mathbb{Q}$, the union of countably many nowhere dense singletons, is not a Baire space.

The Michael line separates the two properties in one more respect: unlike Čech-completeness, strong α-favorability does not pass to closed subspaces, since Player I wins $\mathrm{Ch}(\mathbb{Q})$ (see the example after Lemma 4.2).

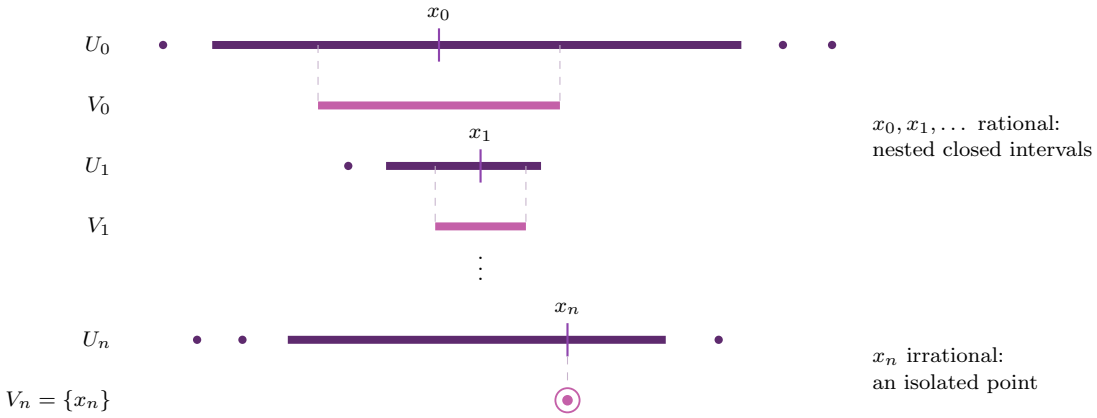


FIGURE 12. Two kinds of answers in the strong Choquet game on the Michael line M of Example 4.9. An open set of M is drawn as a union of open intervals together with isolated irrational points (the small dots). Top: while Player I (plum) plays rational points, Player II (pink) answers with intervals whose closures lie in his sets; these closed intervals are nested and therefore have a common point. Bottom: once Player I plays an irrational point, Player II answers with the singleton of that point, and every later move is the same singleton.

Among paracompact spaces, Čech-completeness also has an exact characterization by a game of another kind, played not on the space but on what the Stone–Čech compactification adds to it: a paracompact space X is Čech-complete if and only if Player II has a winning strategy in the Menger game of Section 14 below, played on the remainder $\beta X \setminus X$. Aurichi and Bella [AB15, Corollary 3.3] deduce this equivalence from a theorem of Telgársky [Tel84].

Exercises.

(1) Let X be a non-empty compact Hausdorff space. Show directly, without Proposition 4.8, that Player II has a winning strategy in $\mathrm{Ch}(X)$, that is, that X is strongly α-favorable.
(2) Let X be a non-empty completely metrizable space with no isolated points. Show that X is uncountable. (Hint: each singleton is closed with empty interior; use Corollary 4.5.)
(3) The interval $(0, 1)$ is not complete in the usual metric. Show directly that Player II has a winning strategy in $\mathrm{Ch}((0, 1))$. (Hint: imitate the proof of Theorem 4.4, choosing $r_n < 2^{-n}$ so that the closed interval $[x_n - r_n, x_n + r_n]$ lies in U_n.)
(4) Let X be the space of Remark 4.6. Write out in full the winning strategy for Player I in $\mathrm{Ch}(X)$ indicated in Figure 9, and check that it is winning. (Hint: enumerate

$\mathbb{Q} \times \{0\}$ as $p_0, p_1, \ldots$, and at step n make sure that p_n lies outside a closed disk of radius less than 2^{-n} that contains U_n and lies in the previous such disk.)

(5) $(*)$ Let $P = \mathbb{R} \setminus \mathbb{Q}$ be the space of irrational numbers, with the usual metric, and enumerate $\mathbb{Q}$ as $q_0, q_1, \ldots$. Show directly that Player II has a winning strategy in $\mathrm{Ch}(P)$. (Hint: answer (x_n, U_n) with $V_n = (x_n - r_n, x_n + r_n) \cap P$, where the closed interval $[x_n - r_n, x_n + r_n]$ misses q_n and its points in P lie in U_n.)

(6) Recall that a space is *countably compact* if every countable open cover has a finite subcover, or, equivalently, if every decreasing sequence of non-empty closed sets has a non-empty intersection. Show that every countably compact regular space is strongly α-favorable (this extends Exercise 1). (Hint: answer (x_n, U_n) with an open set V_n such that $x_n \in V_n$ and $\overline{V_n} \subseteq U_n$.)

(7) $(*)$ Let S be the Sorgenfrey line, the set $\mathbb{R}$ with the topology whose basic open sets are the intervals $[a, b)$. Show that S is strongly α-favorable. (Hint: answer (x_n, U_n) with $V_n = [x_n, x_n + r_n)$, where $r_n > 0$ and $[x_n, x_n + 2r_n) \subseteq U_n$; the points x_n form a non-decreasing sequence, bounded above by $x_0 + r_0$, and its limit lies in every U_n.) The space S is a second strongly α-favorable space that is not Čech-complete: Čech-complete spaces are strongly Baire (see the remark after Theorem 10.1 below), whereas S is not (see Exercise 4 of Section 10).

5. The converse of Choquet's theorem

In this section we prove the hard half of Theorem 4.4: every metrizable space X in which Player II has a winning strategy in $\mathrm{Ch}(X)$ is completely metrizable. The proof is long, so we first explain where a complete metric can come from and what the strategy has to do with it.

The completion. Fix a compatible metric d on X, and let $\widetilde{X}$ be the completion of (X, d). If X is complete, then $\widetilde{X} = X$; in general the completion contains new points, and the position of X inside it decides whether X is completely metrizable. Two subspaces of $\mathbb{R}$ show the difference. The irrationals form the G_δ set $\bigcap_{q \in \mathbb{Q}} (\mathbb{R} \setminus \{q\})$, and they are completely metrizable: continued fractions make them homeomorphic to the space of sequences of positive integers, which is complete in its usual metric. The rationals do not form a G_δ subset of $\mathbb{R}$: otherwise the rationals and the irrationals would be two disjoint dense G_δ subsets of $\mathbb{R}$, whereas, by the Baire category theorem (see Corollary 4.5), the intersection of two dense G_δ subsets of $\mathbb{R}$ is dense, and in particular non-empty. In Lemma 5.2 below we show that every G_δ subset of a complete metric space is completely metrizable. Our goal is therefore to find open sets $G_1, G_2, \ldots$ of $\widetilde{X}$ with $X = \bigcap_n G_n$. In the proof of Proposition 4.8 a representation of this kind, inside βX, produced a winning strategy; here the strategy must produce the representation.

The open sets of the strategy. The strategy of Player II is a machine that answers points with small open sets. At the first stage, Player I may play any point x of X (with the open set X), and the strategy answers with an open set $V \ni x$. These sets cover X; once each of them is enlarged slightly into $\widetilde{X}$, their union is an open subset G_1 of $\widetilde{X}$ that contains X. At the second stage, Player I continues each of these positions with each point of the set just played, and so on. The open set G_n is, roughly, the union of the (enlarged) sets played by Player II at stage n, over all positions, and it therefore contains X. Suppose now that a point y of the completion lies in every G_n. If the sets containing y at the various stages all came from *one* play, then that play would have a common point, because Player II wins it.

The sets will be chosen with diameters tending to 0, so this common point would be y; in particular, the point y would belong to X.

The difficulty. At each stage, however, the point y may lie in sets that come from many different positions, and a choice of one position at each stage need not form a single play: for that, the position chosen at stage $n+1$ must extend the one chosen at stage n. If at every stage only *finitely* many positions are involved, then a compactness argument, Kőnig's lemma (see Lemma 5.4 below), makes the choices fit together. The finiteness is supplied by a theorem of A. H. Stone (see Lemma 5.3 below): every open cover of a metric space has an open refinement, still covering the space, in which each point has a neighborhood that meets only finitely many members. Figure 13 shows how the two ingredients combine.

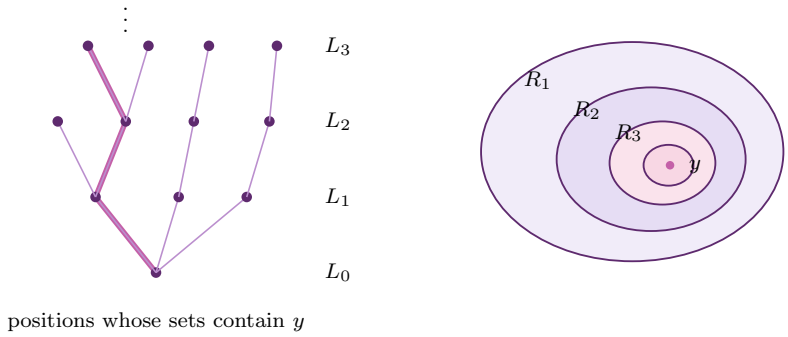


FIGURE 13. The end of the proof of Theorem 4.4. Left: for a point y of the completion lying in every G_n, the set L_n of positions of stage n whose sets contain y is finite and non-empty, and each member of L_{n+1} extends a member of L_n. Kőnig's lemma provides a branch (pink), that is, a single play. Right: along that play the sets R_n containing y shrink to a point; since Player II wins, the play has a common point in X, which must be y.

5.1. **A preliminary lemma and three classical tools.** Throughout this subsection we fix a metric space (X, d) and a metric space $(\widetilde{X}, \tilde{d})$ that contains X as a dense subset, with $\tilde{d}$ extending d. Closures and diameters of subsets of $\widetilde{X}$ are taken in $\widetilde{X}$. In the preliminary lemma we extend each open set of X to an open set of $\widetilde{X}$ with the same points of X and the same diameter (see Figure 14). For an open set $V \subseteq X$, we write $\widetilde{V} = \widetilde{X} \setminus \overline{X \setminus V}$.

Lemma 5.1. *Let V and V' be open subsets of X. Then the following hold:*

(1) *the set $\widetilde{V}$ is open in $\widetilde{X}$, and $\widetilde{V} \cap X = V$;*
(2) *if $V \subseteq V'$, then $\widetilde{V} \subseteq \widetilde{V'}$;*
(3) *the diameters satisfy* $\operatorname{diam} \widetilde{V} = \operatorname{diam} V$.

Proof. (1) The set $\widetilde{V}$ is the complement of a closed set, hence open. Let $x \in V$ be given. Since V is open in X, some ball $B(x, \varepsilon)$ of X lies in V; the ball of radius ε about x in $\widetilde{X}$ meets X exactly in $B(x, \varepsilon)$, and it therefore misses $X \setminus V$, which gives $x \in \widetilde{V}$. If $x \in X \setminus V$, then $x \in \overline{X \setminus V}$, and $x \notin \widetilde{V}$. Thus $\widetilde{V} \cap X = V$.

(2) If $V \subseteq V'$, then $X \setminus V' \subseteq X \setminus V$, and the complements of the closures reverse the inclusion.

(3) We first show that $\widetilde{V} \subseteq \overline{V}$. Let $y \in \widetilde{V}$ and $\varepsilon > 0$ be given. The points of $\widetilde{V}$ at distance less than ε from y form a non-empty open set, which meets the dense set X; by (1), every

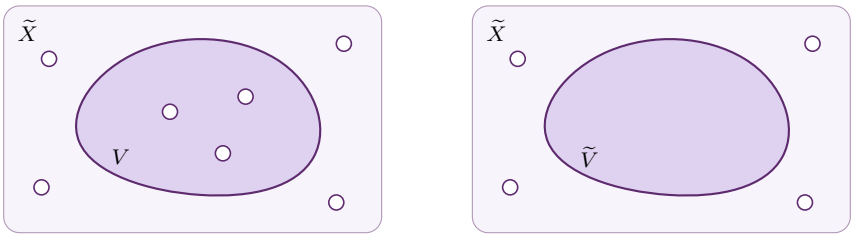


FIGURE 14. Lemma 5.1. The white dots are points of $\widetilde{X}$ that are not in X. Left: an open set V of X, which misses the white dots inside it. Right: the open set $\widetilde{V}$ of $\widetilde{X}$, which also contains them. The two sets have the same points of X and the same diameter.

point of $\widetilde{V} \cap X$ lies in V. Since ε is arbitrary, every ball about y meets V, that is, $y \in \overline{V}$. It follows that $\operatorname{diam} \widetilde{V} \le \operatorname{diam} \overline{V} = \operatorname{diam} V$. The reverse inequality holds because $V \subseteq \widetilde{V}$. ☺

The first tool is the theorem of Alexandrov announced above; it reduces our problem to finding a G_δ set.

Lemma 5.2 (P. S. Alexandrov). *Every G_δ subset of a complete metric space is completely metrizable.*

Proof. Let (Y, ρ_0) be a complete metric space, and let $G = \bigcap_{n\ge1} G_n$ with each G_n open in Y. The idea is to add to ρ_0 terms that blow up near the complements of the sets G_n, so that a Cauchy sequence cannot approach them. We may assume that each closed set $F_n = Y \setminus G_n$ is non-empty, discarding the indices with $G_n = Y$ (if no index remains, then $G = Y$ and there is nothing to prove). For $x \in G$, the distance $\rho_0(x, F_n)$ is positive, because F_n is closed and $x \notin F_n$; we put $f_n(x) = 1/\rho_0(x, F_n)$, a continuous function on G, and for $x, y \in G$ we put

$$\rho(x, y) = \rho_0(x, y) + \sum_{n\ge1} 2^{-n} \min\{1,\, |f_n(x) - f_n(y)|\}.$$

Each summand of ρ is a pseudometric on G, the series converges, and the first summand ρ_0 is a metric; the sum ρ is therefore a metric on G.

The metric ρ is compatible with the topology of G. Indeed, convergence for ρ implies convergence for ρ_0, because $\rho \ge \rho_0$. Conversely, suppose that $x_k \to x$ in G with respect to ρ_0. Each term of the series then tends to 0, by the continuity of f_n, and the nth term never exceeds 2^{-n}; dominated convergence for series yields $\rho(x_k, x) \to 0$.

The metric ρ is complete. Let (x_k) be a ρ-Cauchy sequence in G. Since $\rho \ge \rho_0$, the sequence is ρ_0-Cauchy, and it converges to some $y \in Y$ because ρ_0 is complete. For each n, the sequence $(f_n(x_k))_k$ is Cauchy, because $\min\{1, |f_n(x_k) - f_n(x_l)|\} \le 2^n \rho(x_k, x_l)$; it is therefore bounded by some constant c_n. Thus $\rho_0(x_k, F_n) \ge 1/c_n$ for every k, and in the limit $\rho_0(y, F_n) \ge 1/c_n > 0$. It follows that $y \in G_n$ for every n, that is, $y \in G$, and $\rho(x_k, y) \to 0$ by the compatibility just proved. Consequently, the space G carries the complete compatible metric ρ and is completely metrizable. ☺

The second tool supplies the finiteness. A family $\{V_i : i \in I\}$ of subsets of a topological space Y is *locally finite* if every point of Y has a neighborhood that meets V_i for only finitely many indices $i \in I$. A family $\mathcal{V}$ *refines* a family $\mathcal{U}$ if every member of $\mathcal{V}$ is contained in some member of $\mathcal{U}$.

Lemma 5.3 (A. H. Stone). *Every open cover of a metric space has a locally finite open refinement that covers the space.*

The idea of the proof is to cover the space by small balls, to assign each ball to the first member of the cover that contains it with room to spare, and to stop using small balls around points that larger balls already cover.

Proof. Let (Y, ρ) be a metric space, and let $\{U_s : s \in \Lambda\}$ be an open cover of Y; we write $B(c, r)$ for the open ball of radius r about c. By the well-ordering theorem we may assume that Λ is well ordered. For $s \in \Lambda$ and $n \geq 1$, let $V_{s,n}$ be the union of the balls $B(c, 2^{-n})$ over all centers $c \in Y$ such that:

(i) the index s is the least one with $c \in U_s$;
(ii) the center c lies in no set $V_{t,j}$ with $t \in \Lambda$ and $j < n$;
(iii) the ball $B(c, 3 \cdot 2^{-n})$ is contained in U_s.

The definition is a recursion on n, because (ii) involves only the sets with smaller second index. Each $V_{s,n}$ is open, and $V_{s,n} \subseteq U_s$ by (iii).

The sets $V_{s,n}$ cover Y. Let $c \in Y$ be given, let s be the least index with $c \in U_s$, and let n be so large that $B(c, 3 \cdot 2^{-n}) \subseteq U_s$. Either the point c lies in some $V_{t,j}$ with $j < n$, or it satisfies (i)–(iii), and then $c \in B(c, 2^{-n}) \subseteq V_{s,n}$.

The family is locally finite. Let $c \in Y$ be given, and let s and n be indices with $c \in V_{s,n}$; since $V_{s,n}$ is open, there is $j \geq 1$ with $B(c, 2^{-j}) \subseteq V_{s,n}$. We show that the ball $B(c, 2^{-n-j})$ meets no set $V_{t,i}$ with $i \geq n + j$, and at most one set $V_{t,i}$ for each $i < n + j$; the ball then meets at most $n + j - 1$ sets of the family.

Suppose first that $i \geq n + j$, and let c' be a center used for $V_{t,i}$. Since $n < i$, condition (ii) gives $c' \notin V_{s,n}$, so $\rho(c, c') \geq 2^{-j}$. Every point of $B(c', 2^{-i})$ is therefore at distance at least $2^{-j} - 2^{-i} \geq 2^{-j} - 2^{-n-j} \geq 2^{-n-j}$ from c, and the ball $B(c, 2^{-n-j})$ misses $V_{t,i}$.

Suppose next that $i < n + j$, and that $p \in V_{t,i}$ and $q \in V_{t',i}$ with $t < t'$. There are centers c_1 for $V_{t,i}$ and c_2 for $V_{t',i}$ with $\rho(p, c_1) < 2^{-i}$ and $\rho(q, c_2) < 2^{-i}$. Condition (iii) places the ball $B(c_1, 3 \cdot 2^{-i})$ inside U_t, and condition (i) keeps c_2 out of U_t, because t' is the least index with $c_2 \in U_{t'}$ and $t < t'$. Hence $\rho(c_1, c_2) \geq 3 \cdot 2^{-i}$ and

$$\rho(p, q) \geq \rho(c_1, c_2) - \rho(p, c_1) - \rho(q, c_2) > 2^{-i} \geq 2 \cdot 2^{-n-j}.$$

Two points of $B(c, 2^{-n-j})$ are at distance less than $2 \cdot 2^{-n-j}$, so the ball cannot meet both $V_{t,i}$ and $V_{t',i}$. The sets $V_{s,n}$ therefore form a locally finite open refinement of $\{U_s : s \in \Lambda\}$ that covers Y. ☺

The third tool turns finitely many choices at each stage into one coherent choice.

Lemma 5.4 (D. Kőnig). *Let $L_0, L_1, \ldots$ be non-empty finite sets, and for each n let $\pi_n \colon L_{n+1} \to L_n$ be a function. Then there are $\ell_n \in L_n$, for $n \geq 0$, with $\pi_n(\ell_{n+1}) = \ell_n$ for every n.*

Proof. The idea is to follow, level by level, the elements that are reached from every later level. For $m \geq n$ we write $\pi_{n,m} \colon L_m \to L_n$ for the composite $\pi_n \circ \cdots \circ \pi_{m-1}$ (the identity when $m = n$). We call $\ell \in L_n$ *good* if for every $m \geq n$ some element of L_m is mapped to ℓ by $\pi_{n,m}$. If $\ell = \pi_{n,m}(z)$ for some $z \in L_m$, then $\ell = \pi_{n,k}(\pi_{k,m}(z))$ for every k with $n \leq k \leq m$. Hence an element of L_n is good as soon as it lies in the image of $\pi_{n,m}$ for infinitely many m.

Some element of L_0 is good: every L_m is non-empty, so each m supplies an element of L_0 in the image of $\pi_{0,m}$, and since L_0 is finite, one element is supplied by infinitely many m. If $\ell \in L_n$ is good, then some element of the finite set $\pi_n^{-1}(\ell)$ is good: for each $m > n$, an element of L_m mapped to ℓ passes through some element of $\pi_n^{-1}(\ell)$, and one element of $\pi_n^{-1}(\ell)$ serves infinitely many m. We may therefore choose, recursively, a good $\ell_0 \in L_0$ and,

for each n, a good $\ell_{n+1} \in \pi_n^{-1}(\ell_n)$; the resulting sequence satisfies $\pi_n(\ell_{n+1}) = \ell_n$ for every n, as required. ☺

5.2. **The proof.**

Proof of the converse of Theorem 4.4. Let X be metrizable, and let τ be a winning strategy for Player II in $\mathrm{Ch}(X)$. Following the plan described at the beginning of this section, we build from τ open sets $G_1, G_2, \dots$ in a completion of X, each containing X, and we use Kőnig's lemma to show that their intersection is exactly X. Let d be a compatible metric on X, and let $(\widetilde{X}, \tilde{d})$ be the completion of (X, d), in which X is dense; the notation $\widetilde{V}$ is that of Lemma 5.1. We write $B(x, r)$ for the open ball of radius r about x in X. A *position of stage* n is a position of length $2n$ of $\mathrm{Ch}(X)$ consistent with τ; it ends with the move V_{n-1} of Player II when $n \geq 1$, and it is the empty position when $n = 0$.

Step 1: the families $\mathcal{S}_n$. By recursion on $n \geq 0$ we construct families $\mathcal{S}_n$ of pairs (R, p), where R is a non-empty open subset of $\widetilde{X}$ and p is a position of stage n. The union of the sets R with $(R, p) \in \mathcal{S}_n$ is denoted by G_n. We require the following:

- (P1) if $n \geq 1$ and $(R, p) \in \mathcal{S}_n$, then $R \subseteq \widetilde{V}$, where V is the last move of p, and $\operatorname{diam} R \leq 2^{-n}$;
- (P2) the set G_n contains X;
- (P3) the family of sets R, $(R, p) \in \mathcal{S}_n$, is locally finite in G_n;
- (P4) each $(R, p) \in \mathcal{S}_{n+1}$ has a designated *parent* $(R', p') \in \mathcal{S}_n$ such that $R \subseteq R'$ and p extends p' by two moves: Player I plays $(x, R' \cap X)$ for some point x, and Player II answers as τ prescribes.

We put $\mathcal{S}_0 = \{(\widetilde{X}, \langle\rangle)\}$; conditions (P2) and (P3) hold trivially, and (P1) is vacuous for $n = 0$.

Suppose that $\mathcal{S}_n$ has been constructed. Let $(R', p') \in \mathcal{S}_n$ and $x \in R' \cap X$ be given. The move $(x, R' \cap X)$ is legal for Player I at p': for $n = 0$ it is a first move, and for $n \geq 1$ condition (P1) and Lemma 5.1(1) give $R' \cap X \subseteq \widetilde{V} \cap X = V$, where V is the last move of p'. Let V_x be the answer of τ, which satisfies $x \in V_x \subseteq R' \cap X$, and let q_x be the resulting position of stage $n + 1$. (The sets V_x, the positions q_x and the sets Q_x and O_x below depend also on the pair (R', p'), which we suppress from the notation.) We put

$$Q_x = V_x \cap B(x, 2^{-n-2}) \qquad \text{and} \qquad O_x = R' \cap \widetilde{Q_x}.$$

The set O_x is open in $\widetilde{X}$ and contained in R'. It contains x, because $x \in R'$ and $x \in Q_x \subseteq \widetilde{Q_x}$ by Lemma 5.1(1); it lies in $\widetilde{V_x}$, because $Q_x \subseteq V_x$ and Lemma 5.1(2) applies; and its diameter is at most $\operatorname{diam} \widetilde{Q_x} = \operatorname{diam} Q_x \leq 2^{-n-1}$, by Lemma 5.1(3) and because Q_x lies in a ball of radius 2^{-n-2}. By (P2), every point x of X lies in some R' with $(R', p') \in \mathcal{S}_n$, and then $x \in O_x$; hence the sets O_x, over all pairs $(R', p') \in \mathcal{S}_n$ and all $x \in R' \cap X$, cover X. Their union H is open in $\widetilde{X}$, and the sets O_x form an open cover of the metric space H. By Lemma 5.3, this cover has a locally finite open refinement $\mathcal{R}$ that covers H; we discard its empty members. For each $R \in \mathcal{R}$ we choose a set O_x that contains R, coming from a pair (R', p') and a point x, and we put (R, q_x) into $\mathcal{S}_{n+1}$, with designated parent (R', p').

Conditions (P1)–(P4) hold for $n + 1$. Indeed, $R \subseteq O_x \subseteq \widetilde{V_x}$, where V_x is the last move of q_x, and $\operatorname{diam} R \leq \operatorname{diam} O_x \leq 2^{-n-1}$; the union G_{n+1} equals H, which contains X; the family $\mathcal{R}$ is locally finite in $H = G_{n+1}$; and $R \subseteq O_x \subseteq R'$, with q_x extending p' by the moves $(x, R' \cap X)$ and V_x.

Step 2: a G_δ *set containing* X. The set $G = \bigcap_{n \geq 1} G_n$ is a G_δ subset of $\widetilde{X}$, and $X \subseteq G$ by (P2). It remains to be shown that $G \subseteq X$.

Step 3: every point of G is the common point of a play. Let $y \in G$ be given. For $n \geq 0$, let L_n be the set of pairs $(R, p) \in \mathcal{S}_n$ with $y \in R$. Each L_n is non-empty, because $y \in G_n$ (and L_0 consists of the single pair $(\widetilde{X}, \langle\rangle)$). Each L_n is also finite, by (P3): since $y \in G_n$, some neighborhood of y meets only finitely many of the sets R with $(R, p) \in \mathcal{S}_n$, and every such R that contains y is among them. By (P4), the parent of a member of L_{n+1} contains y, so it lies in L_n; let $\pi_n \colon L_{n+1} \to L_n$ be the parent map. By Lemma 5.4, there are $(R_n, p_n) \in L_n$ such that (R_n, p_n) is the parent of (R_{n+1}, p_{n+1}) for every n.

By (P4), each position p_{n+1} extends p_n by a move $(x_n, R_n \cap X)$ of Player I and the answer of τ. Hence the positions p_n fit together into a single play of $\mathrm{Ch}(X)$, consistent with τ, in which Player I plays the sets $R_n \cap X$. Since τ is winning, there is a point $z \in \bigcap_n (R_n \cap X)$. For every $n \geq 1$, the points y and z both lie in R_n, which has diameter at most 2^{-n} by (P1). Hence $\tilde{d}(y, z) = 0$, and $y = z \in X$ (see Figure 13).

Step 4: conclusion. By Steps 2 and 3, we have $X = G$, a G_δ subset of the complete metric space $\widetilde{X}$; since $\tilde{d}$ extends d, the topology that X inherits from $\widetilde{X}$ is its own. By Lemma 5.2, the space X is completely metrizable. ☺

Remark 5.5. The points of Player I are what make the proof work: in Step 1, Player I plays every point x of every set $R' \cap X$, and Player II must answer with a set that contains x; this is why each G_n contains all of X. Without the points, the theorem fails (see Remark 4.6). For separable spaces, Kechris gives a different proof [Kec95, Section 8].

5.3. **Summary of Part I.** The main results of this part fit into one chain of implications, which, for a metrizable space X, is the following:

X is completely metrizable
$\Updownarrow$ Theorem 4.4
Player II has a winning strategy in $\mathrm{Ch}(X)$
$\Downarrow$ Lemma 4.2
Player II has a winning strategy in $\mathrm{BM}(X)$
$\Downarrow$ Remarks 2.3(3)
Player I has no winning strategy in $\mathrm{BM}(X)$
$\Updownarrow$ Theorem 3.3
X is a Baire space
$\Updownarrow$ Theorem 1.3
no non-empty open subset of X is of the first category.

Followed from top to bottom, the chain yields the Baire category theorem (see Corollary 4.5). The second implication cannot be reversed: the space X of Remark 4.6 is metrizable and Player II wins $\mathrm{BM}(X)$, but the space is not completely metrizable, and Player II therefore has no winning strategy in $\mathrm{Ch}(X)$. The games with a target set measure the size of sets inside X: Player II wins $\mathrm{BM}(X, S)$ exactly when S is comeager, and, in a complete metric space, Player I wins exactly when S is of the first category in some non-empty open set (see Theorem 3.5).

Exercises.

(1) Let X be the space $\mathbb{R} \setminus \{0\}$ with the usual metric, whose completion is $\widetilde{X} = \mathbb{R}$. Compute $\widetilde{V}$ for $V = (-1, 0) \cup (0, 1)$ and for $V = (0, 1)$, and check the three statements of Lemma 5.1 in both cases.

(2) Show that the open cover $\{(-n, n) : n \geq 1\}$ of $\mathbb{R}$ is not locally finite, and find a locally finite open refinement of it that covers $\mathbb{R}$, as promised by Lemma 5.3.
(3) Apply the proof of Lemma 5.2 to $G = (0, 1)$, a single open subset of $Y = \mathbb{R}$, and write down the metric ρ explicitly. Show that the sequence $(1/k)_{k\geq 2}$ is Cauchy for the usual metric but not for ρ.
(4) Show that the conclusion of Lemma 5.4 can fail if the sets L_n are allowed to be infinite. (Hint: let L_0 have one element, let L_n consist of the pairs (i, n) with $i \geq n$ for $n \geq 1$, let π_0 be constant, and let $\pi_n(i, n + 1) = (i, n)$ for $n \geq 1$.)
(5) $(*)$ Let (X, d) be a metric space whose topology is completely metrizable, possibly by a metric other than d. Use the proof of Theorem 4.4 to show that X is a G_δ subset of the completion of (X, d). Deduce that $\mathbb{Q}$ is not completely metrizable.

6. Notes and historical remarks

6.1. **Baire category.** Theorem 1.3 is an elementary reformulation of the definition and carries no attribution. It is not to be confused with the *Banach category theorem* [Ban30], which is deeper: in any topological space, the union of an arbitrary family of open sets of the first category is again of the first category. The title of Oxtoby's paper [Oxt57] refers to this theorem.

6.2. **The Banach–Mazur and strong Choquet games.** As told in the Introduction, the Banach–Mazur game is Problem 43 of the Scottish Book, posed by S. Mazur in 1935 [Mau81] (see also Section 20). In Mazur's formulation, two players alternately choose nested intervals of the real line, and the first player wins if their intersection meets a given set E. Mazur observed two facts: the first player wins if E is comeager in some interval, and the second player wins if E is of the first category. He conjectured the converses. On 4 August 1935 Banach recorded in the Scottish Book that Mazur's conjecture is true. Banach's proof was never published. The first published proof is Oxtoby's [Oxt57]. He showed that in an arbitrary topological space the second player has a winning strategy exactly when E is of the first category. The other converse, that the first player wins exactly when E is comeager in some non-empty open set, requires further hypotheses on the space; it holds, for instance, in complete metric spaces. Together, Oxtoby's theorem and this converse are Theorem 3.5, under the following dictionary:

(1) Mazur's first player is our Player I, and Mazur's second player is our Player II.
(2) Mazur's set E, at which the first player aims, is $X \setminus S$, where S is the set at which Player II aims in $\mathrm{BM}(X, S)$.
(3) "The intersection meets E" becomes "the intersection is not contained in S", so Mazur's first player wins exactly when our Player I wins.

Whether the moves are open or closed intervals makes no difference to the proofs. The game $\mathrm{BM}(X)$, in which Player II needs only a non-empty intersection, differs from $\mathrm{BM}(X, X)$, which Player II wins trivially; the proof of Theorem 3.3 is Oxtoby's argument adapted to $\mathrm{BM}(X)$. Telgársky's survey [Tel87], written for the fiftieth anniversary of the Banach–Mazur game, covers its later history.

The strong Choquet game goes back to Choquet's work on Baire spaces [Cho58]. Choquet expounds the game, together with Theorem 4.4, in his *Lectures on Analysis* [Cho69], where the players are called β and α. Kechris states Theorem 4.4 for separable metrizable spaces, where it characterizes the Polish spaces [Kec95, Section 8]. The proof in Section 5, for arbitrary metrizable spaces, rests on three classical results. Lemma 5.2 is due to Alexandrov

[Ale24]. Lemma 5.3 is A. H. Stone's theorem that metric spaces are paracompact [Sto48]; the short proof given here is M. E. Rudin's [Rud69]. Lemma 5.4 is Kőnig's infinity lemma [Kőn27], first stated for graphs.

6.3. **Čech-complete spaces.** Čech introduced the spaces now named after him in 1937, in the paper in which he constructed the compactification βX independently of Stone [Čec37]; he called them topologically complete, and he proved that a metrizable space is topologically complete exactly when it is completely metrizable. Proposition 4.8 is folklore. The survey of Telgársky mentioned above also treats the many variants of the strong Choquet game and the completeness properties that they describe. Michael introduced the space of Example 4.9 in 1963, as a paracompact space whose product with the irrationals is not normal [Mic63].

Part II. Applications of Banach–Mazur and Choquet games

In Part I we showed that the Banach–Mazur game measures size. In this part we put that measure to work outside topology. Each problem of the part asks, in one form or another, that a set be shown to be large, or that an object be shown to exist when no formula for it is in sight. In each case the answer is a game obtained by changing a rule or two, either in the Banach–Mazur game of Section 3 or in a game of Choquet type in which the players shrink sets (see Section 4).

Roadmap

The idea from Part I that recurs most often is the one recorded in the box of the Introduction: a set is large in the sense of Baire category exactly when Player II can steer the play into it (see Theorem 3.5). The sections follow one another from the concrete to the set-theoretic: badly approximable numbers (Section 7) and bounded orbits of flows (Section 8); category inside complexity classes (Section 9); joint continuity in topological groups (Section 10); the Ellentuck topology and infinite-dimensional Ramsey theory (Section 11); Asplund spaces and Gowers's dichotomy (Section 12); and, last, nonprincipal ultrafilters on ω and the axiom of choice (Section 13).

The sections can be read in any order, with three exceptions: Section 8 builds on the proof in Section 7, and Sections 12 and 13 both draw on the Ramsey theory of Section 11. The sections also differ in what they ask of the reader. In the sections on Diophantine approximation, computational complexity and ultrafilters, which need little beyond Part I, we give complete proofs of the main points. The sections on homogeneous dynamics and Banach spaces require some graduate background (matrix groups, functional analysis). In the sections on topological groups and Ramsey theory we sketch the main arguments rather than prove every step. Subsections marked *optional* may be skipped on a first reading. The games of this part are set side by side in Table 1.

Section	Moves	Player II wins if	Main result
7. Diophantine approximation	closed balls whose radii shrink by the fixed factors α and β	the limit point lies in S	Theorem 7.1
8. Homogeneous dynamics	Schmidt's game on an expanding leaf of the flow	the orbit of the limit point is bounded	Theorem 8.1
9. Complexity	finite binary strings, each extending the last; Player II must compute her moves quickly	the language produced is not in $\mathcal{C}$	Theorem 9.1
10. Topological groups	open sets, as in $\mathrm{BM}(X)$	the intersection is non-empty and sequences from D cluster	Theorem 10.1
11. Ramsey theory	Ellentuck basic sets; Player II adds one element at each move	the set produced lies in $\mathcal{A}$	Theorem 11.1
12. Banach spaces	arbitrary sets against relatively weak*-open sets; block subspaces against vectors	the sets shrink to a point in norm; the vectors lie in Σ	Theorems 12.1 and 12.2
13. Ultrafilters	finite binary strings; sets in $\mathcal{U}$ against points or finite sets	the sequence lies in Z; the set produced lies in $\mathcal{U}$	Theorem 13.1

TABLE 1. The games of Part II. In every row Player I moves first and the payoff set belongs to Player II.

7. Applications to Diophantine approximation

The problem. Every real number can be approximated by fractions, but some numbers are much harder to approximate than others. For $t \in \mathbb{R}$ let $\|t\|$ be the distance from t to the nearest integer. A vector $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ is *badly approximable* if there is a constant $c = c(\mathbf{x}) > 0$ such that

$$\max_{1 \le i \le n} \|q x_i\| \;\ge\; c\, q^{-1/n} \qquad \text{for every integer } q \ge 1.$$

For $n = 1$ the condition becomes $|x - p/q| \ge c/q^2$ for every fraction p/q. Let $\mathbf{Bad}(n)$ be the set of badly approximable vectors. By Dirichlet's theorem (1842), every $\mathbf{x} \in \mathbb{R}^n$ satisfies $\max_i \|q x_i\| < q^{-1/n}$ for infinitely many integers $q \ge 1$. Badly approximable vectors are those for which this bound cannot be improved by more than a constant factor: they are the most stubborn vectors there are. How big is $\mathbf{Bad}(n)$? In particular, is the intersection of countably many translates of $\mathbf{Bad}(n)$ still big?

7.1. **History of the problem and of the solution using topological games.** For $n = 1$ there is a classical description: an irrational number is badly approximable exactly when the partial quotients in its continued fraction expansion are bounded. The golden ratio $(1 + \sqrt{5})/2 = [1; 1, 1, 1, \dots]$, whose partial quotients are as small as they can be, is, as Hurwitz showed in 1891, the most badly approximable number of all. Badly approximable numbers are at once rare and common. Khintchine proved in 1924 that $\mathbf{Bad}(1)$ has Lebesgue measure zero, so a number chosen at random is almost surely *not* badly approximable. Jarník

proved in 1928 that **Bad**(1) has Hausdorff dimension 1, the largest possible. The set **Bad**(1) is thus tiny for measure and huge for dimension.

Neither notion helps with intersections. Measure sees only that **Bad**(1) is a null set, and two sets of full Hausdorff dimension can be disjoint: the set **Bad**(1) and its complement are an example, since the complement has full measure and hence full dimension. A question as innocent as whether there is a number x such that $x+\sqrt{2}$, $x+\sqrt{3}$, $x+\sqrt{5}$, ... are all badly approximable was therefore out of reach.

Wolfgang Schmidt solved the problem in 1966 [Sch66]. To do so he invented a new version of the Banach–Mazur game, now called Schmidt's game. He proved that its winning sets have full Hausdorff dimension and survive countable intersections, and that **Bad**(1) is one of them. In 1969 he extended the last result to **Bad**(n), for every n [Sch69].

The game had more to give. For weights $i, j > 0$ with $i + j = 1$, let **Bad**(i, j) be the set of vectors $(x, y) \in \mathbb{R}^2$ that are badly approximable with weights (i, j), that is, for which there is $c > 0$ with $\max(\|qx\|^{1/i}, \|qy\|^{1/j}) \geq c/q$ for every integer $q \geq 1$. In 1983 Schmidt conjectured that **Bad**$(\frac{1}{3}, \frac{2}{3})$ and **Bad**$(\frac{2}{3}, \frac{1}{3})$ have a point in common. He observed that a counterexample would prove the Littlewood conjecture. Badziahin, Pollington and Velani proved Schmidt's conjecture in 2011 [BPV11]. In 2016 An showed that the sets **Bad**(i, j) are all winning for Schmidt's game, with one and the same parameter for every pair of weights [An16]; the intersection property of winning sets then yields Schmidt's conjecture in a stronger form, namely that every countable intersection of sets **Bad**(i, j) has full dimension.

Recognition. Schmidt received the Cole Prize in Number Theory in 1972. He addressed the International Congress of Mathematicians three times: in Nice in 1970, in Vancouver in 1974 (a plenary address) and in Warsaw in 1983. The paper of Badziahin, Pollington and Velani that settled Schmidt's conjecture appeared in the *Annals of Mathematics*.

7.2. **The game.** Schmidt's game is the Banach–Mazur game of Section 3 with one new rule: the sizes of the moves are prescribed in advance. Let (X, d) be a complete metric space, let $S \subseteq X$ be given, and fix two numbers $\alpha, \beta \in (0, 1)$. The game is played with closed balls $B[x, r] = \{y \in X : d(x, y) \leq r\}$ of prescribed radii. (In a general metric space a ball, as a set, need not determine its center and radius, and Schmidt formulates the game with the pairs (x, r); in $\mathbb{R}^n$, the only space used below, the distinction disappears.) Schmidt called the players Black and White: Black, who moves first, is our Player I, and White is our Player II. In Choquet's terminology our Player I is β and our Player II is α; in Choquet's game, as in Schmidt's, the letter α thus belongs to Player II. The resemblance ends there: the numbers α and β below are Schmidt's shrinking factors, and α-winning sets have nothing to do with the α-favorable spaces of Section 4.

(1) Player I chooses a closed ball $U_0 = B[x_0, r_0]$ of any radius he likes.
(2) Player II chooses a closed ball $V_0 \subseteq U_0$ of radius αr_0.
(3) Player I chooses a closed ball $U_1 \subseteq V_0$ of radius $\beta\alpha r_0$.
(4) The players continue in this way, and the play produces nested balls $U_0 \supseteq V_0 \supseteq U_1 \supseteq V_1 \supseteq \cdots$, where U_k has radius $r_k = (\alpha\beta)^k r_0$ and V_k has radius αr_k.

Since the radii tend to 0 and the space is complete, the balls close down on a single point x_∞ (as in the proof of Theorem 4.4). Player II wins if $x_\infty \in S$. We call this game the (α, β)-*game for* S. The set S is α-*winning* if Player II has a winning strategy in the (α, β)-game for S for every $\beta \in (0, 1)$. The order of the quantifiers matters. Player II declares her handicap α first, promising to shrink by the factor α every time, and only then does Player I pick β, as aggressive as he likes. Her strategy may depend on β, but the number α is fixed once and for all. This order of quantifiers is what lets α-winning sets survive countable intersections

(see the proof of Theorem 7.1(2) below). On the real line the balls are just closed intervals, and we speak of lengths instead of radii (see Figure 15).

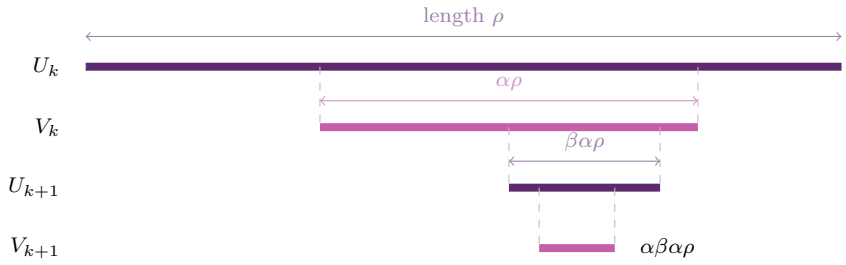


FIGURE 15. Two rounds of Schmidt's (α, β)-game on the line, drawn with $\alpha = \frac{1}{2}$ and $\beta = \frac{2}{5}$. Player I (plum) and Player II (pink) take turns choosing closed intervals, each inside the previous one, and the lengths are prescribed: Player II shrinks by the factor α, Player I by the factor β. Only the positions of the intervals are up to the players.

Because the sizes of the moves are prescribed, neither player can make the balls arbitrarily small in a single move, and this small change lets the game detect Hausdorff dimension (see the proof of Theorem 7.1(1) below). The game also parts company with category. The set **Bad**(1) is winning (see Theorem 7.1 below), and yet it is meager: for each $c > 0$, the numbers x with $|x - p/q| \geq c/q^2$ for every fraction p/q form a closed set that contains no rational number, hence a nowhere dense set, and the union of these sets over $c = 1, \frac{1}{2}, \frac{1}{3}, \ldots$ is **Bad**(1).

7.3. **The Main Result of this section.** Schmidt proved three things about α-winning sets: they are large, they survive countable intersections, and the badly approximable vectors form one of them.

Theorem 7.1 (Schmidt, 1966 and 1969). *Let $0 < \alpha < 1$ be given.*

(1) *If $S \subseteq \mathbb{R}^n$ is α-winning, then the intersection of S with every non-empty open set $U \subseteq \mathbb{R}^n$ has Hausdorff dimension n.*
(2) *The intersection of countably many α-winning subsets of $\mathbb{R}^n$ is α-winning.*
(3) *The set* **Bad**(1) *is $\frac{1}{2}$-winning, and each set* **Bad**(n) *is α-winning for suitable $\alpha > 0$.*

Together, the three parts answer both questions of the problem. A translate of an α-winning set is α-winning (see Exercise 2 at the end of this section). By part (2), the set $\bigcap_k (\mathbf{Bad}(n) + \mathbf{t}_k)$ is therefore α-winning for every sequence of vectors $\mathbf{t}_k$, and by part (1) this intersection has full dimension in every non-empty open set. In particular, every interval contains a set of Hausdorff dimension 1 of numbers x such that $x + \sqrt{p}$ is badly approximable for every prime p.

7.4. **Solving the problem with the game.** We go through the three parts of Theorem 7.1, starting with the one that is pure number theory. For simplicity we prove that **Bad**(1) is α-winning for every $\alpha < \frac{1}{2}$; Schmidt's sharper bookkeeping also handles $\alpha = \frac{1}{2}$ [Sch66].

Badly approximable numbers: the plan. Fix $\alpha \in (0, \frac{1}{2})$ and $\beta \in (0, 1)$, and let Player I open with an interval U_0 of length ρ_0. We put $R = 1/(\alpha\beta) > 1$. Every round shrinks the

intervals by the factor R, and the interval U_k has length $\rho_k = \rho_0 R^{-k}$. We will find a constant $c > 0$ and a strategy for Player II such that the final point x satisfies

$$\left|x - \frac{p}{q}\right| \;\geq\; \frac{c}{q^2} \qquad \text{for every fraction } p/q \text{ with } q \geq 1.$$

It suffices to consider fractions in lowest terms: if $p/q = p'/q'$ with $q \leq q'$, the inequality for p/q implies the one for p'/q', because $c/q^2 \geq c/q'^2$. From now on every fraction is in lowest terms. We call the interval $\left(\frac{p}{q} - \frac{c}{q^2}, \frac{p}{q} + \frac{c}{q^2}\right)$ the *no-go zone* of p/q. Player II's job is to keep the final point out of every no-go zone. There are infinitely many zones, but she has infinitely many moves, and the proof consists in giving each move a manageable amount of work.

Idea 1: fractions with small denominators keep their distance. If $p/q \neq p'/q'$, then

$$\left|\frac{p}{q} - \frac{p'}{q'}\right| = \frac{|pq' - p'q|}{qq'} \;\geq\; \frac{1}{qq'},$$

because the numerator $pq' - p'q$ is a non-zero integer. Two fractions whose denominators are both small therefore cannot be close together, and this little inequality drives the argument.

Idea 2: sort the fractions into generations. Let $A = 1/(3\rho_0)$. We say that p/q belongs to *generation* 0 if $q^2 < A$, and to *generation* $k \geq 1$ if

$$AR^{k-1} \;\leq\; q^2 \;<\; AR^k.$$

Every fraction belongs to exactly one generation, and generation k is the one that Player II will deal with in round k. Grouping the denominators by size in this way keeps the fractions of a single generation far apart: by Idea 1, two different fractions p/q and p'/q' of generation k satisfy $qq' < AR^k$, and they therefore lie at distance more than $1/(AR^k) = 3\rho_0 R^{-k} = 3\rho_k$. If we fatten Player I's interval U_k by ρ_k on each side, we obtain an interval of length $3\rho_k$, which can contain *at most one* fraction of generation k. That fraction, if it exists, is the *dangerous fraction* of round k.

Idea 3: the no-go zones are small. A fraction p/q of generation $k \geq 1$ has $q^2 \geq AR^{k-1}$, so the no-go zone of p/q has length

$$\frac{2c}{q^2} \;\leq\; \frac{2c}{AR^{k-1}} \;=\; 6cR\,\rho_k.$$

We choose $c > 0$ so small that $6cR \leq 1 - 2\alpha$ and $2c \leq (1 - 2\alpha)\rho_0$ (the second condition takes care of generation 0, whose zones have length at most $2c$). Then every zone of generation k has length at most $(1 - 2\alpha)\rho_k$, which is less than ρ_k. In particular, a zone of generation k that reaches into U_k has its center within $\rho_k/2$ of U_k; it therefore belongs to a fraction in the fattened interval, that is, to the dangerous fraction.

The strategy. In round k, Player II looks at Player I's interval U_k and at the no-go zone of the dangerous fraction of round k, if there is one. Removing the zone from U_k leaves at most two closed intervals, of total length at least $\rho_k - (1 - 2\alpha)\rho_k = 2\alpha\rho_k$, and one of them therefore has length at least $\alpha\rho_k$. Player II plays her interval V_k inside that one; if there is no dangerous fraction, she plays any interval of the prescribed length inside U_k. That is the whole strategy.

Proof that the strategy wins. Let p/q be a fraction, and let k be its generation. The final point x lies in $V_k \subseteq U_k$. If the zone of p/q reaches into U_k, the fraction is the dangerous one of round k; the interval V_k avoids its zone, and so does the point x. Otherwise the zone misses U_k altogether, and with it the point x. Either way $|x - p/q| \geq c/q^2$, whence $x \in \mathbf{Bad}(1)$. The constant c depends on α, β and ρ_0, which is harmless, since the definition of $\mathbf{Bad}(1)$ allows each number a constant of its own.

The game has done exactly what we promised in the Introduction: it has turned a global question into a local one. The condition "$|x - p/q| \geq c/q^2$ for all fractions" involves infinitely many fractions at once; the game splits it into infinitely many small jobs, each of them easy (dodge one interval).

Higher dimensions. For $\mathbf{Bad}(n)$ the same plan works, with Idea 1 replaced by the *simplex lemma*: if rational points $\mathbf{p}_0/q_0, \ldots, \mathbf{p}_n/q_n$ of $\mathbb{R}^n$ (with $\mathbf{p}_i \in \mathbb{Z}^n$ and $q_i \geq 1$) do not lie on a common hyperplane, then the simplex they span has volume at least $1/(n!\, q_0 \cdots q_n)$. (The volume is $1/n!$ times the absolute value of the determinant of the $(n+1) \times (n+1)$ matrix whose i-th row is $(1, \mathbf{p}_i/q_i)$. Multiplying the i-th row by q_i produces an integer matrix, and a non-zero determinant of this kind is therefore at least $1/(q_0 \cdots q_n)$ in absolute value, exactly as in Idea 1.) Consequently, with the generations chosen as in Idea 2 (by the size of $q^{1+1/n}$ instead of q^2, since the no-go zone of $\mathbf{p}/q$ now has radius $c\, q^{-1-1/n}$ in the maximum norm), all the rational points of one generation that lie near Player I's ball of the corresponding round sit on a single hyperplane, and in each round Player II has to dodge a thin slab around one hyperplane instead of one interval.

Full dimension. Suppose that Player II has a winning strategy in the (α, β)-game for S. Inside Player II's ball V_k, of radius αr_k, Player I has room for roughly β^{-n} disjoint balls of radius $\beta\alpha r_k$, and he may play any of them. Let him try them all, while Player II keeps following her strategy. The result is a tree of plays: every ball has about β^{-n} children, and the radii shrink by the factor $\alpha\beta$ from one generation to the next. The endpoints of all these plays form a Cantor set, and every point of this Cantor set lies in S, because Player II won every one of those plays. Since the children of each ball can be chosen well separated, a Cantor set built this way has Hausdorff dimension at least roughly

$$\frac{\log(\beta^{-n})}{\log(1/(\alpha\beta))} = \frac{n \log(1/\beta)}{\log(1/\alpha) + \log(1/\beta)}$$

(see Schmidt [Sch66] for the precise bound), and this quantity tends to n as $\beta \to 0$. Since an α-winning set is winning for *every* β, letting $\beta \to 0$ shows that such a set has dimension n. If Player I opens with a ball inside a non-empty open set U, the same construction yields the local version: the set $S \cap U$ has dimension n.

Countable intersections. Let $S_1, S_2, \ldots$ be α-winning sets. Player II now shares her time among them. She splits the rounds into infinitely many infinite arithmetic progressions (for instance, the k-th progression may consist of the rounds $2^{k-1}(2m+1) - 1$, $m \geq 0$), and in the rounds of the k-th progression she plays her strategy for S_k, ignoring everything else. From the point of view of S_k, everything that happens between two consecutive rounds of the k-th progression (her moves for the other sets and Player I's moves) looks like one very big shrinking move by Player I. If the progression has difference d_k, the game for S_k is thus the (α, β_k)-game with $\beta_k = \beta(\alpha\beta)^{d_k - 1}$. Since S_k is α-winning, she has a winning strategy in that game, and the final point, which is the same for every k, lies in every S_k. Her time-sharing strategy is a fugue: each voice enters at a time of its own and follows a line of its own, and in the end all of them sound together. Hausdorff dimension has no such device, and measure, which has one, sees only that $\mathbf{Bad}(n)$ is null; this is the real reason games are useful here.

Exercises.

(1) Let $0 < \alpha < 1$ be given, and let $S \subseteq \mathbb{R}^n$ be α-winning. Show that S meets every non-empty open subset of $\mathbb{R}^n$, that is, that S is dense in $\mathbb{R}^n$.

(2) Let $S \subseteq \mathbb{R}^n$ be α-winning, and let $\lambda \neq 0$ and $\mathbf{t} \in \mathbb{R}^n$ be given. Show that the set $\lambda S + \mathbf{t}$ is α-winning. (Hint: the map $\mathbf{x} \mapsto \lambda\mathbf{x} + \mathbf{t}$ carries $B[\mathbf{x}, r]$ onto $B[\lambda\mathbf{x} + \mathbf{t}, |\lambda| r]$; transport the strategy.)

(3) Let $\alpha \in (0, 1)$ and $\beta \in (0, \frac{1}{2})$ be given, and let $S \subseteq \mathbb{R}$ be countable. Show, without using Theorem 7.1, that Player I has a winning strategy in the (α, β)-game for S. Conclude that $\mathbb{Q}$ is not α-winning for any α. (Hint: since $2\beta < 1$, a closed interval contains two disjoint closed subintervals of β times its length.)

(4) Let $0 < \alpha < \frac{1}{2}$ be given. Show directly that the set of irrational numbers is α-winning, and more generally that the complement of every countable subset of $\mathbb{R}$ is α-winning.

(5) $(*)$ Let $0 < \alpha' < \alpha < 1$ be given, and let $S \subseteq \mathbb{R}^n$ be α-winning. Show that S is α'-winning. (Hint: given β', put $\beta = \alpha'\beta'/\alpha$, so that $\alpha\beta = \alpha'\beta'$; in the (α', β')-game, Player II plays a ball of radius $\alpha' r_k$ inside the ball of radius αr_k prescribed by her strategy for the (α, β)-game.)

8. Applications to homogeneous dynamics

The problem. A *lattice* in $\mathbb{R}^m$ is a set of the form $g\mathbb{Z}^m$ with g an invertible real $m \times m$ matrix, and the lattice $g\mathbb{Z}^m$ is *unimodular* if $\det g = \pm 1$. The unimodular lattices form a space, which can be identified with $\mathrm{SL}(m, \mathbb{R})/\mathrm{SL}(m, \mathbb{Z})$. Flows on this space move lattices around by stretching some directions and squeezing others, for example the flow

$$g_t = \mathrm{diag}\big(e^t, \dots, e^t, e^{-nt}\big), \qquad t \geq 0,$$

on the unimodular lattices in $\mathbb{R}^{n+1}$ (here $m = n+1$). For a flow of this kind almost every orbit is dense, and therefore wanders all over the space, while the lattices whose orbits stay in a bounded region form a set of measure zero. How big is this set of measure zero?

In this section we show that the game of Section 7 is a game about lattices being stretched and squeezed, so a reader of that section already knows the main idea. The words "Lie group", "lattice" and "horospherical subgroup" will appear, but the heart of the matter is a computation with 2×2 matrices acting on the plane, which we do by hand below; the Lie group language is just the coordinate system that lets the same computation run in higher dimensions.

A small glossary. The reader may keep the example $G = \mathrm{SL}(m, \mathbb{R})$, $\Gamma = \mathrm{SL}(m, \mathbb{Z})$ in mind throughout.

- A *Lie group* is a group that is also a smooth manifold on which multiplication and inversion are smooth, such as $\mathrm{SL}(m, \mathbb{R})$, the group of real $m \times m$ matrices of determinant 1.
- A *lattice* Γ in a Lie group G is a discrete subgroup such that the quotient G/Γ has finite volume. The word is used twice here: a lattice in $\mathbb{R}^m$ is a grid of points (a lattice, in the sense just defined, in the group $\mathbb{R}^m$), and a lattice in G is a subgroup of matrices. The two are linked by the identification of the coset $g\,\mathrm{SL}(m, \mathbb{Z})$ with the grid $g\mathbb{Z}^m$, under which $\mathrm{SL}(m, \mathbb{R})/\mathrm{SL}(m, \mathbb{Z})$ becomes the space of unimodular grids.
- "Almost every" and "measure zero" refer to the natural measure on G/Γ inherited from G (its Haar measure), which the flow preserves.
- A set of points of G/Γ is *bounded* if its closure is compact; for grids, boundedness is made concrete by Mahler's criterion in the history below.
- A flow g_t is *quasi-unipotent* if all the eigenvalues of the matrices g_t have absolute value 1; in other words, no direction is stretched exponentially.

- A lattice Γ in G is *irreducible* if no subgroup of finite index in Γ is a product of lattices in two non-trivial factors of G; in a simple group such as $\mathrm{SL}(m,\mathbb{R})$, every lattice is irreducible.

Other terms (nilpotent, semisimple, mixing) appear only where we describe the general case, and a reader may pass over them.

8.1. **History of the problem and of the solution using topological games.** The space of lattices is not compact: one can squash a lattice to create a very short non-zero vector, and then squash further. "Going to infinity" in this space is usually called "going into the cusp", and a lattice deep in the cusp is a very thin needle-like grid with a vector of length nearly zero. By Mahler's compactness criterion (1946), this is the only way to escape to infinity: a set of unimodular lattices is bounded exactly when the lengths of their shortest non-zero vectors are bounded away from zero.

The link with the previous section was found by Dani. He showed in 1985 [Dan85] that badly approximable vectors correspond exactly to bounded orbits of the flow above. In 1986 he used Schmidt's games to show that bounded orbits are abundant for several families of flows [Dan86]. In his plenary address at the International Congress of Mathematicians in Kyoto in 1990 [Mar91], Margulis conjectured that this abundance is universal: for every one-parameter flow that is not quasi-unipotent (roughly, every flow that really stretches something), the set of points with bounded orbits is *thick*, that is, it meets every non-empty open set in a set of full Hausdorff dimension.

Kleinbock and Margulis proved the conjecture in 1996 [KM96]. They worked on the expanding leaves of the flow, but their argument rests on the mixing of the flow, and it produces sets of full dimension rather than winning strategies. Kleinbock and Weiss introduced *modified Schmidt games*, adapted to flows that stretch different directions at different rates. In 2013 they proved that sets of bounded orbits are winning for these games [KW13], a result that adds the countable intersection property to thickness. In 2018 An, Beresnevich and Velani established that the points of $\mathbf{Bad}(i,j)$ on every nondegenerate planar curve form a winning set [ABV18].

Recognition. Margulis received the Fields Medal in 1978, the Wolf Prize in 2005 and the Abel Prize in 2020. Dani received the Shanti Swarup Bhatnagar Prize in 1990. He was an invited speaker at the International Congress of Mathematicians in Zürich in 1994. The dynamical approach to Diophantine approximation that grew out of this work led Kleinbock and Margulis to a proof of the Baker–Sprindžuk conjectures, published in the *Annals of Mathematics* in 1998 [KM98]; that proof uses non-divergence estimates, not games.

8.2. **The game.** The game is Schmidt's game of Section 7, played not on the whole space but on an *expanding leaf* of the flow. The expanding horospherical subgroup H consists of the elements h with $g_{-t}hg_t \to e$ as $t \to \infty$; these are the directions that the flow stretches. Distances in these directions grow exponentially, and a tiny ball soon becomes a ball of unit size; this growth lets Player II deal with the problem one scale at a time (see the dictionary between scales and times below). The group H is a nilpotent Lie group with a natural metric, and the orbit Hx is the image of H under the map $h \mapsto hx$, which need not be injective. The players play Schmidt's game with closed balls in H, and Player II wins if the final point h is such that hx has a bounded forward orbit. For the diagonal flow above, the subgroup H consists of the matrices $\left(\begin{smallmatrix} I_n & \mathbf{y} \\ 0 & 1 \end{smallmatrix}\right)$, and the leaf through $\mathbb{Z}^{n+1}$ is the set of lattices $\Lambda_{\mathbf{y}} = \left(\begin{smallmatrix} I_n & \mathbf{y} \\ 0 & 1 \end{smallmatrix}\right)\mathbb{Z}^{n+1}$ with $\mathbf{y} \in \mathbb{R}^n$; the game on this leaf is therefore literally Schmidt's game on $\mathbb{R}^n$.

8.3. **The Main Result of this section.** The main result has two parts: a dictionary between numbers and orbits, due to Dani, and the theorem of Kleinbock and Margulis.

Theorem 8.1 (Dani, 1985; Kleinbock–Margulis, 1996).

(1) Dani's correspondence. *A vector* $\mathbf{x} \in \mathbb{R}^n$ *lies in* $\mathbf{Bad}(n)$ *if and only if the forward orbit* $\{g_t\Lambda_{\mathbf{x}} : t \geq 0\}$ *is bounded.*

(2) Thickness of bounded orbits. *For every non-quasi-unipotent one-parameter subgroup of a connected semisimple Lie group* G *without compact factors, and every irreducible lattice* Γ *in* G*, the set of points of* G/Γ *with bounded orbits is thick.*

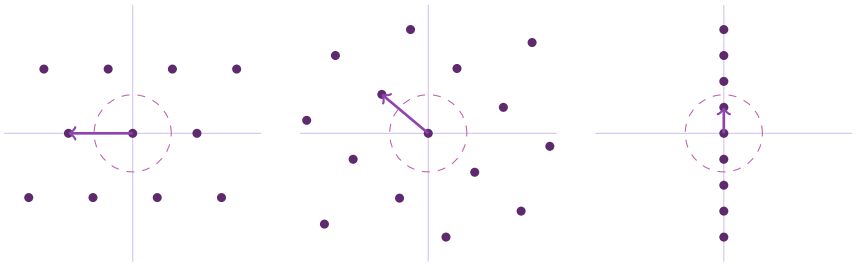
By part (1), Schmidt's theorem of Section 7 is a theorem about one particular flow; part (2) extends the abundance of bounded orbits to every flow that stretches.

8.4. **Solving the problem with the game.** We first follow the whole mechanism in the simplest case $n = 1$, where every step can be computed by hand, and then explain what changes in general.

Lattices in the plane. For $n = 1$ we have $g_t = \operatorname{diag}(e^t, e^{-t})$ and $u_x = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$, so

$$\Lambda_x = u_x\mathbb{Z}^2 = \{(p + qx,\ q) : p, q \in \mathbb{Z}\}, \qquad g_t\Lambda_x = \{(e^t(p + qx),\ e^{-t}q) : p, q \in \mathbb{Z}\}.$$

The flow stretches the first coordinate and squeezes the second, and the *product* of the two coordinates of a vector, namely $q(p + qx)$, does not change along the orbit. The whole argument rests on this invariant product (see Figure 16).



(a) $x = (\sqrt{5} - 1)/2$, $t = 0$ (b) $x = (\sqrt{5} - 1)/2$, $t = 1.6$ (c) $x = 1/2$, $t = 1.6$

FIGURE 16. The grid $g_t\Lambda_x$ near the origin, with a shortest non-zero vector drawn as an arrow and a disk of radius 0.6 dashed. The flow stretches the grid horizontally by e^t and squeezes it vertically by e^{-t}. For the badly approximable number $x = (\sqrt{5} - 1)/2$, in (a) and (b), the grid is sheared but no non-zero point ever comes close to the origin, so the orbit stays bounded. For the rational number $x = 1/2$, in (c), the points $(0, 2qe^{-t})$ with $q \in \mathbb{Z}$ crowd toward the origin, and the orbit goes into the cusp.

Dani's correspondence, by hand. Suppose first that $x \in \mathbf{Bad}(1)$: there is a constant $c > 0$ with $|p + qx| \geq c/|q|$ for all integers p and q with $q \neq 0$. Let $(e^t(p + qx), e^{-t}q)$ be a non-zero vector of $g_t\Lambda_x$, with $t \geq 0$. If $q \neq 0$, the product of the coordinates of this vector has absolute value $|q|\,|p + qx| \geq c$, so the two coordinates cannot both be smaller than $\sqrt{c}$, and the vector has length at least $\sqrt{c}$. If $q = 0$, the vector is $(e^t p, 0)$ with $p \neq 0$, of length at least $e^t \geq 1$. Hence no lattice on the forward orbit has a non-zero vector shorter than $\min(1, \sqrt{c})$, and by Mahler's criterion the orbit is bounded.

Conversely, suppose that $x \notin \mathbf{Bad}(1)$. Then for every $\varepsilon \in (0,1)$ there are integers p and $q \geq 1$ with $|p + qx| < \varepsilon/q$. Let t be the time with $e^{-t}q = \sqrt{\varepsilon}$, which is positive because $q \geq 1 > \sqrt{\varepsilon}$. At that time the vector coming from (p, q) has second coordinate $\sqrt{\varepsilon}$ and first coordinate of absolute value $e^t|p + qx| < (q/\sqrt{\varepsilon})(\varepsilon/q) = \sqrt{\varepsilon}$, and it is therefore shorter than $\sqrt{2\varepsilon}$. Since ε is arbitrary, the orbit dives deeper and deeper into the cusp, and it is unbounded by Mahler's criterion.

A good rational approximation p/q of x is thus *the same thing* as an excursion of the orbit into the cusp at a time $t \approx \log q$.

The dictionary between scales and times. A one-line computation gives

$$g_t \, u_x \, g_{-t} = u_{e^{2t}x},$$

so for every unimodular lattice Λ the flow carries the expanding leaf $\{u_x\Lambda : x \in \mathbb{R}\}$ onto the leaf through $g_t\Lambda$, stretching it by the factor e^{2t}. An interval of length ρ in the leaf is stretched to length about 1 at the time t with $e^{2t} \approx 1/\rho$. We now return to the proof in Section 7 that $\mathbf{Bad}(1)$ is α-winning. In round k, Player I's interval has length $\rho_k = \rho_0 R^{-k}$, and the fractions Player II worries about are those of generation k, with $q^2 \approx R^k$; by Dani's correspondence as worked out above, these are precisely the fractions that send the orbit into the cusp at the time t with $e^{2t} \approx R^k \approx 1/\rho_k$. In other words, *round k of the game corresponds to the time at which the current interval has been stretched to unit size, and in that round Player II dodges the points whose orbits visit the cusp at that time.* Read in this dictionary, Schmidt's proof is a proof about orbits, and the formula $g_t u_x g_{-t} = u_{e^{2t}x}$ translates each step of one proof into the corresponding step of the other.

The general case. Kleinbock and Margulis follow the same plan for an arbitrary flow that stretches, working on an expanding leaf Hx. At each scale they look at a ball in the leaf, wait (in their heads) until the flow has stretched that ball to unit size, and determine the points of the ball that are deep in the cusp at that time. For the diagonal flow on the plane this bad set was a single short interval, the no-go zone of one fraction. In general there is no single dangerous fraction, and here lies the real work: using the mixing of the flow, Kleinbock and Margulis show [KM96] that the bad set occupies only a small proportion of the ball. Removing the bad set at every scale leaves a Cantor-like set whose points never go far into the cusp, and therefore have bounded orbits, and whose dimension is close to that of the leaf. The set of points with bounded orbits thus has full dimension on every expanding leaf. Their argument does not produce a winning strategy, since a set that occupies a small proportion of a ball may still meet every ball of the prescribed size inside it; the winning property came later, with the modified games of Kleinbock and Weiss (see the history above).

From leaves to the whole space. Roughly, near any point the space looks like a product of a piece of an expanding leaf with the directions that the flow does not stretch, and whether an orbit stays bounded depends only mildly on those other directions. Full dimension on every expanding leaf therefore yields full dimension in the whole space, that is, thickness (see Kleinbock and Margulis [KM96]).

Exercises.

(1) Let $x \in \mathbb{R}$ and $t \geq 0$ be given. Check by multiplying matrices that $g_t u_x g_{-t} = u_{e^{2t}x}$. Show that Λ_x is unimodular, and that the product of the two coordinates of a vector of $g_t\Lambda_x$ coming from (p, q) does not depend on t.

(2) Let $x \in \mathbb{R}$ be given. Show that $\Lambda_{x+1} = \Lambda_x$. Explain why this is consistent with Theorem 8.1(1), by checking directly that $x \in \mathbf{Bad}(1)$ if and only if $x + 1 \in \mathbf{Bad}(1)$.

(3) Let $x = a/b$ be rational, with $b \geq 1$. Find a non-zero vector of $g_t\Lambda_x$ of length be^{-t}, and deduce that the orbit $\{g_t\Lambda_x : t \geq 0\}$ is unbounded. Check that this agrees with Theorem 8.1(1), by showing that no rational number is badly approximable.

(4) (∗) Let $\varphi = (1+\sqrt{5})/2$ and $\varphi' = (1-\sqrt{5})/2$. Show that $(p+q\varphi)(p+q\varphi') = p^2+pq-q^2$ for all integers p, q, and deduce that $|q|\,|p+q\varphi| \geq 1/(1+\sqrt{5})$ whenever $q \neq 0$. Conclude that, for every $t \geq 0$, every non-zero vector of $g_t\Lambda_\varphi$ has length at least $(1+\sqrt{5})^{-1/2}$.

9. Applications to computational complexity

The problem. A *language* is a set L of finite binary strings, and a complexity class is a set of languages, such as P (decidable in polynomial time), the class $\mathrm{E} = \mathrm{DTIME}(2^{O(n)})$ (decidable in time $2^{O(n)}$ on inputs of length n) or $\mathrm{ESPACE} = \mathrm{DSPACE}(2^{O(n)})$. We would like to say that "most" languages in E are hard, or that "most" languages in ESPACE need big circuits. Ordinary Baire category and Lebesgue measure on the space of all languages are useless for this: every complexity class is countable, hence meager and of measure zero, and therefore P and E look equally small. The problem is to find a notion of "small" that works *inside* a complexity class.

9.1. **History of the problem and of the solution using topological games.** The problem was solved by Kurt Mehlhorn in 1973 [Meh73], who introduced resource-bounded Baire category, in which the descriptions of meager sets must themselves be computable with limited resources. Jack Lutz reformulated the theory in terms of the Banach–Mazur game played with resource-bounded strategies, in his thesis of 1987 and in a paper published in 1990 [Lut90]. Among other results, he proved that languages with high circuit complexity are typical in ESPACE in this sense. He also introduced the companion notion of resource-bounded measure, which he developed in a paper of 1992 [Lut92]. Both notions became standard tools for quantitative questions about complexity classes.

Recognition. Mehlhorn received the Leibniz Prize of the Deutsche Forschungsgemeinschaft in 1986.

9.2. **The game.** We list all binary strings in the standard order, namely $\lambda, 0, 1, 00, 01, 10, 11, 000, \ldots$ (shorter strings first, and strings of the same length in dictionary order). A language L is then the same thing as an infinite 0–1 sequence, the *characteristic sequence* of L, whose N-th bit says whether the N-th string belongs to L. A finite 0–1 string w is a partial description of a language, since w decides membership for the first $|w|$ strings. The 2^n strings of length n come after the $2^n - 1$ shorter ones (see Exercise 1 below), so a description that decides all strings of length less than n has length at least $2^n - 1$. This exponential gap is what turns time polynomial in the length of a description into time $2^{O(n)}$ on inputs of length n.

Let $\mathcal{C}$ be a class of languages. The game for $\mathcal{C}$ is played as follows.

(1) The players alternately extend finite binary strings, $w_0 \sqsubset w_1 \sqsubset w_2 \sqsubset \cdots$, with Player I playing w_{2k} and Player II playing w_{2k+1}.

(2) The play produces the language $L = \bigcup_k w_k$.

(3) Player II wins if $L \notin \mathcal{C}$.

This is the Banach–Mazur game on the Cantor space, in which the move w stands for the basic open set N_w of languages whose characteristic sequence begins with w. By the theorem of Banach and Oxtoby, Player II has a winning strategy exactly when $\mathcal{C}$ is meager (see Theorem 3.5; its proof works just as well when the moves are restricted to basic open sets). The resource-bounded version adds one rule: *Player II is not allowed to think for too long.*

Her strategy must compute each move in time polynomial in the length of the current string (or in space polynomial in that length, when we work in ESPACE). Player I has no restrictions at all. A class is *meager in* E if Player II has a winning strategy that obeys the time bound, and *meager in ESPACE* if she has one that obeys the space bound.

9.3. **The Main Result of this section.** A notion of smallness is useful only if it does not declare everything small. The theory rests on a resource-bounded version of the Baire category theorem, part (1) of Theorem 9.1 below, which we prove in the next subsection. Since E is not meager in itself, its meager subclasses are genuinely small, and the notion can tell its subclasses apart; the same holds for ESPACE.

Theorem 9.1 (Mehlhorn, 1973; Lutz, 1990).

(1) Baire category theorem. *The class* E *is not meager in itself in the resource-bounded sense.*
(2) *The class* P *is meager in* E *in the resource-bounded sense.*
(3) *The class* $\mathrm{SIZE}(2^n/n)$ *of languages that, for all sufficiently large* n*, are decided on inputs of length* n *by circuits with at most* $2^n/n$ *gates is meager in* ESPACE *in the resource-bounded sense.*

By part (3), big circuit complexity is typical in ESPACE. Lutz also proved the measure version: in resource-bounded measure, almost every language in ESPACE has circuit-size complexity above $2^n/n$ [Lut92]. Here category and measure agree, although, as on the real line, they can disagree in general (see Remark 1.2(2)).

9.4. **Solving the problem with the game.** The game formulation makes the definition of category effective in the most natural way imaginable: one simply limits the resources of the winning player. The proofs then become constructions of efficient strategies, and they are pleasantly concrete.

The Baire category theorem: the class E **is not meager in itself.** The idea is that Player I can play so predictably that the whole play is computable from Player II's strategy. Suppose that Player II has a polynomial-time strategy. We show that Player I can make the play produce a language in E, whence the strategy does not win the game for the class E. His strategy is the simplest possible: at each of his moves he appends a single 0. The whole play is then computable: to decide whether the N-th string is in L, we run the game until the current string has length at least N and read off bit N. Each round adds at least one bit, and therefore at most N rounds are needed. Each of Player II's moves is made from a string of length at most N, so it takes time polynomial in N; hence bit N is computed in time polynomial in N. Since the strings of length n sit at positions $N \approx 2^n$, membership of a string of length n is decided in time polynomial in 2^n, that is, in time $2^{O(n)}$. Thus $L \in \mathrm{E}$, and the class E is not meager in itself.

The class P **is meager in** E**: diagonalization, one move at a time.** Let $M_1, M_2, \ldots$ be a list of Turing machines such that M_i halts within $n^i + i$ steps on inputs of length n and every language in P is decided by some M_i. Player II wants to make sure that L disagrees with every M_i on at least one string, and each of her moves takes care of a few more machines. Let w be the current string, and let s be the least length such that w decides no string of length s; then the string w decides only strings of length less than s, so $|w| < 2^s$, and all 2^s strings of length s are still free. For each $i \leq \log s$, Player II picks the i-th string x_i of length

s, runs M_i on x_i, and extends w so that

$$x_i \in L \iff M_i \text{ rejects } x_i,$$

filling all the other new positions with 0. Each simulation takes at most $s^{\log s} + \log s$ steps, which is less than 2^s once s is large, and therefore the whole move takes time polynomial in 2^s. (Here $|w| \geq 2^{s-1} - 1$, because w decides all strings of length less than $s - 1$; a bound polynomial in 2^s is thus polynomial in $|w|$, as the rules require.) Player I may then extend the string as much as he likes, but he cannot undo what she wrote. Once $s \geq 2$, each of her moves decides a string of length s, whence s grows from each of her moves to the next. Since Player II moves infinitely often, every M_i is eventually defeated on some string, and $L \notin \mathrm{P}$ follows.

The one clever point is the restriction $i \leq \log s$. Player II cannot afford to simulate M_i on short strings when i is large: the s^i steps of the simulation could exceed her time budget, which is a polynomial in 2^s fixed once and for all. To defeat M_{1000} on strings of length 10, for instance, she might need 10^{1000} steps while writing about 2^{10} bits, and a strategy that did this for every i at lengths that grow too slowly would obey no polynomial bound at all. She waits instead until $s \geq 2^{1000}$, when the strings are long enough that simulating M_{1000} costs little relative to $|w|$. This "diagonalization with delay" is a standard trick of complexity theory, and the game makes it look natural.

Circuit size: a counting argument, one length at a time. Here the resource bound is space instead of time, and the key fact is Shannon's counting argument (1949): there are 2^{2^n} Boolean functions of n variables, one for each truth table. A circuit can be described by listing the operation and the two inputs of each gate, and a careful count (taking into account that renaming the gates does not change the circuit) shows that for large n there are fewer circuits of size $2^n/n$ than there are truth tables. Hence some Boolean function of n variables has no circuit of size $2^n/n$.

Player II turns this count into a strategy. At each of her moves she takes the smallest length n such that no string of length n has been decided yet. If n is too small for the count above, she fills the new positions with 0. Otherwise she finds, by brute force, the first truth table on n variables that has no circuit of size $2^n/n$: she goes through the truth tables one by one, and for each of them she checks every circuit of size $2^n/n$ to see whether it computes that truth table. She then fills the remaining positions before the strings of length n with 0, and defines L on the 2^n strings of length n to be that truth table. The search takes an astronomical amount of *time*, but very little *space*: at any moment she needs to remember only one truth table and one circuit, which takes space polynomial in 2^n, hence polynomial in the length of the current string. Space can be reused and time cannot, and Player II is charged only for space. Since she does this for infinitely many n, the final language has no circuits of size $2^n/n$ at infinitely many lengths, and therefore $L \notin \mathrm{SIZE}(2^n/n)$.

Exercises.

(1) Number the binary strings in the standard order starting from 0, with the empty string λ as string number 0. Show that the strings of length n are those numbered $2^n - 1$ through $2^{n+1} - 2$, and that a string w of length n has number $m - 1$, where m is the number whose binary expansion is $1^\frown w$. Deduce that a finite 0–1 string decides all strings of length less than n exactly when its length is at least $2^n - 1$.

(2) Consider the game of this section without resource bounds. Show that Player II has a winning strategy for every countable class $\mathcal{C}$, and that Player I has a winning strategy for the class of languages containing the empty string λ.

(3) Fill in the proof that E is not meager in itself. Let p be an increasing polynomial such that Player II computes her answer to a string w in time at most $p(|w|)$, and let Player I always append a single 0. Show that bit N of the resulting language can be computed in time polynomial in N. (Hint: every move that has to be simulated is made from a string of length at most N.)

(4) Show that $s^{\log_2 s} + \log_2 s < 2^s$ for every integer $s \geq 17$, but not for $s = 16$. Explain why the proof that P is meager in E needs such an inequality only for large s.

(5) $(*)$ A circuit with s gates on n variables is a list of s gates, each labeled AND, OR or NOT and given two inputs among the variables and the earlier gates (a NOT gate ignores its second input), the last gate being the output. Show that there are at most $s\,(3(n+s)^2)^s$ circuits with between 1 and s gates, and deduce that, for all sufficiently large n, some Boolean function of n variables has no circuit with at most $2^n/(10n)$ gates.

10. Applications to topological groups

The problem. A *semitopological group* is a group with a topology in which multiplication is *separately* continuous: the translations $x \mapsto xa$ and $x \mapsto ax$ are continuous. A semitopological group is a *topological group* if multiplication is jointly continuous and inversion is continuous. Separate continuity is much weaker than joint continuity (the function $f(x,y) = xy/(x^2+y^2)$, with $f(0,0) = 0$, is continuous in each variable separately but not jointly continuous at the origin; see Figure 17). When does separate continuity of multiplication force a semitopological group to be a topological group? Behind this lies a more general question: when is a separately continuous function $f\colon X \times Y \to \mathbb{R}$ jointly continuous at many points?

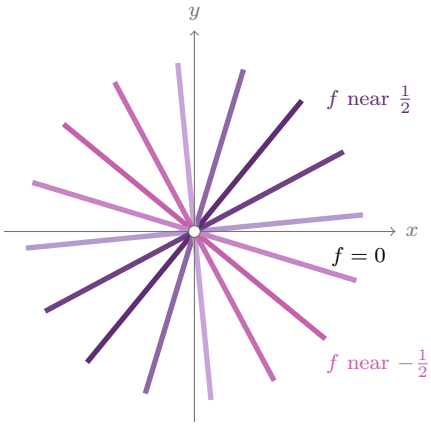


Figure 17. The function $f(x,y) = xy/(x^2+y^2)$, with $f(0,0) = 0$, is constant on each ray from the origin, where it equals $\frac{1}{2}\sin 2\theta$; the color shows the value. The function vanishes on the axes, whence it is continuous in each variable separately. On every neighborhood of the origin, the function takes all the values between $-\frac{1}{2}$ and $\frac{1}{2}$; hence it is not jointly continuous there.

10.1. **History of the problem and of the solution using topological games.** Baire proved in his thesis of 1899 that a separately continuous function on $\mathbb{R}^2$ is jointly continuous at the points of a dense G_δ set. In 1957 Ellis proved that every locally compact Hausdorff semitopological group is a topological group [Ell57], and in 1974 Namioka proved a general joint continuity theorem for functions on $X \times K$, with X complete in a suitable sense and K compact, from which he deduced Ellis's theorem [Nam74].

The game-theoretic solution came in three steps. In 1981 Christensen replaced the completeness hypothesis on X in Namioka's theorem by the existence of a winning strategy in a topological game [Chr81]. In 1983 Saint-Raymond showed that the weaker hypothesis that the opposing player has no winning strategy suffices [SR83]. He deduced that the conclusion of Namioka's theorem holds whenever X is a metrizable Baire space. The third step concerns groups. Bouziad proved in 1996 that every Čech-analytic Baire semitopological group is a topological group [Bou96]. In 2001 Kenderov, Kortezov and Moors introduced a game that describes a class of spaces, the strongly Baire spaces, on which every semitopological group is a topological group [KKM01].

10.2. **The game.** Let X be a topological space, and let $D \subseteq X$ be a dense set. The game $\mathrm{SB}(D)$ is due to Kenderov, Kortezov and Moors, who call the players β and α; here β becomes Player I and α becomes Player II.

(1) Player I chooses a non-empty open set $U_1 \subseteq X$.
(2) Player II chooses a non-empty open set $V_1 \subseteq U_1$.
(3) In general, Player I chooses a non-empty open $U_{k+1} \subseteq V_k$ and Player II a non-empty open $V_{k+1} \subseteq U_{k+1}$.
(4) Player II wins if $\bigcap_k V_k \neq \emptyset$ and every sequence (a_k) with $a_k \in V_k \cap D$ has a cluster point in X.

A regular space is *strongly Baire* if, for some dense D, Player I has no winning strategy in $\mathrm{SB}(D)$. This game is the Banach–Mazur game of Section 3 with a more demanding winning condition for Player II; the games of Christensen and Saint-Raymond are variants in which Player II also picks points as she goes.

The motivating examples are the Čech-complete spaces of Subsection 4.1: on such a space Player II wins $\mathrm{SB}(X)$ with the strategy of the proof of Proposition 4.8, stripped of the points, and only the condition on cluster points needs a new argument. Let X be Čech-complete, say with $X = \bigcap_k G_k$ for sets G_k open in a compactification bX, and for each move U_k of Player I let $\widetilde{U}_k$ be an open subset of bX with $\widetilde{U}_k \cap X = U_k$. Player II wins with $D = X$ by choosing V_k whose closure in bX lies inside $G_k \cap \widetilde{U}_k$; such a set exists because bX is regular. The closures form a decreasing sequence of non-empty compact sets. Their intersection lies in $\bigcap_k G_k = X$, and it equals $\bigcap_k V_k$ because the closure of V_{k+1} meets X inside $U_{k+1} \subseteq V_k$. Finally, a sequence with $a_k \in V_k$ has a cluster point in bX; since $a_j \in V_k$ for $j \geq k$, this point lies in every closure, hence in $\bigcap_k V_k$.

10.3. **The Main Result of this section.** The main result goes from functions to groups: Namioka's joint continuity theorem, Ellis's theorem on locally compact groups, and the theorem of Kenderov, Kortezov and Moors, which contains Ellis's.

Theorem 10.1 (Ellis, 1957; Namioka, 1974; Kenderov–Kortezov–Moors, 2001).

(1) Namioka's theorem. *Let X be a Čech-complete space, let K be a compact space, and let $f\colon X \times K \to \mathbb{R}$ be separately continuous. Then there is a dense G_δ set $A \subseteq X$ such that f is jointly continuous at every point of $A \times K$.*

(2) Ellis's theorem. *Every locally compact Hausdorff semitopological group is a topological group.*

(3) *Every semitopological group whose underlying space is strongly Baire is a topological group.*

By the argument of the preceding subsection, every regular Čech-complete space, in particular every locally compact Hausdorff space and every complete metric space, is strongly Baire; hence part (3) contains part (2). Part (1) has a useful reformulation (see Exercise 5 below). For $x \in X$, the function f is jointly continuous at every point of $\{x\} \times K$ exactly when the map $x' \mapsto f(x', \cdot)$ from X to $C(K)$, which is continuous for the topology of pointwise convergence, is continuous at x for the norm topology. Namioka's theorem is therefore about upgrading pointwise convergence to uniform convergence, at least at many points.

10.4. **Solving the problem with the game: a sketch.** The arguments of Christensen, Saint-Raymond and Kenderov–Kortezov–Moors share a pattern worth learning, of which we give only a sketch; for the details, see the original papers [Chr81, SR83, KKM01]. Suppose, for instance, that a semitopological group with identity e has a strongly Baire underlying space X, and that multiplication is not jointly continuous at (e, e). Then there is a neighborhood W of e such that, for all neighborhoods O_1 and O_2 of e, there are $u \in O_1$ and $v \in O_2$ with $uv \notin W$. Player I uses these pairs to build a strategy: at each move he chooses an open set inside Player II's last move that records a new pair of witnesses, chosen so that separate continuity at any cluster point of the witnesses would be violated. Since X is strongly Baire, this strategy is not winning, and some play consistent with it is won by Player II. That play provides a cluster point of the witnesses, and separate continuity at that point leads to a contradiction. (The witnesses must be taken from the dense set D, so that the winning condition of the game, which only speaks of sequences from D, applies to them; arranging this is the technical heart of the proofs, and we omit it.) In short, the failure of joint continuity is turned into a strategy for Player I, and the topological hypothesis says exactly that such strategies can be beaten. Continuity of inversion is handled separately. For locally compact groups the argument yields Ellis's theorem, and Namioka's original deduction of that theorem, which uses translations to spread a single point of joint continuity over the whole group, follows the same idea.

Exercises.

(1) Let $f(x, y) = xy/(x^2 + y^2)$ for $(x, y) \neq (0, 0)$, and $f(0, 0) = 0$. Show that f is continuous in each variable separately, that f is not jointly continuous at $(0, 0)$, and that f is jointly continuous at every other point of $\mathbb{R}^2$.

(2) Show that every strongly Baire space is a Baire space. (Hint: a winning strategy for Player I in $\mathrm{BM}(X)$ is also winning in $\mathrm{SB}(D)$ for every dense set D.) Deduce that $\mathbb{Q}$ is not strongly Baire.

(3) Let X be a complete metric space. Show directly that Player II has a winning strategy in $\mathrm{SB}(X)$, by answering U_k with a non-empty open set V_k such that $\overline{V_k} \subseteq U_k$ and $\operatorname{diam} V_k < 2^{-k}$.

(4) Let $\mathbb{R}$ carry the Sorgenfrey topology, whose basic open sets are the intervals $[a, b)$. Show that addition is jointly continuous, that the translations are homeomorphisms, and that $x \mapsto -x$ is not continuous. Assuming that the Sorgenfrey line is regular, deduce from Theorem 10.1(3) that it is not strongly Baire.

(5) (∗) Let X be a topological space, let K be a compact space, and let $f\colon X \times K \to \mathbb{R}$ be separately continuous. Show that f is jointly continuous at every point of $\{x\} \times K$ if and only if the map $x' \mapsto f(x', \cdot)$ from X to $C(K)$ is continuous at x for the supremum norm.

11. Applications to infinite-dimensional Ramsey theory

The problem. Ramsey's theorem (1930) says that, whenever the n-element subsets of $\omega = \{0, 1, 2, \dots\}$ are colored with two colors, there is an infinite $H \subseteq \omega$ all of whose n-element subsets have the same color. We now color the *infinite* subsets instead. Call $\mathcal{A} \subseteq [\omega]^\omega$, the set of infinite subsets of ω, *Ramsey* if there is an infinite $H \subseteq \omega$ such that $[H]^\omega \subseteq \mathcal{A}$ or $[H]^\omega \cap \mathcal{A} = \emptyset$. Using the axiom of choice, one can build sets that are not Ramsey: if $\mathcal{U}$ is a nonprincipal ultrafilter on ω, the set of all $X = \{x_0 < x_1 < \cdots\}$ for which the union of the intervals $[x_{2k}, x_{2k+1})$ of ω belongs to $\mathcal{U}$ is one (see Exercise 6 below). Which sets are Ramsey?

There is a Rothko hiding in every Pollock: however wildly two colors are splashed on the pairs of natural numbers, some infinite set has all of its pairs painted in a single color. In this section we show that the picture survives the passage from pairs to infinite sets whenever the coloring is regular enough: the sets that are Ramsey in a hereditary sense are exactly those with the Baire property in a suitable topology, and every analytic set is among them. The tool is a game of Banach–Mazur type.

11.1. **History of the problem and of the solution using topological games.** Ramsey's theorem appeared in 1930 [Ram30]. In 1952 Erdős and Rado showed, using the axiom of choice, that the infinite-dimensional analogue of Ramsey's theorem fails [ER52]. Nash-Williams proved in 1965 that clopen sets, in the topology inherited from the Cantor space, are Ramsey [NW65]. Galvin and Prikry extended this result to Borel sets [GP73], and Silver extended it to analytic sets [Sil70], by a proof that used metamathematical methods. Silver's paper appeared in 1970, three years before that of Galvin and Prikry. Mathias developed the theory further, with applications to forcing, in his work on happy families [Mat77].

Erik Ellentuck solved the problem in 1974 [Ell74]. He introduced a new topology on $[\omega]^\omega$, finer than the usual one, and proved that the sets that are Ramsey in a hereditary sense are exactly those with the Baire property in this topology. Silver's theorem follows from Ellentuck's in a few lines. In 1983 Kastanas [Kas83] characterized the same sets by the determinacy of a game, which is a Banach–Mazur game in Ellentuck's topology. In Section 12 we show that Gowers's game is the analogue of Kastanas's game for block sequences in Banach spaces.

11.2. **The game.** The game is played in a new topology whose open sets look strange at first, so we begin with the idea behind them. In the usual (Cantor space) topology on $[\omega]^\omega$, two infinite sets are close if they begin in the same way, that is, if they share a long finite initial segment s. Ellentuck's open sets ask for more: they fix the beginning s *and* restrict the rest of the set to come from a given reservoir M. For a finite $s \subseteq \omega$ and an infinite $M \subseteq \omega$

with $\max s < \min M$, let

$$[s, M] = \{X \in [\omega]^\omega : s \text{ is an initial segment of } X \text{ and } X \subseteq s \cup M\}.$$

Think of s as a stem that has already been decided and of M as the pool of candidates for the rest. The sets $[s, M]$ form a basis of a topology, the *Ellentuck topology* (see Exercise 1 below). A set $\mathcal{A}$ is *completely Ramsey* if for every $[s, M]$ there is an infinite $N \subseteq M$ with $[s, N] \subseteq \mathcal{A}$ or $[s, N] \cap \mathcal{A} = \emptyset$.

Let $\mathcal{A} \subseteq [\omega]^\omega$ be given, let $s \subseteq \omega$ be finite, and let $N \subseteq \omega$ be infinite with $\max s < \min N$. The Kastanas game $K(\mathcal{A}, s, N)$ is played as follows.

(1) Player I chooses an infinite $M_0 \subseteq N$.
(2) Player II chooses $n_0 \in M_0$ and an infinite $N_0 \subseteq M_0$ with $n_0 < \min N_0$.
(3) In general, Player I chooses an infinite $M_k \subseteq N_{k-1}$, and Player II chooses $n_k \in M_k$ and an infinite $N_k \subseteq M_k$ with $n_k < \min N_k$.
(4) Player II wins if $s \cup \{n_0, n_1, \dots\} \in \mathcal{A}$.

In Kastanas's paper the payoff set is Player I's; since the theorem is about determinacy and $\mathcal{A}$ is completely Ramsey exactly when the complement of $\mathcal{A}$ is (see Exercise 4 below), the switch does not matter. The moves are basic open sets in disguise: Player I's move M_k stands for $[s \cup \{n_0, \dots, n_{k-1}\}, M_k]$, and Player II answers with the smaller basic open set $[s \cup \{n_0, \dots, n_k\}, N_k]$. The Kastanas game is thus a Banach–Mazur game in the Ellentuck topology, in which Player II must commit to one more element at each move (see Figure 18).

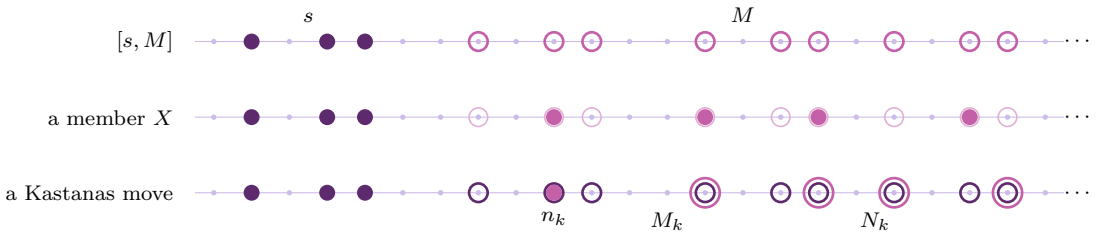


FIGURE 18. The Ellentuck topology on $[\omega]^\omega$. Top: a basic open set $[s, M]$ is given by a finite stem s (filled plum) and an infinite set M above it (circles). Middle: a member of $[s, M]$ is the stem together with an infinite subset of M (filled pink). Bottom: one round of the Kastanas game after the stem: Player I offers an infinite set M_k (plum circles), and Player II commits to one element $n_k \in M_k$ and keeps an infinite set $N_k \subseteq M_k$ above it (pink circles) for the rest of the play.

11.3. **The Main Result of this section.** The main result is Ellentuck's theorem, in the game form due to Kastanas.

Theorem 11.1 (Ellentuck, 1974; Kastanas, 1983). *For $\mathcal{A} \subseteq [\omega]^\omega$ the following are equivalent:*

(1) *the set $\mathcal{A}$ is completely Ramsey;*
(2) *the set $\mathcal{A}$ has the Baire property in the Ellentuck topology;*
(3) *for every finite $s \subseteq \omega$ and every infinite $N \subseteq \omega$ with $\max s < \min N$, the Kastanas game $K(\mathcal{A}, s, N)$ is determined.*

In any topological space, the sets with the Baire property are closed under the Souslin operation, and the analytic sets are obtained from closed sets by that operation (see [Kec95]). Every closed set of the usual topology is closed in the finer Ellentuck topology. Hence every analytic set, and in particular every Borel set, has the Baire property in the Ellentuck topology, and it is completely Ramsey by Theorem 11.1. Since completely Ramsey sets are Ramsey (see Exercise 4 below), the argument recovers the theorems of Galvin–Prikry and Silver in one stroke.

11.4. **Solving the problem with the game.** The proofs of the Galvin–Prikry theorem and of Ellentuck's theorem rest on one device, the *combinatorial forcing* of Galvin and Prikry [GP73]; Silver's original proof was metamathematical. We say that M *accepts* s if $[s, M] \subseteq \mathcal{A}$, and that M *rejects* s if no infinite $N \subseteq M$ accepts s. In plain words, the set M accepts the stem s if every completion of s by infinitely many elements of M lies in $\mathcal{A}$, and M rejects s if no thinning out of M produces a set that accepts s. By a diagonal argument, one finds, inside any M, an infinite N that decides every finite $s \subseteq N$: for each such s, the set of elements of N above $\max s$ accepts or rejects s. Ellentuck's insight was that these diagonal arguments are exactly the proof that the Ellentuck space is a Baire space, and that the nowhere dense sets of the Ellentuck topology are exactly the sets $\mathcal{A}$ such that every $[s, M]$ contains some $[s, N]$ disjoint from $\mathcal{A}$ [Ell74]. Kastanas's game turns the diagonalization into a play: a winning strategy for Player II, run against all possible moves of Player I and glued together by a diagonal fusion, produces an infinite $M \subseteq N$ with $[s, M] \subseteq \mathcal{A}$, and a winning strategy for Player I produces one with $[s, M] \cap \mathcal{A} = \emptyset$ [Kas83]. The property of being completely Ramsey is thus to Kastanas's game what the Baire property is to the Banach–Mazur game.

11.5. **The notation $[\omega]^\omega$ and the partition calculus.** The games of this section are played on the Ellentuck space of infinite subsets of $\omega = \{0, 1, 2, \dots\}$, and the notation for that space differs from the notation of Section 2. It comes from the partition calculus, as we now explain. For a set S and a cardinal κ, one writes $[S]^\kappa = \{X \subseteq S : |X| = \kappa\}$ and $[S]^{<\kappa} = \{X \subseteq S : |X| < \kappa\}$; thus $[\omega]^\omega$ and $[\omega]^{<\omega}$ are the families of infinite and of finite subsets of ω, as opposed to the sequence spaces A^ω and $A^{<\omega}$ of Section 2. This is the notation of the partition calculus of P. Erdős and R. Rado [ER56], in which $\kappa \to (\lambda)^r_\mu$ means that, whenever $|S| = \kappa$ and $[S]^r$ is partitioned into μ pieces, there is a set $H \subseteq S$ with $|H| = \lambda$ such that $[H]^r$ lies inside a single piece. Ramsey's theorem becomes $\omega \to (\omega)^r_k$ for finite r and k. For the infinite exponent the relation fails: using the axiom of choice, Erdős and Rado partitioned $[\omega]^\omega$ into two pieces with no infinite homogeneous set [ER52]. Positive results recover $\omega \to (\omega)^\omega_2$ for definable partitions: Galvin and Prikry for Borel partitions [GP73], Silver for analytic ones [Sil70], and Ellentuck for partitions with the Baire property in the Ellentuck topology [Ell74]; in this section, Kastanas's game is the bridge to these theorems. The partition calculus, and the school of infinite combinatorics that Erdős and his collaborators built around it, are set out in the monograph of Erdős, Hajnal, Máté and Rado [EHMR84].

Exercises.

(1) Show that the sets $[s, M]$ form a basis of a topology on $[\omega]^\omega$. (Hint: if $[s, M] \cap [t, N]$ is non-empty, then one of s and t, say s, is an initial segment of the other, and the intersection is $[t, P]$, where P is the set of elements of $M \cap N$ greater than $\max t$.)

(2) Show that the Ellentuck topology is finer than the usual one: for a finite $t \subseteq \{0, \dots, n-1\}$, the set of $X \in [\omega]^\omega$ with $X \cap \{0, \dots, n-1\} = t$ is a basic set $[s, M]$.

Show also that, if $\omega \setminus M$ is infinite, the set $[\emptyset, M]$ has empty interior in the usual topology.

(3) Let $\mathcal{A}$ be the set of $X \in [\omega]^\omega$ containing infinitely many even numbers, and let s and N be as in the definition of the Kastanas game. Show that Player II has a winning strategy in $K(\mathcal{A}, s, N)$ if N contains only finitely many odd numbers, and that Player I has one otherwise. Find in each case an infinite $N' \subseteq N$ with $[s, N'] \subseteq \mathcal{A}$ or $[s, N'] \cap \mathcal{A} = \emptyset$.

(4) Show that a set $\mathcal{A} \subseteq [\omega]^\omega$ is completely Ramsey if and only if $[\omega]^\omega \setminus \mathcal{A}$ is, and that every completely Ramsey set is Ramsey.

(5) $(*)$ Color the two-element subsets of ω red or blue, and let $\mathcal{A}$ be the set of $X \in [\omega]^\omega$ whose two least elements form a red pair. Show that $\mathcal{A}$ is completely Ramsey. (Hint: treat separately the cases in which s has at least two elements, one element, or none; in the last case use Ramsey's theorem.)

(6) Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω, viewed as a subset of $[\omega]^\omega$. Show that $\mathcal{U}$ itself is Ramsey, by finding an infinite $H \notin \mathcal{U}$ and checking that $[H]^\omega \cap \mathcal{U} = \emptyset$. For $X = \{x_0 < x_1 < \cdots\}$ in $[\omega]^\omega$, let Y_X be the union of the intervals $[x_{2k}, x_{2k+1}) = \{m \in \omega : x_{2k} \leq m < x_{2k+1}\}$, and let $\mathcal{A}$ be the set of X with $Y_X \in \mathcal{U}$. Show that Y_X and $Y_{X \setminus \{x_0\}}$ are disjoint and that their union is $\{m \in \omega : m \geq x_0\}$. Deduce that $\mathcal{A}$ is not Ramsey.

12. Applications to Banach space theory

The problems.

(1) *Differentiability of convex functions.* In calculus, a convex function of one variable is differentiable at all but countably many points. In infinite dimensions things are much less tidy. A Banach space X is an *Asplund space* if every continuous convex function on an open convex subset of X is Fréchet differentiable on a dense G_δ set. Which spaces are Asplund spaces?

(2) *The unconditional basic sequence problem.* Does every infinite-dimensional Banach space contain an unconditional basic sequence, that is, a sequence (e_n) that is a basis of the closed subspace spanned by (e_n) and for which every convergent expansion $\sum a_n e_n$ keeps converging when the signs of its terms are changed or its terms are rearranged?

In this section we put two games to work in Banach spaces. The fragmentability game of Kenderov and Moors detects the Asplund property, and Gowers's game drives his dichotomy theorem, which explains the answer to the unconditional basic sequence problem. Functional analysis lies outside the background we assume, so we begin with a short glossary.

A small glossary. Let X be a Banach space.

- The *dual* X^* is the Banach space of bounded linear functionals on X, and its closed unit ball is denoted by B_{X^*}. The *weak* topology* on X^* is the topology of pointwise convergence on the elements of X.
- A function f on an open subset of X is *Gâteaux differentiable* at x if there is $\varphi \in X^*$ with $f(x+th) = f(x)+t\varphi(h)+o(t)$ as $t \to 0$ for each fixed h, and *Fréchet differentiable* at x if $f(x+h) = f(x) + \varphi(h) + o(\|h\|)$, uniformly in the direction of h.
- A *basis* of X is a sequence (e_n) such that every $x \in X$ has a unique expansion $x = \sum_n a_n e_n$, and a *basic sequence* is a basis of its closed linear span. A basic sequence is *C-unconditional* if $\| \sum_n \varepsilon_n a_n e_n \| \leq C \| \sum_n a_n e_n \|$ for all finitely supported sequences

of scalars (a_n) and all signs $\varepsilon_n = \pm 1$, and *unconditional* if it is C-unconditional for some C.

- The space X is the *direct sum* of closed subspaces Y_1 and Y_2 if $Y_1 \cap Y_2 = \{0\}$ and $Y_1 + Y_2 = X$.
- Let K be a set with a topology $\mathcal{T}$ and a metric d. The set K is *fragmented* by d if every non-empty subset of K has non-empty relatively $\mathcal{T}$-open subsets of arbitrarily small d-diameter.

12.1. **History of the problems and their solutions using topological games.** Mazur proved in 1933 that continuous convex functions on separable Banach spaces are Gâteaux differentiable on dense G_δ sets, and Asplund introduced the spaces now named after him in 1968, in a paper in *Acta Mathematica* [Asp68]. The characterization problem was solved in 1975 by Namioka and Phelps [NP75] and by Stegall [Ste75]. The property behind the Asplund spaces is fragmentability, and Kenderov and Moors described it by a game, first in 1996 and then, in the form used below, in a paper published in 1999 [KM99].

The unconditional basic sequence problem was one of the central open problems of Banach space theory for decades. The problem was solved, negatively, by Gowers and Maurey in 1991 (published in 1993) [GM93]. They built a *hereditarily indecomposable* space: a Banach space so tangled that no infinite-dimensional subspace splits as a direct sum of two infinite-dimensional subspaces. Gowers then showed that such strange spaces are no accident. In papers of 1996 and 2002, by means of an infinite game played with subspaces and vectors, he proved that every infinite-dimensional Banach space contains an infinite-dimensional subspace that either has an unconditional basis or is hereditarily indecomposable [Gow96, Gow02]. The second paper, which contains the Ramsey theorem behind the game, appeared in the *Annals of Mathematics*.

Recognition. Gowers received the Fields Medal in 1998 for research connecting functional analysis and combinatorics; this research includes the solution of the unconditional basic sequence problem and the dichotomy theorem. He was an invited speaker at the International Congress of Mathematicians in Zürich in 1994, and he received the Prize of the European Mathematical Society in 1996.

12.2. **The games.**

12.2.1. *The fragmentability game.* Let K be a set with two topologies $\mathcal{T}_1$ and $\mathcal{T}_2$. The case to keep in mind is the dual ball B_{X^*} of a Banach space X, with the weak* topology as $\mathcal{T}_1$ and the norm topology as $\mathcal{T}_2$. The game is played as follows.

(1) Player I chooses a non-empty subset $U_0 \subseteq K$.
(2) Player II chooses a non-empty relatively $\mathcal{T}_1$-open subset $V_0 \subseteq U_0$.
(3) In round $k \geq 1$, Player I chooses a non-empty subset $U_k \subseteq V_{k-1}$, and Player II a non-empty relatively $\mathcal{T}_1$-open subset $V_k \subseteq U_k$.
(4) Player II wins if $\bigcap_k U_k$ is empty, or if $\bigcap_k U_k$ is a single point x and every $\mathcal{T}_2$-neighborhood of x contains some U_k.

This is a Banach–Mazur game with an asymmetry: Player I may play any non-empty subset of the previous move, while Player II must play relatively $\mathcal{T}_1$-open ones; she wins if the sets close in on at most one point, and do so in the topology $\mathcal{T}_2$.

12.2.2. *Gowers's game.* Let X have a basis (e_n). A *block vector* is a non-zero finite linear combination of the e_n, and a *block subspace* is the closed span of a sequence of block vectors $x_1, x_2, \ldots$ with *successive supports*: every basis vector used in x_k comes before every basis

vector used in x_{k+1}. Let Σ be a set of sequences of normalized block vectors. In round $k \geq 1$, Player I chooses an infinite-dimensional block subspace Y_k, and Player II chooses a normalized block vector $x_k \in Y_k$. Player II wins if $(x_1, x_2, \dots) \in \Sigma$. Player I's subspaces play the role of open sets. In the Banach–Mazur game a small neighborhood limits where the other player may look, and a block subspace does the same. Player I may choose it to live as far out along the basis as he likes; he thereby forces Player II's next vector to be supported beyond any finite stage, and he can also restrict the kind of vectors available to her.

12.3. **The Main Results of this section.** Differentiability is tied to the dual space for a simple reason: the derivative of a function on X at a point is a linear functional, that is, an element of X^*. Differentiability is therefore a question about how the candidate derivatives sit inside X^*; for a convex function f and a point x, these are the *subgradients*, the functionals $\varphi \in X^*$ with $f(y) \geq f(x) + \varphi(y - x)$ for all y. Roughly, Fréchet differentiability at a point means that subgradients at nearby points are close to the derivative *in norm*, and the norm is what fragmentability controls. In the first result below we collect several characterizations of Asplund spaces, one of which is exactly what the game of Kenderov and Moors detects; the second is Gowers's dichotomy.

Theorem 12.1 (Asplund, 1968; Namioka–Phelps, 1975; Stegall, 1975). *For a Banach space X the following are equivalent:*

(1) *the space X is an Asplund space;*
(2) *every separable subspace of X has a separable dual;*
(3) *the dual X^* has the Radon–Nikodým property, a property from the theory of vector-valued measures that we do not use below;*
(4) *the dual ball B_{X^*}, with the weak* topology, is fragmented by the norm (see the glossary at the beginning of this section); that is, every non-empty subset of B_{X^*} has non-empty relatively weak*-open subsets of arbitrarily small norm diameter.*

Kenderov and Moors [KM99] proved that the fragmentability game detects exactly this property: the set K, with the topology $\mathcal{T}_1$, is fragmented by a metric whose topology contains $\mathcal{T}_2$ if and only if Player II has a winning strategy in the fragmentability game. In particular, by condition (4), Player II wins the game on B_{X^*}, with the weak* and norm topologies, whenever X is an Asplund space.

Theorem 12.2 (Gowers's dichotomy, 1996 and 2002). *Every infinite-dimensional Banach space contains an infinite-dimensional subspace that either has an unconditional basis or is hereditarily indecomposable.*

The two alternatives exclude each other, because a hereditarily indecomposable space contains no unconditional basic sequence [GM93]: the closed spans of the even-numbered and of the odd-numbered terms of such a sequence would form a direct sum. Every infinite-dimensional Banach space, however large, thus contains a subspace that is either perfectly civilized or thoroughly rebellious; there is no third option.

12.4. **Solving the problems with the games.**

12.4.1. *Asplund spaces.* Problem (1) was solved in 1975 without games, by Namioka and Phelps and by Stegall. What the game of Kenderov and Moors adds is a new way to establish

fragmentability, and with it the Asplund property: instead of producing a fragmenting metric, one describes how Player II should play. We give the easy half of their theorem.

Suppose that B_{X^*} is fragmented by the norm. Then Player II has a simple strategy: she answers each move U_k with a relatively weak*-open subset $V_k \subseteq U_k$ of norm diameter less than 2^{-k}. Since $U_{k+1} \subseteq V_k$ for every k, the intersection $\bigcap_k U_k$ has norm diameter 0, and it is either empty or a single point x. In the second case, the set U_{k+1} lies in the norm ball of radius 2^{-k} about x, whence every norm neighborhood of x contains some U_k, and Player II wins. Conversely, Kenderov and Moors build a fragmenting metric out of a winning strategy for Player II; roughly, two points are close when the strategy takes a long time to separate them. The advantage of the game is that a strategy is a local object, built one move at a time, while a fragmenting metric is a global one, and describing how to play is often much easier than writing down a metric.

12.4.2. *Gowers's dichotomy.* Gowers's infinite Ramsey theorem [Gow02] says that for every analytic set Σ and every sequence of positive tolerances, there is a block subspace Y such that either no block sequence of Y lies in Σ, or Player II has a winning strategy in Gowers's game played below Y (Player I choosing block subspaces of Y) for a slight enlargement of Σ, obtained by wiggling the members of Σ within the given tolerances. The theorem is the counterpart, for block sequences, of the theorems of Galvin–Prikry and Ellentuck on Ramsey sets (see Section 11), and Gowers's game plays the part of Kastanas's game. An exact Ramsey statement is false for vectors; what survives is the game, played up to the given tolerances. Gowers's proof uses a "combinatorial forcing" argument, in which subspaces are said to accept or reject finite block sequences.

Roughly, the dichotomy follows from an application of the Ramsey theorem to the set of block sequences that fail to be C-unconditional [Gow96, Gow02]. If, for some constant C, some block subspace avoids these sequences, then every normalized block sequence in that subspace is C-unconditional, and the subspace has an unconditional basis. Otherwise, for every C and in every block subspace, Player II can produce block sequences that fail to be C-unconditional, and from her strategies Gowers extracts a hereditarily indecomposable subspace: none of its infinite-dimensional closed subspaces is the direct sum of two infinite-dimensional closed subspaces. The game locates this subspace; that such spaces exist at all is the theorem of Gowers and Maurey.

Exercises.

(1) Show that the standard unit vector bases of c_0 and of ℓ_p, for $1 \le p < \infty$, are 1-unconditional.

(2) Let $f(x,y) = x^3y/(x^4+y^2)$ for $(x,y) \neq (0,0)$, and $f(0,0) = 0$. Show that f is Gâteaux differentiable at the origin, with $\varphi = 0$, but not Fréchet differentiable there. (Hint: look at $f(x,x^2)$.)

(3) Let (K,d) be a metric space, and let $\mathcal{T}_1 = \mathcal{T}_2$ be the topology of d. Show that Player II has a winning strategy in the fragmentability game, by answering U_k with the intersection of U_k and a ball of radius 2^{-k-1} about a point of U_k.

(4) Let (e_n) be a 1-unconditional basis of X, and let $x_1, x_2, \dots$ be block vectors with successive supports. Show that $\|\sum_k \varepsilon_k b_k x_k\| \le \|\sum_k b_k x_k\|$ for all finite sequences of scalars b_k and signs $\varepsilon_k = \pm 1$.

(5) $(*)$ Let X have a basis (e_n), and consider Gowers's game in X. Show that Player II has a winning strategy when Σ is the set of all sequences of normalized block vectors with successive supports, and that Player I has a winning strategy when Σ is the set

of sequences whose first term is $e_1/\|e_1\|$. (Hint: the coefficient functionals of a basis are continuous.)

13. Applications to ultrafilters

Definitions. A *filter* on ω is a non-empty family $\mathcal{F}$ of subsets of ω that does not contain $\emptyset$, is closed under finite intersections, and contains every superset of each of its members. An *ultrafilter* is a filter $\mathcal{U}$ such that for every $x \subseteq \omega$ either x or $\omega \setminus x$ belongs to $\mathcal{U}$. The ultrafilter of all sets containing a fixed point n is called *principal*; every other ultrafilter is *nonprincipal*. A nonprincipal ultrafilter contains every cofinite set (see Exercise 1 at the end of this section). Let $\mathcal{U}$ be a nonprincipal ultrafilter, and consider a partition of ω into pieces that do not belong to $\mathcal{U}$. The ultrafilter is a *P-point* if for every such partition some element of $\mathcal{U}$ meets every piece in a finite set, and it is *selective* if for every such partition some element of $\mathcal{U}$ meets every piece in at most one point. The set of all ultrafilters on ω, with a natural compact topology, is the space $\beta\omega$, and its subspace $\omega^* = \beta\omega \setminus \omega$ consists of the nonprincipal ultrafilters; these spaces appear only in passing.

The problem. Identify every subset of ω with its characteristic function; the power set $\mathcal{P}(\omega)$ then becomes the Cantor space 2^ω, and a filter on ω becomes a set of reals. The axiom of choice provides nonprincipal ultrafilters on ω in abundance (there are $2^{2^{\aleph_0}}$ of them, the points of ω^*), and yet nobody has ever exhibited one. The problem is to explain this: to determine how irregular a nonprincipal ultrafilter must be as a set of reals, how much choice its existence requires, and which ultrafilters are tame enough to carry the Ramsey theory of Section 11.

A principal ultrafilter is a point of ω in disguise: it collects the sets that contain the point, and its two-valued measure is the Dirac measure at the point. A nonprincipal ultrafilter is a point at infinity that passes judgment on every subset of ω, consistently and irrevocably, and always against the finite ones. Seen through Ulam's binary version of the Banach–Mazur game, such a judge is invisible to topology: in Theorem 13.1(1) below we show that the game it defines has no winner. Seen through the games of Galvin, the well-mannered judges are the selective ultrafilters and the P-points, and in Theorem 13.1(4) below we show that they are exactly the judges that cannot be outplayed.

13.1. **History of the problem and of the solution using topological games.** Ultrafilters came into mathematics through two doors, the algebra of sets and general topology. Tarski proved in 1930, using the axiom of choice, that every filter of subsets of a set extends to an ultrafilter [Tar30]; Stone made the ultrafilters of a Boolean algebra into the points of a compact space in 1936 and 1937 [Sto36]; and Cartan introduced filters and ultrafilters in 1937 to describe convergence in general topology [Car37]. In 1938 Sierpiński observed that a nonprincipal ultrafilter on ω, regarded as a set of reals, is not Lebesgue measurable [Sie38]; the symmetry behind his argument, the flip $x \mapsto \omega \setminus x$, also shows that it does not have the Baire property (we prove both facts in Theorem 13.1(2) below). In 1962 Mycielski and Steinhaus proposed the Axiom of Determinacy, which makes every set of reals regular and therefore rules out nonprincipal ultrafilters on ω altogether [MS62] (see Theorem 13.1(3) below). Feferman showed in 1965, by forcing, that the existence of a definable nonprincipal ultrafilter on ω cannot be proved in ZFC [Fef65], and Shelah showed in 1984 that ZF together with the axiom of dependent choices is consistent with the statement that every set of reals

has the Baire property, relative to ZF alone [She84]. Some choice beyond dependent choice is therefore needed to produce a single nonprincipal ultrafilter on ω.

Meanwhile the ultrafilters themselves were being sorted. Walter Rudin proved in 1956 that under the continuum hypothesis there are P-points, and deduced that ω^* is not homogeneous [Rud56]; Frolík proved in 1967 that ω^* is not homogeneous in ZFC [Fro67]. Booth and Kunen studied the selective ultrafilters [Boo70], and Kunen proved that an ultrafilter is selective exactly when every coloring of $[\omega]^2$ in two colors has a homogeneous set in the ultrafilter. Mathias showed in 1977 that selective ultrafilters carry the Ramsey theory of analytic sets [Mat77]. Kunen proved in 1976 that it is consistent with ZFC that there are no selective ultrafilters [Kun76], and Shelah proved shortly afterwards that it is consistent that there are no P-points at all (see the account of Wimmers [Wim82]).

The topological side of the problem was settled by Jalali-Naini in his thesis of 1976 [JN76] and, independently, by Talagrand in 1980 [Tal80]. Both characterized the meager filters by a combinatorial condition on partitions of ω into intervals. Talagrand's motivation came from pointwise compact sets of measurable functions, a subject closely tied to the theory of Rosenthal compacta, begun by Rosenthal and developed by Bourgain, Fremlin and Talagrand. The games enter twice. The irregularity of a nonprincipal ultrafilter follows from a strategy-stealing argument in Ulam's binary game, which rests on the same symmetry $x \mapsto \omega \setminus x$ as Sierpiński's proof (we give the argument in the proof of Theorem 13.1(1) below); and Galvin and McKenzie characterized the selective ultrafilters and the P-points by the games described below (see the book of Bartoszyński and Judah [BJ95]).

Recognition. Talagrand received the Abel Prize in 2024, and Shelah the Wolf Prize in 2001.

13.2. **The games.**

Ulam's game. Let $Z \subseteq 2^\omega$ be given. In the game $\mathrm{Ul}(Z)$, Player I chooses a non-empty finite binary string s_0, Player II answers with a non-empty finite binary string t_0, Player I continues with a non-empty string s_1, and so on, forever. Player II wins if the concatenation

$$x = s_0{}^\frown t_0{}^\frown s_1{}^\frown t_1{}^\frown \cdots$$

belongs to Z. Each move determines the basic clopen set $N_s = \{x \in 2^\omega : s \subseteq x\}$ of the position s it reaches, and this set lies inside the previous one; the game is therefore the Banach–Mazur game on the Cantor space, in the binary form proposed by Ulam (see Section 20). Since the Cantor space is a compact metric space, both parts of the theorem of Banach and Oxtoby apply (see Theorem 3.5; its proof works equally well with moves restricted to basic open sets), and here they read as follows: Player II has a winning strategy in $\mathrm{Ul}(Z)$ if and only if Z is comeager, and Player I has one if and only if $Z \cap N_s$ is meager for some finite binary string s.

Galvin's games. Let $\mathcal{U}$ be an ultrafilter on ω. In the *selective game* $S(\mathcal{U})$, Player I chooses $A_0 \in \mathcal{U}$, Player II chooses $n_0 \in A_0$, Player I chooses $A_1 \in \mathcal{U}$, Player II chooses $n_1 \in A_1$, and so on; Player II wins if $\{n_0, n_1, \dots\} \in \mathcal{U}$. In the *P-point game* $P(\mathcal{U})$, Player II chooses a finite set $F_k \subseteq A_k$ instead of a single element, and wins if $\bigcup_k F_k \in \mathcal{U}$. Player I plays the skeptic: at each turn he names a set that $\mathcal{U}$ approves of, and he challenges Player II to assemble, one element (or one finite batch) at a time, a set that $\mathcal{U}$ approves of as well.

The tame ultrafilters of the main result are the P-points and the selective ultrafilters defined at the beginning of this section. Every selective ultrafilter is a P-point, since meeting each piece of a partition in at most one point is a special case of meeting it in a finite set; the counterpart of this inclusion for the games is the content of Exercise 4 below.

13.3. **The Main Result of this section.** In the main result we gather the behavior of nonprincipal ultrafilters in the two kinds of games, together with its consequences for category, measure and determinacy: in Ulam's game they are invisible, and in Galvin's games the tame ones are exactly those that cannot be outplayed.

Theorem 13.1 (Sierpiński, 1938; Mycielski and Steinhaus, 1962; Galvin and McKenzie). *Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω, regarded as a subset of 2^ω. Then the following hold.*

(1) *The game $\mathrm{Ul}(\mathcal{U})$ is not determined.*
(2) *The set $\mathcal{U}$ is neither meager nor comeager, it does not have the Baire property, and it is not measurable for the coin-tossing measure on 2^ω.*
(3) *If the game $\mathrm{Ul}(Z)$ is determined for every $Z \subseteq 2^\omega$, then every subset of 2^ω has the Baire property, and there is no nonprincipal ultrafilter on ω.*
(4) *The ultrafilter $\mathcal{U}$ is selective if and only if Player I has no winning strategy in $S(\mathcal{U})$, and it is a P-point if and only if Player I has no winning strategy in $P(\mathcal{U})$.*

By part (2), it is no accident that nobody has written down a nonprincipal ultrafilter on ω: analytic and coanalytic sets have the Baire property (see [Kec95]), so no nonprincipal ultrafilter is analytic or coanalytic. The proofs of (1) and (2), given below, use only two properties of $\mathcal{U}$:

- membership in $\mathcal{U}$ does not depend on finitely many digits, because $\mathcal{U}$ is nonprincipal;
- the complement $\bar{x} = \omega \setminus x$ belongs to $\mathcal{U}$ if and only if x does not, because $\mathcal{U}$ is an ultrafilter.

13.4. **Solving the problem with the games.**
Stealing a strategy. The idea is to let Player I play two games against a strategy of Player II at once, the second always the mirror image of the first, so that the strategy is forced to put both a set and its complement into $\mathcal{U}$. For a finite binary string s, we write $\bar{s}$ for the string obtained by flipping every digit of s. Suppose that τ is a winning strategy for Player II in $\mathrm{Ul}(\mathcal{U})$. Player I opens the first game with the string 0, and the strategy τ answers t_0. He opens the second game with $\overline{0^\frown t_0}$ and receives the answer u_0. Back in the first game, he plays $\bar{u}_0$ and receives t_1; in the second, he plays $\bar{t}_1$ and receives u_1; and so on (Figure 19). By construction, each block of either play is the flip of the corresponding stretch of the other, and therefore the two plays x and y satisfy $y = \bar{x}$. Both plays are won by τ, whence $x \in \mathcal{U}$ and $\bar{x} \in \mathcal{U}$, which is impossible. Hence Player II has no winning strategy.

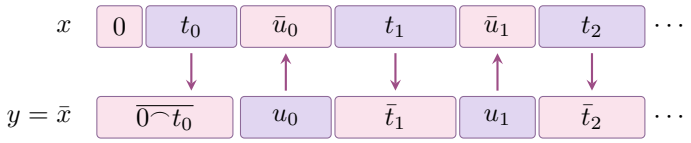


FIGURE 19. Stealing Player II's strategy in $\mathrm{Ul}(\mathcal{U})$. Lavender blocks are answers of the strategy τ, blush blocks are moves of Player I, and each arrow copies an answer of τ in one game, flipped, as a move of Player I in the other.

Suppose now that Player I has a winning strategy σ, and let s be its first move. From the position s on, the strategy σ answers the moves of Player II and forces $x \notin \mathcal{U}$, that is, $\bar{x} \in \mathcal{U}$; flipping every move, we obtain a strategy that, from the position $\bar{s}$, answers its opponent and forces the play into $\mathcal{U}$. Player II can transplant this strategy into $\mathrm{Ul}(\mathcal{U})$. She makes her first

move at least as long as s, so that at her next turn the position is longer than s. She then pretends that the position is $\bar{s}$ followed by a single move of her opponent, namely the digits of the actual position beyond the length of s, and from then on she answers as the flipped strategy does. The resulting play differs in finitely many digits from a play that the flipped strategy wins, and therefore she wins $\mathrm{Ul}(\mathcal{U})$, which we have just seen to be impossible. This proves (1).

Baire property and measure. If Z is comeager, then Player II wins $\mathrm{Ul}(Z)$ by playing, at her k-th move, into the k-th of countably many dense open sets whose intersection lies in Z; if Z is meager, then Player I wins in the same way. By (1), the set $\mathcal{U}$ is therefore neither meager nor comeager. Next, a set $Z \subseteq 2^\omega$ that is invariant under finite changes and has the Baire property is meager or comeager. Indeed, suppose that Z is not meager. Since Z has the Baire property, it differs by a meager set from an open set, which must be non-empty; hence the difference $N_s \setminus Z$ is meager for some string s. For every string t of the same length as s, the homeomorphism of 2^ω that replaces the initial segment s by t carries N_s onto N_t and preserves Z; it follows that $N_t \setminus Z$ is meager as well. The finitely many sets N_t cover 2^ω, whence the complement of Z is meager. As $\mathcal{U}$ is invariant under finite changes and is neither meager nor comeager, it does not have the Baire property. Measure is no better. The flip $x \mapsto \bar{x}$ preserves the coin-tossing measure λ and carries $\mathcal{U}$ onto its complement, so, if $\mathcal{U}$ were measurable, we would have $\lambda(\mathcal{U}) = 1 - \lambda(\mathcal{U}) = \frac{1}{2}$, while Kolmogorov's zero–one law, applied to a measurable set that is invariant under finite changes, gives $\lambda(\mathcal{U}) \in \{0, 1\}$. This proves (2).

Determinacy kills ultrafilters. We show that, if all of Ulam's games are determined, then every subset of 2^ω has the Baire property; part (2) then leaves no room for a nonprincipal ultrafilter. Assume that every game $\mathrm{Ul}(Z)$ is determined, and let $Z \subseteq 2^\omega$ and a string s be given. Applying determinacy to the set $\{z : s^\frown z \in Z\}$ and the theorem of Banach and Oxtoby (see Theorem 3.5), we find a string $s' \supseteq s$ such that Z is comeager in $N_{s'}$ or meager in $N_{s'}$. Let O be the union of the sets $N_{s'}$ in which Z is comeager, and let O' be the union of those in which Z is meager. The union $O \cup O'$ is then a dense open set, and the symmetric difference $Z \mathbin{\triangle} O$ is covered by $O \setminus Z$, by $O' \cap Z$, and by the nowhere dense set $2^\omega \setminus (O \cup O')$, all of which are meager. Since Z was arbitrary, every subset of 2^ω has the Baire property, and by (2) there is no nonprincipal ultrafilter on ω. Applied to the sets of a pointclass that is closed under continuous preimages, the same argument leads to the remarks on definable ultrafilters at the end of this section.

Selective ultrafilters. The proof of part (4) uses a theorem of Kunen, recorded in the history above, which we quote without proof.

Lemma 13.2 (Kunen). *If $\mathcal{U}$ is a selective ultrafilter on ω, then for every coloring of the two-element subsets of ω in two colors there is a set $H \in \mathcal{U}$ all of whose two-element subsets have the same color.*

The idea of the first half is to color each pair according to whether the strategy of Player I still allows its larger element after its smaller one; a homogeneous set in $\mathcal{U}$, provided by Lemma 13.2, is then a set that Player II can play in full. Suppose that $\mathcal{U}$ is selective, and let σ be a strategy for Player I in $S(\mathcal{U})$. For a finite set $s \subseteq \omega$, listed increasingly, let $A_s \in \mathcal{U}$ be the move of σ after Player II has played the elements of s (we put $A_s = \omega$ if this sequence of moves is not legal), and let $B_n = \bigcap\{A_s : s \subseteq \{0, \dots, n-1\}\}$. As a finite intersection of members of $\mathcal{U}$, the set B_n belongs to $\mathcal{U}$. We color a pair $m < n$ pink if $n \in B_{m+1}$, and purple otherwise. By Lemma 13.2 there is a homogeneous set $H \in \mathcal{U}$; replacing H by $H \cap A_\emptyset$, which is still homogeneous and belongs to $\mathcal{U}$, we may assume that $H \subseteq A_\emptyset$. The color of H is not purple: for $m = \min H$, the sets $H \setminus \{m\}$ and B_{m+1} both belong to $\mathcal{U}$ and therefore meet,

and a common point n yields a pink pair $\{m, n\}$ inside H. The color of H is thus pink. We list H increasingly as $h_0 < h_1 < \cdots$. Since $H \subseteq A_\emptyset$ and every pair in H is pink, we have $h_0 \in A_\emptyset$, and $h_{k+1} \in B_{h_k+1} \subseteq A_{\{h_0,\dots,h_k\}}$ for every k. Hence Player II may answer σ with $h_0, h_1, h_2, \dots$ in turn, and the set she produces is $H \in \mathcal{U}$. Consequently, the strategy σ is not winning.

Conversely, suppose that $\mathcal{U}$ is not selective, and fix a partition $\{P_j\}$ of ω into pieces outside $\mathcal{U}$ such that no element of $\mathcal{U}$ meets every piece in at most one point. Player I wins by always removing the pieces already visited. When Player II has played elements of $P_{j_0}, \dots, P_{j_{k-1}}$, Player I plays $\omega \setminus (P_{j_0} \cup \cdots \cup P_{j_{k-1}})$, which belongs to $\mathcal{U}$ because a finite union of sets outside an ultrafilter lies outside it. The set produced by Player II then meets each piece in at most one point, and it is not in $\mathcal{U}$. The P-point game is handled by the same two arguments, with finite sets in place of single points and the P-point property in place of homogeneity; for the details we refer to the book of Bartoszyński and Judah [BJ95]. This proves (4).

Selectivity is exactly what the Ramsey theory of Section 11 needs. Mathias proved that if $\mathcal{U}$ is selective, then every analytic $\mathcal{A} \subseteq [\omega]^\omega$ has a homogeneous set in $\mathcal{U}$: there is $H \in \mathcal{U}$ with $[H]^\omega \subseteq \mathcal{A}$ or $[H]^\omega \cap \mathcal{A} = \emptyset$ [Mat77]. A nonprincipal ultrafilter $\mathcal{U}$ also serves the other side of the story: it produces a set that is not Ramsey. The ultrafilter itself, regarded as a subset of $[\omega]^\omega$, is not such a set. Indeed, for every basic set $[s, M]$, the set M contains an infinite $N \notin \mathcal{U}$ (one of two complementary infinite halves of M), and no member of $[s, N]$ belongs to $\mathcal{U}$, because membership in $\mathcal{U}$ ignores finitely many points; thus the ultrafilter is even completely Ramsey. A small twist does the job. We list each $X \in [\omega]^\omega$ increasingly as $x_0 < x_1 < \cdots$, and we let $\mathcal{A}$ be the set of those X for which $\bigcup_n [x_{2n}, x_{2n+1})$ belongs to $\mathcal{U}$. Removing x_0 from X exchanges this union with $\bigcup_n [x_{2n+1}, x_{2n+2})$, its complement in the cofinite set $[x_0, \infty)$; hence exactly one of X and $X \setminus \{x_0\}$ belongs to $\mathcal{A}$. Both sets lie in $[H]^\omega$ whenever $X \subseteq H$, and consequently no infinite H satisfies $[H]^\omega \subseteq \mathcal{A}$ or $[H]^\omega \cap \mathcal{A} = \emptyset$. A single nonprincipal ultrafilter thus provides a set that is not Ramsey.

13.5. **Further remarks (optional).**

Definable ultrafilters. The proof of part (3) of Theorem 13.1 works one pointclass at a time, and under projective determinacy (a consequence of the existence of infinitely many Woodin cardinals, by the theorem of Martin and Steel recalled in Section 15) every projective set has the Baire property, so no nonprincipal ultrafilter is projective either. Without determinacy, the bound recorded after the main result, that no nonprincipal ultrafilter is analytic or coanalytic, is sharp. In Gödel's constructible universe, a transfinite recursion along the Σ^1_2 well-ordering of the reals produces a Σ^1_2 nonprincipal ultrafilter $\mathcal{U}$, which is even Δ^1_2, since $x \notin \mathcal{U}$ if and only if $\omega \setminus x \in \mathcal{U}$.

Meager filters. The game argument shows that a nonprincipal ultrafilter is not meager; in the following theorem we characterize the meager filters, and thereby isolate the part of the ultrafilter property that is responsible.

Theorem 13.3 (Jalali-Naini, 1976; Talagrand, 1980). *A filter $\mathcal{F}$ on ω that contains every cofinite set is meager, as a subset of 2^ω, if and only if there is a partition of ω into consecutive finite intervals $I_0 < I_1 < \cdots$ such that every element of $\mathcal{F}$ meets all but finitely many of the I_n.*

Proof. One direction is a direct check. For the other, we build a single set e such that the points of the k-th nowhere dense set disagree with e on every later block of a partition into intervals, and these blocks are the intervals we need. If such a partition exists, then the filter is contained in the union over m of the sets $\{x : x \cap I_n \neq \emptyset \text{ for all } n \geq m\}$. These sets are closed, and nowhere dense, because any finite string can be extended by zeros across a whole

interval I_n. For the converse, let $C_0 \subseteq C_1 \subseteq \cdots$ be closed nowhere dense sets whose union contains $\mathcal{F}$. We choose $0 = n_0 < n_1 < \cdots$ and binary strings w_k of length $n_{k+1} - n_k$ such that $N_{u^\frown w_k} \cap C_k = \emptyset$ for every string u of length n_k. This is possible because there are only finitely many such u, and we may extend a candidate for w_k once for each of them, each time avoiding C_k. Let $I_k = [n_k, n_{k+1})$ and let $e \subseteq \omega$ be the set whose characteristic function is $w_0{}^\frown w_1{}^\frown \cdots$. Since $C_m \subseteq C_k$ for $k \geq m$, every point of C_m disagrees with e somewhere on I_k for each $k \geq m$. Now let $x \in \mathcal{F}$ be given. The set $x \cup e$ then belongs to $\mathcal{F}$ and therefore lies in C_m for some m; for every $k \geq m$, the sets $x \cup e$ and e then differ somewhere in I_k, which means that x meets I_k. Hence the intervals I_k witness the condition of the theorem. ☺

A nonprincipal ultrafilter $\mathcal{U}$ fails the condition of Theorem 13.3 for every partition: the union of the even intervals and the union of the odd intervals are complementary, so one of them belongs to $\mathcal{U}$, and it misses infinitely many intervals. This is a second proof that $\mathcal{U}$ is not meager. By Theorem 13.3, nonmeagerness also survives finite-to-one images. Let $\mathcal{F}$ be a filter on ω that contains every cofinite set, let $f\colon \omega \to \omega$ be finite-to-one, and let $f(\mathcal{F}) = \{y \subseteq \omega : f^{-1}(y) \in \mathcal{F}\}$ be the image filter. If $f(\mathcal{F})$ is meager, witnessed by a partition $\{I_n\}$, then every $x \in \mathcal{F}$ meets $f^{-1}(I_n)$ for all but finitely many n, because the image $f[x]$ belongs to $f(\mathcal{F})$. The sets $f^{-1}(I_n)$ are finite and pairwise disjoint, and infinitely many of them are non-empty; hence any partition of ω into consecutive intervals each of which contains a non-empty set of the form $f^{-1}(I_n)$ witnesses that $\mathcal{F}$ is meager.

Points that nobody can point at. The space ω^* is the Stone space of the Boolean algebra $\mathcal{P}(\omega)/\mathrm{fin}$, and its points are the nonprincipal ultrafilters on ω. Pospíšil showed in 1937 that there are $2^{2^{\aleph_0}}$ of them, by means of an independent family of $2^{\aleph_0}$ subsets of ω in the sense of Fichtenholz, Kantorovich and Hausdorff: every intersection of finitely many members of the family and of the complements of finitely many other members is infinite. Hence every choice, for each member, of either the member or its complement extends to a nonprincipal ultrafilter, and distinct choices give distinct ultrafilters. A P-point is exactly a point $p \in \omega^*$ such that every G_δ subset of ω^* containing p is a neighborhood of p. If $D \subseteq \omega^*$ is countable and p lies in the closure of D but not in D, then the complement $\omega^* \setminus D$ is a G_δ set containing p that is not a neighborhood of p; no homeomorphism of ω^* can therefore move a P-point to such a point. Points of this kind exist in ZFC, and P-points exist under the continuum hypothesis; this is Rudin's proof that ω^* is not homogeneous [Rud56]. In 1978 Kunen proved in ZFC that there are *weak P-points*, points of ω^* that are not limits of any countable subset of the rest of ω^* [Kun80]; they give a proof that ω^* is not homogeneous without the continuum hypothesis. All of these points are produced by choice, and by the remarks on definable ultrafilters above, none of them can be produced by a definition of low complexity. The large-cardinal picture is sharper still: Todorcevic proved that, in the presence of large cardinals, the selective ultrafilters are exactly the filters that are generic over $L(\mathbb{R})$ for the partial order $\mathcal{P}(\omega)/\mathrm{fin}$ (see Farah [Far98]). A selective ultrafilter $\mathcal{U}$ is thus a generic object for the very Boolean algebra whose Stone space it inhabits, and every statement about it that holds in $L(\mathbb{R})[\mathcal{U}]$ is forced by one of its own members, as every true statement about a Cohen real is forced by a finite initial segment of that real.

Exercises.

(1) Let $\mathcal{U}$ be an ultrafilter on ω that contains a finite set. Show that $\mathcal{U}$ is principal. Deduce that a nonprincipal ultrafilter contains every cofinite set, and that if $x \in \mathcal{U}$ and $y \subseteq \omega$ differs from x in finitely many points, then $y \in \mathcal{U}$.

(2) For each of the following sets $Z \subseteq 2^\omega$, decide which player has a winning strategy in $\mathrm{Ul}(Z)$, and check the answer against Theorem 3.5: (a) the sequences with infinitely many terms equal to 1; (b) the sequences that are eventually 0; (c) the sequences x with $x(n) = 1$ for infinitely many even n; (d) the sequences x with $x(0) = 0$.
(3) Let $\mathcal{F}$ be the filter of cofinite subsets of ω. Show that $\mathcal{F}$ is meager in 2^ω, first directly and then by means of Theorem 13.3.
(4) Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω such that Player I has no winning strategy in $S(\mathcal{U})$. Show, without using Theorem 13.1, that Player I has no winning strategy in $P(\mathcal{U})$ either. (Hint: Player II may play one-element sets in $P(\mathcal{U})$.)
(5) $(*)$ Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω with the property in the conclusion of Lemma 13.2. Show that $\mathcal{U}$ is selective. (Hint: given a partition of ω into pieces outside $\mathcal{U}$, color a pair according to whether its two elements lie in the same piece.)

Part III. Applications of other topological games

In the games of this part the players no longer shrink a set. In the Menger game they choose open covers and finite subfamilies of them, and in the games of descriptive set theory they call out natural numbers, one at a time. What survives from Part I is the framework of Section 2 (see Definition 2.2): a game is a tree of positions together with a set of plays won by Player II, and the theorems of this part turn on which player, if any, has a winning strategy.

Roadmap

A property of a space can be turned into a statement about one player of a game, and a stronger property into a statement about the other player. When this happens, the two properties coincide on a space exactly when the game on that space is determined. In Section 14 we follow this idea through Menger's conjecture: the Menger property and σ-compactness are statements about the two players of the Menger game (see Theorem 14.1), and the theorem of Fremlin and Miller shows that the game can be undetermined on a set of reals. In Section 15 the roles are reversed, and determinacy becomes a tool rather than a question. There the games are played with natural numbers, and Martin's theorem says that every game with a Borel payoff set is determined (see Theorem 15.1). Wadge's comparison of Borel sets then follows from the determinacy of a single game, in which a winning strategy for either player is a continuous reduction of one set to the other set or to its complement. The two sections are independent of each other.

14. Applications to set-theoretic topology

The problem. Compact sets are the "small" sets of topology, and countable unions of compact sets (σ-compact sets) are almost as tame. In 1924 Karl Menger proposed a covering property that looked like a clean characterization of σ-compactness. A space X has the *Menger property* if for every sequence $\mathcal{U}_1, \mathcal{U}_2, \dots$ of open covers of X there are finite sets $\mathcal{F}_n \subseteq \mathcal{U}_n$ such that $\bigcup_n \mathcal{F}_n$ covers X. Every σ-compact space has the Menger property: if $X = \bigcup_n K_n$ with each K_n compact, it suffices to take for $\mathcal{F}_n$ a finite subset of $\mathcal{U}_n$ that covers K_n. Menger conjectured the converse for metric spaces. The problem is to decide Menger's conjecture, in particular for sets of real numbers.

14.1. **History of the problem and of the solution using topological games.** Menger introduced the property in 1924 [Men24], in a different but equivalent form, while studying a basis property of metric spaces. He conjectured that the property characterizes σ-compactness. Hurewicz gave the property the form above in a paper of 1926 [Hur26]. In the same paper he proved a theorem that is now stated as a characterization of the property by an infinite game; the game form is due to later authors, notably Telgársky and Scheepers. Hurewicz also introduced a stronger property, for which he made the analogous conjecture. Under the continuum hypothesis Menger's conjecture fails: a Luzin set, an uncountable set of reals meeting every meager set in a countable set, has the Menger property but is not σ-compact. In 1984 Telgársky characterized σ-compactness of metrizable spaces by the same game [Tel84]. Menger's conjecture thereby became the statement that the Menger game is determined on every set of reals.

In 1988 David Fremlin and Arnold Miller solved the problem [FM88]: they proved, without any hypothesis beyond ZFC, that there is a set of reals with the Menger property that is not σ-compact. The closely related property of Rothberger (1938), in which a single member of each cover is chosen, has a parallel story in terms of games. Galvin introduced the point-open game in 1978 [Gal78]. In 1994 Pawlikowski proved that a set of reals is undetermined in this game if and only if it is uncountable and has Rothberger's property [Paw94]. Borel conjectured in 1919 that every set of reals of strong measure zero is countable; Laver proved in 1976 that the conjecture is consistent [Lav76]. Since Rothberger's property implies strong measure zero, every set of reals is determined in the point-open game in Laver's model. Under the continuum hypothesis, on the other hand, there are Luzin sets, and every Luzin set is undetermined in this game, by a theorem of Recław [Rec94].

14.2. **The games.** The *Menger game* on X is played in rounds (see Figure 20). In round n, Player I chooses an open cover $\mathcal{U}_n$ of X, and Player II chooses a finite set $\mathcal{F}_n \subseteq \mathcal{U}_n$. Player II wins if $\bigcup_n \mathcal{F}_n$ covers X. In the *Rothberger game* Player II is more frugal still: she chooses a single member of each cover, and she wins if the chosen sets cover X. In the literature Player I is called ONE and Player II is called TWO. In Galvin's *point-open game*, Player I chooses a point $x_n \in X$ and Player II an open set $U_n \ni x_n$, and Player II wins if the sets U_n do not cover X; the point-open game is dual to the Rothberger game, in the sense that a player has a winning strategy in one of the two games exactly when the other player has one in the other game [Gal78]. The point-open game reverses the winning condition: in the Menger and Rothberger games Player II wants her chosen sets to cover X, while in Galvin's game she chooses the open sets but wants them not to cover X.

14.3. **The Main Result of this section.** Hurewicz and Telgársky turned the Menger property and σ-compactness into statements about the two players of the Menger game, and Fremlin and Miller proved that the gap between these two statements is real.

Theorem 14.1 (Hurewicz, 1926; Telgársky, 1984; Fremlin–Miller, 1988).

(1) *A space X has the Menger property if and only if Player I has no winning strategy in the Menger game on X.*
(2) *A separable metrizable space X is σ-compact if and only if Player II has a winning strategy in the Menger game on X.*
(3) *There is a set of reals with the Menger property that is not σ-compact. Consequently, the Menger game on this set is not determined, and Menger's conjecture is false.*

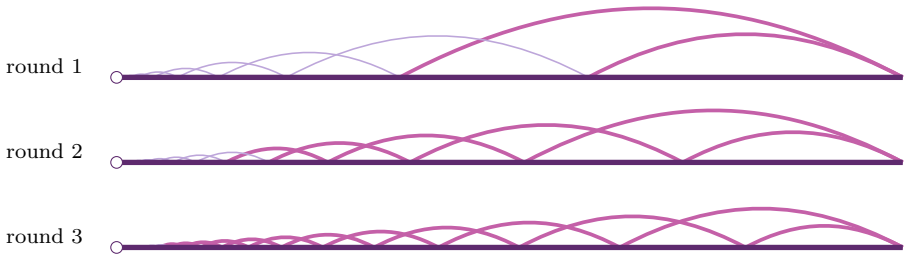


FIGURE 20. The Menger game on the half-open interval $X = (0, 4]$. In each round Player I chooses an open cover of X by intervals that accumulate at 0, drawn as arcs; Player II keeps finitely many of them (pink). Her choice in a single round never covers X, since finitely many of the intervals miss a neighborhood of 0. The space X is nevertheless σ-compact: Player II covers a larger compact piece of X in each round, and her choices together cover all of X.

14.4. **Solving the problem with the games.**

Menger's property. One direction is immediate. If Player I has no winning strategy and covers $\mathcal{U}_1, \mathcal{U}_2, \dots$ are given, then the strategy "play $\mathcal{U}_n$ in round n" is not winning: some play against it is won by Player II, and her moves witness the Menger property. The converse is the substantial direction, and we give only an outline. Applied to a constant sequence of covers, the Menger property shows that the space is Lindelöf. Player II may therefore restrict herself, at each position, to a fixed countable subcover of the cover that Player I's strategy plays. Only countably many finite positions can then arise, and the strategy plays only countably many covers. The Menger property, applied to a suitable arrangement of these countably many covers, yields a play that defeats the strategy. We omit the details, which go back to Hurewicz [Hur26].

σ-compactness. If $X = \bigcup_n K_n$ with K_n compact and increasing, Player II wins by choosing, in round n, a finite subfamily of $\mathcal{U}_n$ that covers K_n. Conversely, Telgársky unravels a winning strategy of Player II, position by position, into countably many compact sets that cover X [Tel84].

Menger's conjecture. By parts (1) and (2), a separable metrizable space with the Menger property that is not σ-compact is exactly one on which neither player has a winning strategy. Fremlin and Miller obtain such a set of reals from combinatorial properties of families of functions in ω^ω under eventual domination, in an argument that splits into cases [FM88]. The link is a later theorem of Hurewicz, from 1927: a subset of ω^ω has the Menger property exactly when no continuous image of it in ω^ω is a dominating family. Seen as a statement about games, Menger's conjecture is attractive and false for one and the same reason: it asserts a form of determinacy. On analytic sets of reals the Menger game is determined, since an analytic set with the Menger property is σ-compact; on wild sets it need not be, and Fremlin and Miller proved that a wild enough set exists without any extra axioms.

Exercises.

(1) Let X be a countable space. Show that Player II has a winning strategy in the Rothberger game on X, and that Player I has a winning strategy in the point-open game on X.

(2) Show that a space with the Menger property is Lindelöf. Deduce that an uncountable discrete space does not have the Menger property, and describe a winning strategy for Player I in the Menger game on such a space.
(3) Show that the Menger property is inherited by closed subspaces and preserved by continuous surjections.
(4) Show that Player II has a winning strategy in the Menger game on $[0,1]$, but that Player I has a winning strategy in the Rothberger game on $[0,1]$. (Hint: in round n, let Player I cover $[0,1]$ by relatively open intervals of length at most 2^{-n-2}; a finite family of intervals covering $[0,1]$ has total length at least 1.)
(5) $(*)$ Show that ω^ω does not have the Menger property, and hence is not σ-compact. (Hint: let $\mathcal{U}_n$ consist of the sets $\{x \in \omega^\omega : x(n) = k\}$ for $k \in \omega$.) Deduce from Theorem 14.1(1) that Player I has a winning strategy in the Menger game on ω^ω, and find one directly.

15. Applications to descriptive set theory

Definitions. The space ω^ω of infinite sequences of natural numbers carries the product topology: for a finite sequence s, the basic open set $N_s = \{x \in \omega^\omega : s \subseteq x\}$ consists of the sequences that begin with s, and a set is open if it is a union of such sets. (By tradition ω^ω is called "the Baire space". The name is not to be confused with the notion of Definition 1.1, although the space ω^ω, being completely metrizable, is a Baire space in that sense as well.) The *Borel sets* form the smallest family of subsets that contains the open sets and is closed under complements and countable unions. They are arranged in levels: the open sets, the closed sets, then the F_σ sets (countable unions of closed sets) and the G_δ sets, and so on through the countable ordinals; the level at which a Borel set first appears is its *Borel rank*. A set is *analytic* if it is a continuous image of ω^ω. Measurable and Woodin cardinals appear in the history below for orientation alone.

The problem. Here is the simplest infinite game imaginable: fix a set $E \subseteq \omega^\omega$ of infinite sequences of natural numbers. In the *Gale–Stewart game* $G(E)$, Player I and Player II take turns calling out natural numbers, Player I playing $x(0), x(2), \dots$ and Player II playing $x(1), x(3), \dots$, and Player II wins if the resulting sequence x belongs to E. In the language of Definition 2.2, the game $G(E)$ is the pair $(\omega^{<\omega}, E)$, with moves in $A = \omega$. Recall that a game is determined if one of the players has a winning strategy (see Section 2). For finite games this is always the case, by Zermelo's theorem; in chess, where draws are possible, the same argument shows that one of the players can force a win or both can force a draw, although nobody knows which. For infinite games determinacy can fail: using the axiom of choice one can build sets E for which neither player has a winning strategy. A nonprincipal ultrafilter on ω gives one: Ulam's game $\mathrm{Ul}(\mathcal{U})$ of Section 13 is not determined (see Theorem 13.1), and it becomes a Gale–Stewart game once the countably many non-empty finite binary strings are numbered: calling out a number then means playing a string. Which sets, and in particular which Borel sets, give determined games, and what does the answer tell us about the structure of the Borel sets?

15.1. **History of the problem and of the solution using infinite games.** The story of this problem is unusual: the game came first, and the applications came later. Gale and Stewart introduced these games in 1953 [GS53]. They proved that $G(E)$ is determined when E is open or closed, constructed an undetermined game from a well-ordering of the reals,

and asked how far determinacy extends. Wolfe proved in 1955 that games with F_σ or G_δ payoff sets are determined [Wol55], and Davis extended the result in 1964 to the next level of the Borel hierarchy [Dav64]. In 1970 Martin proved that analytic games are determined if there is a measurable cardinal [Mar70]. In 1971 Friedman showed that any proof of Borel determinacy must use uncountably many iterations of the power set operation [Fri71]. The problem therefore lay beyond the methods of ordinary analysis. That a statement about sets of real numbers should need, for its proof, sets of sets of sets of ... of reals, iterated uncountably many times, was something new.

In 1975 Donald A. Martin solved the problem [Mar75]: every Borel game is determined. His proof uses iterated power sets of the reals of every countable length, and by Friedman's theorem no proof can do with less. Even before Martin's proof, Wadge had compared sets of reals through a game of his own. In work of 1972, written up in his thesis of 1983 [Wad83], he showed that determinacy makes the Borel sets almost linearly ordered under continuous reducibility; Martin and Monk showed that this ordering is well-founded. Determinacy was later extended far beyond the Borel sets: Martin and Steel proved in 1989 that projective games are determined if there are infinitely many Woodin cardinals [MS89].

Recognition. Martin's proof appeared in the *Annals of Mathematics*. Martin, Steel and Woodin received the Karp Prize of the Association for Symbolic Logic in 1988 for their work on determinacy, and the proof of projective determinacy by Martin and Steel appeared in the *Journal of the American Mathematical Society*.

15.2. **The games.** Besides the Gale–Stewart game $G(E)$ of the problem, we need a game for comparing two sets. In the *Wadge game* $G_{\mathrm{W}}(E, F)$, Player II tries to imitate Player I: whenever his sequence lands in E, hers should land in F, and vice versa. Player I plays $x(0), x(1), \ldots$ and Player II plays $y(0), y(1), \ldots$, alternately, Player I first, and Player II wins if

$$x \in E \iff y \in F.$$

The play is the single sequence $(x(0), y(0), x(1), y(1), \ldots)$; the Wadge game is therefore a Gale–Stewart game, and its payoff set is Borel when E and F are. A strategy of Player II turns the sequence of Player I into her own, one term at a time (see Figure 21).

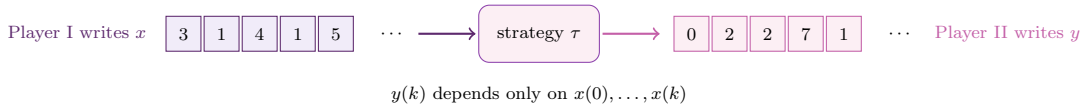


FIGURE 21. A strategy τ of Player II in the Wadge game $G_{\mathrm{W}}(E, F)$ is a machine that reads the sequence x of Player I and writes her sequence y, one term at a time; each term depends only on what she has seen so far. Such a machine defines a continuous map $x \mapsto y$, and if τ is winning, then the sequence x lies in E exactly when y lies in F, so the map reduces E to F.

15.3. **The Main Result of this section.** The main result is Martin's theorem, together with Wadge's consequence for the comparison of Borel sets. For $E, F \subseteq \omega^\omega$ write $E \leq_W F$, and say that E is *Wadge reducible* to F, if there is a continuous $f\colon \omega^\omega \to \omega^\omega$ with $E = f^{-1}(F)$.

Theorem 15.1 (Martin, 1975; Wadge, 1972).

(1) Borel determinacy. *For every Borel set $E \subseteq \omega^\omega$, the game $G(E)$ is determined.*
(2) Wadge's lemma. *For all Borel sets $E, F \subseteq \omega^\omega$, either $E \leq_W F$ or $F \leq_W \omega^\omega \setminus E$.*

By part (2), any two Borel sets are comparable under $\leq_W$, up to complements. Martin and Monk showed that $\leq_W$ is also well-founded on the Borel sets. The Borel sets, up to Wadge equivalence, therefore form a well-founded hierarchy that refines the classical Borel hierarchy and is almost linear: two Borel sets that are not comparable under $\leq_W$ are, up to Wadge equivalence, a set and its complement, whence every antichain has at most two elements. The position of a Borel set in this hierarchy measures its complexity far more finely than its Borel rank does.

15.4. **Solving the problem with the games.**

Open games. The idea is the maxim "never step into a position from which you are already lost", and the openness of E guarantees that this cautious play is enough. Suppose that E is open and that Player II has no winning strategy in $G(E)$. We shall call a finite position *good for Player I* if Player II has no winning strategy from that position; by hypothesis, the empty position is good. If a position with Player I to move is good for him, some move leads to a good position, for otherwise Player II would have a winning strategy from every successor position and hence from the position itself. If a position with Player II to move is good for Player I, so is every position reached in one move, since a move of Player II to a position that is not good would give her a winning strategy from the position she moved from. Player I can therefore play so that every position of the play is good; let x be a play in which he does so. If x were in E, the openness of E would provide an initial segment s of x with $N_s \subseteq E$. Player II would then win from the position s whatever the players did, and this position would not be good for Player I. Consequently, the play x lies outside E: the rule "move to a good position" is a winning strategy for Player I, and the game is determined.

Borel games. Martin's proof [Mar75] reduces every Borel game to a clopen game on a larger tree. In outline, for each Borel set E he builds a game G' whose moves are more complicated objects than natural numbers, together with a continuous map from plays of G' to plays of $G(E)$ that sends strategies to strategies; the payoff set of G', the preimage of E under this map, is clopen. The argument for open games works on any tree, and the game G' is therefore determined; the transfer of strategies makes $G(E)$ determined as well. The construction proceeds by induction on the Borel rank of E, and at each step the moves of the new game are sets of positions of the old one; this is why Martin's proof climbs through the iterated power sets of $\mathbb{R}$, in agreement with Friedman's theorem.

Wadge's lemma. By Borel determinacy, one of the players has a winning strategy in $G_{\mathrm{W}}(E, F)$. If Player II has one, her strategy is a blueprint for a continuous function: Player I feeds in an input sequence x one digit at a time, and Player II's step-by-step answers are the output y. Each output digit depends only on finitely many input digits, so the map $x \mapsto y$ is continuous; since the strategy is winning, the input x lies in E if and only if the output y lies in F, whence $E \leq_W F$. If Player I has one, his strategy turns each sequence y of moves of Player II continuously into a sequence x, with $y \in F$ if and only if $x \notin E$; this map witnesses $F \leq_W \omega^\omega \setminus E$. The game thus produces the reduction itself, and the comparability of Borel sets is literally the determinacy of a game.

Exercises.

(1) For each of the following sets $E \subseteq \omega^\omega$, decide which player has a winning strategy in $G(E)$, and describe one: (a) the sequences with $x(1) > x(0)$; (b) the sequences with $x(1) < x(0)$; (c) the sequences for which $\{x(2n) : n \in \omega\}$ is finite; (d) the sequences with $x(2n+1) = x(0)$ for infinitely many n.

(2) Show that $\leq_W$ is reflexive and transitive, and that $E \leq_W F$ implies $\omega^\omega \setminus E \leq_W \omega^\omega \setminus F$.

(3) Let $E \subseteq \omega^\omega$ be closed. Show that $G(E)$ is determined, by adapting the argument for open games with the roles of the players exchanged. (Hint: call a position good for Player II if Player I has no winning strategy from it.)
(4) Let E be the set of $x \in \omega^\omega$ with $x(n) = 0$ for some n, and let $F = N_{(0)}$ be the set of sequences beginning with 0. Show that Player I has a winning strategy in $G_{\mathrm{W}}(E, F)$, and that E is not Wadge reducible to F. Find a continuous f with $F = f^{-1}(\omega^\omega \setminus E)$, as Theorem 15.1(2) predicts.
(5) $(*)$ Let $E \subseteq \omega^\omega$ be open but not closed. Show that $F \leq_W E$ for every open set $F \subseteq \omega^\omega$. (Hint: fix $z \in \overline{E} \setminus E$; in $G_{\mathrm{W}}(F, E)$, let Player II copy z until an initial segment p of Player I's sequence satisfies $N_p \subseteq F$.)

Part IV. Applications of non-topological games

The games in this part are not played with open sets at all: they are finite games on boards, or games played with elements of two structures. They are included because they show the two features of games described in the Introduction, local decisions and combinable strategies, at work far from topology.

Roadmap

In this part we prove three classical theorems by games. In each of them, a statement about a large family of objects (all two-colorings of a set, all points of a square, all sentences of a given quantifier rank) is reduced to a statement about a finite game. In Section 16 a random coloring shows that a good coloring exists; the potential function of Erdős and Selfridge takes the place of the random choice and builds such a coloring one element at a time (see Theorem 16.1). In Section 17 we show that the board game Hex cannot end in a draw, and we derive from this fact about finite boards the Brouwer fixed point theorem in dimension two (see Theorem 17.1). In Section 18 we prove that two structures satisfy the same sentences of quantifier rank at most n exactly when the second player can survive n rounds of the Ehrenfeucht–Fraïssé game (see Theorem 18.6); a question about all sentences thereby becomes a question about strategies. The three sections can be read in any order, and none of them needs more than Section 2 from Part I.

16. Applications to the probabilistic method

The problem. The probabilistic method proves that combinatorial objects exist by showing that a random object has the required property with positive probability. For example, Erdős proved in 1963 that if a family $\mathcal{F}$ of n-element sets has fewer than 2^{n-1} members, then the elements can be colored red and blue so that no member of $\mathcal{F}$ is monochromatic: a random coloring makes a given set monochromatic with probability 2^{1-n}, so the expected number of monochromatic sets is less than 1. Such proofs give no construction. The problem is to turn them into explicit, deterministic, efficient constructions.

16.1. **History of the problem and of the solution using combinatorial games.** The probabilistic method goes back to a 1943 paper of Szele on tournaments. Erdős made it famous in 1947 with his lower bound for diagonal Ramsey numbers [Erd47]. The coloring theorem above is from 1963 [Erd63]. In 1973 Erdős and Selfridge studied the games now

called Maker–Breaker games, in which the players alternately claim elements of a board and Maker tries to claim all the elements of some winning set [ES73]. They gave a strategy for Breaker based on a potential function: the expected number of winning sets that Maker would complete if the rest of the game were played at random. Breaker, they showed, can keep this expectation from growing. With one player instead of two, the same potential argument solves the problem for the coloring theorem and related results: it colors the elements one at a time and never lets the expected number of monochromatic sets reach 1.

The argument was recognized as a general derandomization technique, the method of conditional probabilities. Spencer systematized the method in 1987 [Spe87], and Raghavan extended it in 1988 with pessimistic estimators [Rag88]. Beck built a theory of positional games on the Erdős–Selfridge theorem and presented it in his book of 2008 [Bec08].

Recognition. Erdős received the Wolf Prize in 1983/84.

16.2. **The game.** Let $\mathcal{F}$ be a family of subsets of a finite set V. In the *Maker–Breaker game* on $(V, \mathcal{F})$, Maker is Player I and Breaker is Player II. The players alternately claim unclaimed elements of V, Player I first, until all elements are claimed. Player I wins if he has claimed all the elements of some $A \in \mathcal{F}$, and Player II wins otherwise. Tic-tac-toe is the example to keep in mind: the set V is the board and the family $\mathcal{F}$ is the set of lines, with the difference that Breaker wins merely by preventing Maker from completing a line.

16.3. **The Main Result of this section.** The main result is the Erdős–Selfridge theorem, together with two classical consequences: a derandomized coloring and a result on multidimensional tic-tac-toe.

Theorem 16.1 (Erdős–Selfridge, 1973). *Let $\mathcal{F}$ be a family of subsets of a finite set V.*

(1) *If $\sum_{A\in\mathcal{F}} 2^{-|A|} < \frac{1}{2}$, then in the Maker–Breaker game on $(V,\mathcal{F})$ in which Maker moves first, Breaker has a winning strategy.*
(2) *If $\sum_{A\in\mathcal{F}} 2^{1-|A|} < 1$, then a coloring of V in two colors with no monochromatic member of $\mathcal{F}$ can be computed in time polynomial in $|V|$ and $|\mathcal{F}|$.*
(3) *In tic-tac-toe on the board $\{1,\dots,n\}^d$, the first player cannot force a win when $\frac{1}{2}\big((n+2)^d - n^d\big) < 2^{n-1}$.*

Part (3) follows from part (1) because the board (see Figure 22 for the case $n = 3$, $d = 2$) has $\frac{1}{2}\big((n+2)^d - n^d\big)$ lines of n cells each: the second player uses Breaker's strategy and occupies a cell of every line. By the Hales–Jewett theorem (1963), on the other hand, the first player wins when d is large compared with n [HJ63].

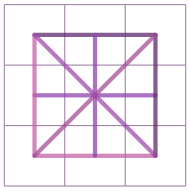


FIGURE 22. The eight winning lines of ordinary tic-tac-toe, each of three cells. Here $\sum_A 2^{-|A|} = 8 \cdot 2^{-3} = 1$, which is not less than $\frac{1}{2}$, so Theorem 16.1(1) does not apply; in fact Maker wins the Maker–Breaker game on this board (see Exercise 3 at the end of this section).

16.4. **Solving the problem with the game.** The proof turns the expectation of the problem into a potential function that Player II never lets grow. At any stage we call $A \in \mathcal{F}$ *alive* if Player II has claimed none of the elements of A, we denote by $u(A)$ the number of unclaimed elements of A, and we set

$$\Phi = \sum_{A \text{ alive}} 2^{-u(A)}, \qquad w(v) = \sum_{A \text{ alive},\, v \in A} 2^{-u(A)}.$$

If the remaining elements were assigned to the two players by fair coin tosses, the quantity Φ would be the expected number of sets completed by Player I. A claim of v by Player I increases Φ by $w(v)$, and a claim of v by Player II decreases it by $w(v)$. By hypothesis the potential starts below $\frac{1}{2}$, and after the first move of Player I it is below 1 (see Exercise 1 at the end of this section). Player II's strategy is to claim, at each move, an element of maximal weight. Her claim of an element kills the alive sets that contain the element and leaves the other terms unchanged, so it increases no weight; Player I's next claim therefore raises Φ by at most the weight that Player II has just removed. Over each pair of moves the potential does not increase, and it stays below 1. If Player I completed a set A, the term of A in Φ would be $2^0 = 1$, which is impossible, since $\Phi < 1$ and all terms are nonnegative; Breaker thus wins. For part (2) one player colors the elements one at a time, choosing at each step a color that does not increase the expected number of monochromatic sets (such a color exists: see Exercise 4). The expectation starts below 1, never increases, and at the end is an integer, hence 0. Each conditional expectation is a sum of $|\mathcal{F}|$ explicit terms, and the whole computation takes time polynomial in $|V|$ and $|\mathcal{F}|$. The proof of part (2) is more than an existence proof: it is an explicit algorithm, short enough to write in a few lines of Python.

The game adds robustness to the probabilistic method. A probabilistic argument shows that a random play is good on average; a strategy forces a good outcome one move at a time, even against an adversary who chooses half of the moves. Forcing the outcome move by move is what makes the argument constructive, and the simplicity of each move is what makes it efficient.

Exercises.

(1) Check the bookkeeping in the proof of Theorem 16.1(1): a claim of v by Player I increases Φ by exactly $w(v)$, a claim of v by Player II decreases Φ by exactly $w(v)$, and, under the hypothesis of part (1), we have $\Phi < 1$ after the first move of Player I.

(2) Show that the board $\{1, \dots, n\}^2$ has $2n + 2 = \frac{1}{2}\big((n+2)^2 - n^2\big)$ lines, and find the least n for which Theorem 16.1(3) applies with $d = 2$. Do the same for $d = 3$, where the number of lines is $3n^2 + 6n + 4$.

(3) Consider the Maker–Breaker game on the 3×3 tic-tac-toe board, with $\mathcal{F}$ the set of its 8 lines. Show that $\sum_{A \in \mathcal{F}} 2^{-|A|} = 1$, so Theorem 16.1(1) does not apply, and show that Maker has a winning strategy. (Hint: Maker takes the center and then creates two threats at once.) Explain why Maker's win does not contradict the fact that ordinary tic-tac-toe is a draw.

(4) Let $\mathcal{F}$ be a family of n-element subsets of a finite set V with fewer than 2^{n-1} members. Explain why, in the proof of Theorem 16.1(2), one of the two colors for the next element never increases the conditional expected number of monochromatic members of $\mathcal{F}$.

(5) $(*)$ Let V be the set of vertices of a complete binary tree with n levels, and let $\mathcal{F}$ be the family of 2^{n-1} paths from the root to a leaf. Show that $\sum_{A \in \mathcal{F}} 2^{-|A|} = \frac{1}{2}$ and that

Maker has a winning strategy. Thus the strict inequality in Theorem 16.1(1) cannot be weakened.

17. Applications to fixed point theory

The problem. The *Brouwer fixed point theorem* says that every continuous map $f\colon [0,1]^n \to [0,1]^n$ has a fixed point. The original proofs used algebraic topology. The problem is to find an elementary, combinatorial proof that exposes the finite core of the theorem and, ideally, yields a method for locating approximate fixed points.

17.1. **History of the problem and of the solution using combinatorial games.** The theorem was proved in general by Hadamard in 1910 and Brouwer in 1912 [Bro12], with the methods of the new algebraic topology. A combinatorial route was opened by Sperner's lemma of 1928 on labeled triangulations [Spe28], from which Knaster, Kuratowski and Mazurkiewicz derived the fixed point theorem in 1929. Scarf turned this line of thought into an algorithm for computing approximate fixed points in 1967 [Sca67]. Meanwhile the game of Hex was invented by Piet Hein in 1942 and again, apparently independently, by John Nash at Princeton in 1948 or 1949. Nash observed that the game cannot end in a draw and showed, with a strategy-stealing argument, that the first player has a winning strategy.

David Gale found the game-theoretic solution in 1979 [Gal79]. He proved that the Brouwer fixed point theorem is an easy consequence of the fact that Hex cannot end in a draw, and he extended both statements to n dimensions.

Recognition. Gale's paper, in the *American Mathematical Monthly*, received a Lester R. Ford Award of the Mathematical Association of America in 1980, the year in which Gale shared the John von Neumann Theory Prize. Nash received the Nobel Memorial Prize in Economic Sciences in 1994 and the Abel Prize in 2015.

17.2. **The game.** Hex is played on a rhombus-shaped board of hexagonal cells, and anyone who has played it knows the feeling that somebody *must* win: a complete blockade of one player seems to be, at the same time, a connection for the other. To make this precise, we fix $k \geq 1$ and describe the board by the centers of its cells, the points (i,j) with $1 \leq i,j \leq k$. The cell (i,j) is adjacent to those of the cells $(i\pm1,j)$, $(i,j\pm1)$, $(i+1,j+1)$ and $(i-1,j-1)$ that lie on the board. Player I owns the bottom and top rows (where $j=1$ and $j=k$), and Player II the left and right columns (where $i=1$ and $i=k$). The players alternately place stones on empty cells, Player I first. Player I wins if his stones contain a chain of adjacent cells from bottom to top, and Player II wins if hers contain one from left to right.

17.3. **The Main Result of this section.** The main result is the Hex theorem and, as a consequence, the fixed point theorem in dimension two.

Theorem 17.1 (Gale, 1979).

(1) The Hex theorem. *If every cell of a $k \times k$ Hex board is occupied by a stone of Player I or of Player II, then exactly one of the players has a chain of his or her stones joining his or her two sides of the board.*

(2) *The Hex theorem implies the Brouwer fixed point theorem for* $[0,1]^2$.

Part (1) has an n-dimensional counterpart, which Gale also proved [Gal79]: in n-dimensional Hex, played by n players on the grid $\{1,\dots,k\}^n$, a full board has at least one winner, and this statement implies the Brouwer fixed point theorem for $[0,1]^n$.

17.4. **Solving the problem with the game.**
No draws. Gale's idea is to follow the boundary between the two colors until it can go no further. We add an extra row of Player I's stones above and below the board, and an extra column of Player II's stones on either side. Starting at a corner of the enlarged board, between the two colors, we walk along the edges of the hexagons that separate a stone of Player I from one of Player II. At each vertex of the walk three cells meet, two of one color and one of the other; the walk therefore continues in exactly one way, and it never returns to a vertex, since a vertex already visited has no unused separating edge. The board is finite, so the walk ends, and it can end only at another corner of the enlarged board; the stones of one color along one side of the walk then form a winning chain. The two players cannot both win: a bottom-to-top chain and a left-to-right chain would have to cross, and on the hexagonal board crossing chains share a cell. Figure 23 shows the walk on a full board.

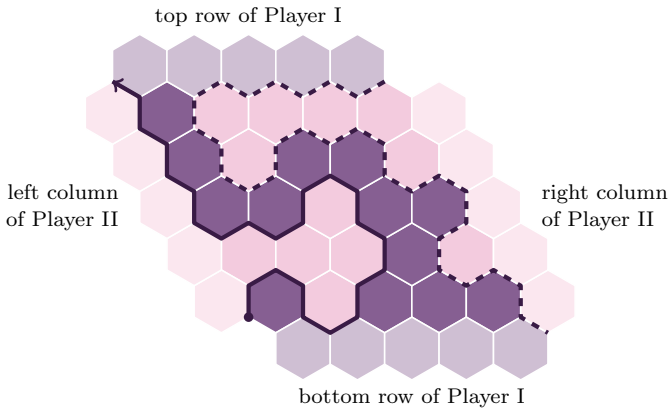


FIGURE 23. The no-draw argument on a full 5×5 board (see Theorem 17.1). The stones of Player I are plum and those of Player II are pink; the paler cells are the extra rows and columns. The walk starts at the lower left corner, between the two colors, and follows the edges that separate plum from pink. It cannot stop or turn back, so it leaves the board at another corner, and the plum stones along its right side form a chain from bottom to top. The dashed line is the walk that starts at the lower right corner.

Brouwer's theorem. The idea is that a winning chain must cross from a side where f pushes one way to a side where it pushes the other way, and continuity forbids such a reversal between neighboring cells. Suppose that a continuous map $f = (f_1, f_2)\colon [0,1]^2 \to [0,1]^2$ has no fixed point. By compactness there is $\varepsilon > 0$ with $\max_i |f_i(z) - z_i| \geq \varepsilon$ for all z, and by uniform continuity there is $\delta < \varepsilon$ with $|f_i(z) - f_i(z')| < \varepsilon$ whenever $|z - z'|_\infty \leq \delta$. We choose k with $1/(k-1) \leq \delta$, place at each cell (i,j) the point $z = ((i-1)/(k-1), (j-1)/(k-1))$, and give the cell to Player II if $|f_1(z) - z_1| \geq \varepsilon$, and to Player I otherwise, in which case $|f_2(z) - z_2| \geq \varepsilon$. According to the Hex theorem, one of the players has a winning chain. Suppose first that Player II has a chain from left to right. On the left side we have $z_1 = 0 \leq f_1(z)$, whence $f_1(z) - z_1 \geq \varepsilon$; on the right side, the inequality $z_1 = 1 \geq f_1(z)$ forces $z_1 - f_1(z) \geq \varepsilon$. The sign of $f_1(z) - z_1$ must change along the chain, so two adjacent cells of the chain, carrying points z and z', satisfy $f_1(z) - z_1 \geq \varepsilon$ and $z_1' - f_1(z') \geq \varepsilon$. Adding these inequalities, we obtain $(f_1(z) - f_1(z')) + (z_1' - z_1) \geq 2\varepsilon$, which is impossible because both terms are less than

ε. A chain of Player I leads to the same contradiction, with f_2 in place of f_1. Hence every continuous map of $[0,1]^2$ into itself has a fixed point.

Seen through the game, the topological content of the theorem is the fact that Hex cannot end in a draw. The argument also has a finite form. Taken in the contrapositive, it shows that for every $\varepsilon > 0$ some point z of a sufficiently fine grid satisfies $|f(z) - z|_\infty < \varepsilon$; by compactness these approximate fixed points have a limit point as $\varepsilon \to 0$, and by continuity it is a fixed point. Combinatorial proofs of fixed point theorems, from Sperner's lemma to Scarf's algorithm, enter computation in exactly this finite form.

Exercises.

(1) List the adjacent pairs of cells of the 2×2 Hex board, and check directly that in each of the 16 ways of filling the board exactly one player has a winning chain, as Theorem 17.1(1) asserts.

(2) Show that on the 3×3 board Player I has a winning strategy. (Hint: open with the center cell $(2,2)$; its neighbors $(1,1)$ and $(2,1)$ lie in the bottom row and its neighbors $(2,3)$ and $(3,3)$ in the top row, and Player I answers a move of Player II in one of these pairs with the other cell of the pair.)

(3) In the proof of Theorem 17.1(2), check that the points z and z' placed at adjacent cells satisfy $|z - z'|_\infty \leq \delta$, and carry out the argument for a winning chain of Player I, using f_2.

(4) $(*)$ Show that Player I has a winning strategy in Hex on the $k \times k$ board for every $k \geq 1$. (Hint: the reflection $(i,j) \mapsto (j,i)$ preserves adjacency and exchanges the sides of the two players; combine Theorem 17.1(1), the determinacy of finite games, and the fact that an extra stone never hurts its owner, to show that Player II cannot have a winning strategy.)

18. Applications to model theory

The problem. Two structures $\mathcal{A}$ and $\mathcal{B}$ in the same signature are *elementarily equivalent*, written $\mathcal{A} \equiv \mathcal{B}$, if they satisfy the same first-order sentences. The problem is to find a method, not requiring the inspection of all sentences, for deciding whether two given structures are elementarily equivalent and, with the same tool, for proving that a given property of structures cannot be expressed by a first-order sentence.

18.1. **History of the problem and of the solution using combinatorial games.** Elementary equivalence was introduced by Tarski. From the 1930s to the 1950s his program, the classification of the complete first-order theories of familiar structures, required tools for proving that two structures satisfy the same sentences. The answer was suggested by the back-and-forth method, associated with Cantor's theorem of 1895 that any two countable dense linear orders without endpoints are isomorphic, although Cantor's own proof used only the forth half of the method.

Roland Fraïssé solved the problem in his thesis, defended in Paris in 1953 and published in 1954 [Fra54]: he characterized elementary equivalence by systems of partial isomorphisms. Andrzej Ehrenfeucht recast the solution as a game in 1961 [Ehr61], and he used the game to prove that certain theories are complete. In finite model theory, which developed from the 1970s onward in connection with databases and computational complexity, these games became one of the principal methods for proving that properties are not definable in a given logic. Fagin, for instance, used games of this kind in 1975 to show that connectivity of finite graphs is not expressible in monadic existential second-order logic [Fag75].

18.2. **The game.** The game is played with two structures, and we begin by fixing the kind of structure involved.

Definition 18.1. A *relational signature* is a set Σ of relation symbols, each with an *arity*, which is an integer $r \geq 1$; the signature is *finite* if the set Σ is finite. A Σ-*structure* $\mathcal{A}$ consists of a non-empty set A, the *universe* of $\mathcal{A}$, together with a relation $R^{\mathcal{A}} \subseteq A^r$ for each symbol $R \in \Sigma$ of arity r.

Linear orders are regarded as structures in the signature $\{<\}$, with a single binary symbol interpreted as a strict linear order, and a graph is a structure $\mathcal{G}$ in the signature $\{E\}$ whose relation $E^{\mathcal{G}}$ is symmetric and irreflexive. Tuples are finite throughout this section. For a tuple $\bar{a} = (a_1, \ldots, a_k) \in A^k$ and an element $a \in A$, we write $\bar{a}a$ for the tuple $(a_1, \ldots, a_k, a)$, and similarly $\bar{a}\bar{c}$ for the concatenation of two tuples. The winning condition of the game compares a tuple chosen in one structure with a tuple chosen in the other, by means of the following notion.

Definition 18.2. Let $\mathcal{A}$ and $\mathcal{B}$ be Σ-structures, let $k \geq 0$ be an integer, and let $\bar{a} \in A^k$ and $\bar{b} \in B^k$ be given. The map $\bar{a} \mapsto \bar{b}$ is a *partial isomorphism* if the following two conditions hold:

(1) for all $i, l \in \{1, \ldots, k\}$, we have $a_i = a_l$ if and only if $b_i = b_l$;
(2) for every symbol $R \in \Sigma$ of arity r and all $i_1, \ldots, i_r \in \{1, \ldots, k\}$, the tuple $(a_{i_1}, \ldots, a_{i_r})$ lies in $R^{\mathcal{A}}$ if and only if the tuple $(b_{i_1}, \ldots, b_{i_r})$ lies in $R^{\mathcal{B}}$.

The name is explained by the two conditions. Condition (1) says that the pairs (a_i, b_i) form the graph of a bijection between the sets $\{a_1, \ldots, a_k\}$ and $\{b_1, \ldots, b_k\}$, and condition (2) says that this bijection is an isomorphism between the substructures of $\mathcal{A}$ and $\mathcal{B}$ with these universes; in a relational signature, every non-empty subset of a universe is the universe of a substructure. For $k = 0$ both conditions are vacuous, and the empty map is a partial isomorphism. Partial isomorphisms survive any reindexing of the tuples. For a map $\pi\colon \{1, \ldots, h\} \to \{1, \ldots, k\}$, we write $\bar{a}_\pi = (a_{\pi(1)}, \ldots, a_{\pi(h)})$ and $\bar{b}_\pi = (b_{\pi(1)}, \ldots, b_{\pi(h)})$; if the map $\bar{a} \mapsto \bar{b}$ is a partial isomorphism, then so is the map $\bar{a}_\pi \mapsto \bar{b}_\pi$, because each condition of Definition 18.2 for the new tuples is one of the conditions for the old ones.

In the game, Player I wants to expose a difference between the two structures, and Player II wants to show that they look alike; the names used in the literature, *Spoiler* for Player I and *Duplicator* for Player II, describe the game well. The game may begin after some elements of the two structures have already been matched. This generality costs nothing, and it is what makes induction on the number of rounds possible.

Definition 18.3. Let $\mathcal{A}$ and $\mathcal{B}$ be Σ-structures, let $k, n \geq 0$ be integers, and let $\bar{a} = (a_1, \ldots, a_k) \in A^k$ and $\bar{b} = (b_1, \ldots, b_k) \in B^k$ be given. The *Ehrenfeucht–Fraïssé game* $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$ has n rounds. In round t, where $1 \leq t \leq n$, Player I chooses an element of A or an element of B, and Player II answers with an element of the other structure. We write a_{k+t} for the element of A and b_{k+t} for the element of B chosen in round t, whichever player chose it. Player II wins the play if the map $(a_1, \ldots, a_{k+n}) \mapsto (b_1, \ldots, b_{k+n})$ is a partial isomorphism, and Player I wins it otherwise. When $k = 0$, we write $\mathrm{EF}_n(\mathcal{A}, \mathcal{B})$ for the game.

The game fits the format of Definition 2.2. A move of Player I is a pair $(1, c)$ with $c \in A$ or a pair $(2, c)$ with $c \in B$, the first coordinate recording the structure in which he plays; Player II answers a move $(1, c)$ with a pair $(2, d)$ such that $d \in B$, and a move $(2, c)$ with a pair $(1, d)$ such that $d \in A$. The coordinates keep the two structures apart even when their universes meet, or when $\mathcal{A}$ and $\mathcal{B}$ are the same structure. After the $2n$-th move both players play a dummy move $*$ forever, and the payoff set W consists of the plays whose first $2n$ moves satisfy the winning condition of Definition 18.3. Strategies and winning strategies are

then those of Definition 2.2, and only their values at positions of length less than $2n$ matter. Two features of this format are used repeatedly below. First, the tree of positions depends only on $\mathcal{A}$, $\mathcal{B}$ and n, while the tuples $\bar{a}$ and $\bar{b}$ enter only through the payoff set. Second, whether a play is won depends only on its first $2n$ moves; the game is therefore determined (see Exercise 4 at the end of Section 2), and exactly one of the players has a winning strategy (see Remarks 2.3(3)). For $n = 0$ there are no rounds at all: Player II has a winning strategy in $\mathrm{EF}_0(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$ if and only if the map $\bar{a} \mapsto \bar{b}$ is a partial isomorphism.

18.3. **The Main Result of this section.** The main result is the theorem of Fraïssé and Ehrenfeucht, which matches the rounds of the game with the nested quantifiers of first-order formulas. We use the standard syntax and semantics of first-order logic, and we fix only the conventions that matter when formulas are counted.

Definition 18.4. Let Σ be a relational signature. The variables are $v_1, v_2, v_3, \ldots$, and the letters x, y, z, u, w, with or without indices, stand for variables.

(1) The *atomic formulas* are the constant $\top$ (true), the equations $x = y$, and the formulas $R(x_1, \ldots, x_r)$ with $R \in \Sigma$ of arity r. The *formulas* are built from the atomic formulas by negation $\neg$, conjunction $\wedge$ and existential quantification $\exists x$; the formulas $\varphi \vee \psi$, $\varphi \to \psi$ and $\forall x\, \varphi$ are abbreviations of $\neg(\neg\varphi \wedge \neg\psi)$, $\neg(\varphi \wedge \neg\psi)$ and $\neg\exists x\, \neg\varphi$.

(2) The *quantifier rank* $\mathrm{qr}(\varphi)$ of a formula φ is defined by recursion: we set $\mathrm{qr}(\alpha) = 0$ for atomic α, $\mathrm{qr}(\neg\varphi) = \mathrm{qr}(\varphi)$, $\mathrm{qr}(\varphi \wedge \psi) = \max\{\mathrm{qr}(\varphi), \mathrm{qr}(\psi)\}$, and $\mathrm{qr}(\exists x\, \varphi) = \mathrm{qr}(\varphi) + 1$.

The quantifier rank measures the depth of nesting of the quantifiers, not their number: the sentence $\exists x\, \exists y\, (x < y)$ has rank 2, while $\exists x\, (x < x) \wedge \exists y\, (y = y)$ has rank 1. The abbreviations behave as expected, with $\mathrm{qr}(\forall x\, \varphi) = \mathrm{qr}(\varphi) + 1$ and $\mathrm{qr}(\varphi \vee \psi) = \max\{\mathrm{qr}(\varphi), \mathrm{qr}(\psi)\}$. If $\bar{x} = (x_1, \ldots, x_k)$ is a list of distinct variables containing every free variable of a formula φ, we write $\varphi(\bar{x})$, and for a Σ-structure $\mathcal{A}$ and a tuple $\bar{a} \in A^k$ we write $\mathcal{A} \models \varphi(\bar{a})$ if φ holds in $\mathcal{A}$ when each x_i is interpreted as a_i. In this notation, Tarski's clause for the existential quantifier reads as follows. Suppose that $\varphi = \exists z\, \psi$, let $\bar{x}'$ be the list obtained from $\bar{x}$ by deleting z if it occurs there, and let $\bar{a}'$ be the corresponding subtuple of $\bar{a}$. The free variables of ψ then lie in the list $\bar{x}'z$, and the formula φ holds of $\bar{a}$ in $\mathcal{A}$ if and only if $\mathcal{A} \models \psi(\bar{a}', a)$ for some $a \in A$. Two formulas $\varphi(\bar{x})$ and $\psi(\bar{x})$ are *logically equivalent* if, for every Σ-structure $\mathcal{A}$ and every tuple $\bar{a} \in A^k$, the formula φ holds of $\bar{a}$ in $\mathcal{A}$ exactly when ψ does.

Definition 18.5. Let $\mathcal{A}$ and $\mathcal{B}$ be Σ-structures, let $k, n \geq 0$ be integers, and let $\bar{a} \in A^k$ and $\bar{b} \in B^k$ be given. We write $(\mathcal{A}, \bar{a}) \equiv_n (\mathcal{B}, \bar{b})$ if every formula $\varphi(\bar{x})$ of quantifier rank at most n, with $\bar{x}$ a list of k distinct variables, satisfies

$$\mathcal{A} \models \varphi(\bar{a}) \iff \mathcal{B} \models \varphi(\bar{b}).$$

For $k = 0$ we write $\mathcal{A} \equiv_n \mathcal{B}$: the two structures satisfy the same sentences of quantifier rank at most n.

Every sentence has a quantifier rank, and two structures are therefore elementarily equivalent exactly when $\mathcal{A} \equiv_n \mathcal{B}$ for every n.

Theorem 18.6 (Fraïssé, 1954; Ehrenfeucht, 1961). *Let $\mathcal{A}$ and $\mathcal{B}$ be structures in a finite relational signature, and let $n \geq 0$ be given.*

(1) *For every $k \geq 0$ and all tuples $\bar{a} \in A^k$ and $\bar{b} \in B^k$, the relation $(\mathcal{A}, \bar{a}) \equiv_n (\mathcal{B}, \bar{b})$ holds if and only if Player II has a winning strategy in the game $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$.*
(2) *The relation $\mathcal{A} \equiv_n \mathcal{B}$ holds if and only if Player II has a winning strategy in the game $\mathrm{EF}_n(\mathcal{A}, \mathcal{B})$.*
(3) *The structures $\mathcal{A}$ and $\mathcal{B}$ are elementarily equivalent if and only if Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{A}, \mathcal{B})$ for every n.*

In the next subsection we prove the theorem and three classical applications. The first is that the orders $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$ are elementarily equivalent, although only one of them is countable. The second concerns the linear order L_m with m elements: we show that $L_m \equiv_n L_{m'}$ whenever $m, m' \geq 2^n - 1$, and we deduce that no first-order sentence is true exactly in the finite linear orders of even size. The third is that connectivity of finite graphs is not first-order expressible either. From a linear order $1 < 2 < \cdots < m$ with $m \geq 3$ we define, by a first-order formula, a graph that is connected exactly when m is odd (see Figure 25); a sentence expressing connectivity would then turn into a sentence expressing odd size, and the second application rules this out.

18.4. **Solving the problem with the game.** The proof of Theorem 18.6 falls into two halves of different kinds. A winning strategy of Player II preserves every formula of quantifier rank at most n: we prove this by induction on formulas, and each quantifier is matched by a round of the game. Conversely, the relation $\equiv_n$ produces a strategy. Here the finiteness of the signature enters: in Lemma 18.10 below we show that everything that formulas of rank at most n can say about a tuple is said by a single formula, the Hintikka formula of the tuple. Both halves rest on two lemmas about the game itself.

Two lemmas on the game. According to the first lemma, a winning strategy survives the passage to subtuples, repetitions and rearrangements of the tuples matched before the game starts.

Lemma 18.7. *Let $\mathcal{A}$ and $\mathcal{B}$ be structures in a relational signature, let $k, n \geq 0$ be integers, and let $\bar{a} \in A^k$ and $\bar{b} \in B^k$ be given. Suppose that Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$. Then the following hold.*

(1) *The map $\bar{a} \mapsto \bar{b}$ is a partial isomorphism.*
(2) *For every $h \geq 0$ and every map $\pi\colon \{1, \ldots, h\} \to \{1, \ldots, k\}$, Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{A}, \bar{a}_\pi, \mathcal{B}, \bar{b}_\pi)$.*

Proof. Both parts follow from the fact, noted after Definition 18.2, that partial isomorphisms survive reindexing. Let τ be a winning strategy for Player II in $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$. The n rounds of a play consistent with τ produce tuples $\bar{c} \in A^n$ and $\bar{d} \in B^n$, and the map $\bar{a}\bar{c} \mapsto \bar{b}\bar{d}$ is a partial isomorphism because τ is winning. For (1), plays consistent with τ exist, since a player always has a legal move; the map $\bar{a} \mapsto \bar{b}$ arises from $\bar{a}\bar{c} \mapsto \bar{b}\bar{d}$ by reindexing along the inclusion of $\{1, \ldots, k\}$ in $\{1, \ldots, k + n\}$, and it is therefore a partial isomorphism. For (2), we recall that the tree of positions does not depend on the tuples matched before the game starts; the function τ is thus a strategy for Player II in $\mathrm{EF}_n(\mathcal{A}, \bar{a}_\pi, \mathcal{B}, \bar{b}_\pi)$ as well, with the same consistent plays. In such a play, the tuples $\bar{a}_\pi\bar{c}$ and $\bar{b}_\pi\bar{d}$ arise from $\bar{a}\bar{c}$ and $\bar{b}\bar{d}$ by reindexing along the map that sends $i \leq h$ to $\pi(i)$ and $h + t$ to $k + t$. The map $\bar{a}_\pi\bar{c} \mapsto \bar{b}_\pi\bar{d}$ is then a partial isomorphism as well, Player II wins the play in the new game, and the strategy τ is winning there. ☺

The second lemma lets us play the game one round at a time: after the first round, what remains is a game with one round fewer, started from the extended tuples. Its two conditions are the *forth* and *back* steps of the back-and-forth method.

Lemma 18.8. *Let $\mathcal{A}$ and $\mathcal{B}$ be structures in a relational signature, let $n \geq 1$ and $k \geq 0$ be integers, and let $\bar{a} \in A^k$ and $\bar{b} \in B^k$ be given. Then Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$ if and only if the following two conditions hold:*

(forth) *for every $a \in A$ there is $b \in B$ such that Player II has a winning strategy in $\mathrm{EF}_{n-1}(\mathcal{A}, \bar{a}a, \mathcal{B}, \bar{b}b)$;*

(back) *for every $b \in B$ there is $a \in A$ such that Player II has a winning strategy in $\mathrm{EF}_{n-1}(\mathcal{A}, \bar{a}a, \mathcal{B}, \bar{b}b)$.*

Proof. The proof consists in cutting a play after its first round. Suppose that the first round of $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$ has produced the elements $a \in A$ and $b \in B$, in either order. The positions of $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$ that begin with this round are the two moves of the round followed by a position of $\mathrm{EF}_{n-1}(\mathcal{A}, \bar{a}a, \mathcal{B}, \bar{b}b)$. A play that begins with the round is won by Player II in the first game if and only if the rest of the play is won by her in the second, since both conditions say that one and the same map, $\bar{a}a\bar{c} \mapsto \bar{b}b\bar{d}$, is a partial isomorphism.

Suppose first that τ is a winning strategy for Player II in $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$, and let $a \in A$ be given. The strategy τ answers the opening move a with some $b \in B$. At each position p of $\mathrm{EF}_{n-1}(\mathcal{A}, \bar{a}a, \mathcal{B}, \bar{b}b)$ at which Player II is to move, she may play the move that τ assigns to the round (a, b) followed by p. This rule is a strategy for her in the shorter game, and each play consistent with it, preceded by the round (a, b), is consistent with τ and is won by Player II; by the preceding paragraph, the rule is a winning strategy. Thus (forth) holds, and the opening moves of Player I in B establish (back) in the same way.

Conversely, suppose that (forth) and (back) hold. For each $a \in A$ we choose an element $b_a \in B$ and a winning strategy τ_a for Player II in $\mathrm{EF}_{n-1}(\mathcal{A}, \bar{a}a, \mathcal{B}, \bar{b}b_a)$, and for each $b \in B$ we choose an element $a_b \in A$ and a winning strategy τ_b in $\mathrm{EF}_{n-1}(\mathcal{A}, \bar{a}a_b, \mathcal{B}, \bar{b}b)$. Player II answers an opening move $a \in A$ with b_a and then follows τ_a; she answers an opening move $b \in B$ with a_b and continues with τ_b. (At positions in which her first answer was different, the strategy may be defined arbitrarily, since they never occur in plays consistent with it.) Every play consistent with this strategy consists of a first round followed by a play consistent with τ_a or τ_b. The latter is won by Player II, hence so is the whole play, and the strategy is winning. ☺

Hintikka formulas. For the half of Theorem 18.6 that turns the relation $\equiv_n$ into a strategy, we need, for each tuple, a single formula of rank at most n that says everything formulas of rank at most n can say about it. Its definition follows the game. The formula of rank $n+1$ lists the formulas of rank n of all the one-element extensions of the tuple twice: once under $\exists$, saying that each extension in one structure is matched in the other (the forth step), and once under $\forall$, saying that nothing else occurs (the back step). For this list to be a formula it must be finite, and this is where the finiteness of the signature enters.

We fix an enumeration of all formulas. For a finite set Φ of formulas, we write $\bigwedge \Phi$ for the conjunction of its members in the order of the enumeration ($\top$ when Φ is empty) and $\bigvee \Phi$ for the formula $\neg \bigwedge \{\neg\varphi : \varphi \in \Phi\}$. Thus the expressions $\bigwedge \Phi$ and $\bigvee \Phi$ denote definite formulas, whose quantifier rank is the largest rank of a member of Φ (0 when Φ is empty), and they hold of a tuple exactly when every member, respectively some member, of Φ holds of it. For $k \geq 0$, let Λ_k denote the set of atomic formulas whose variables lie among $v_1, \ldots, v_k$, together with their negations.

Definition 18.9. Let Σ be a finite relational signature. For integers $n, k \geq 0$, the sets $H_{n,k}$ of formulas are defined by recursion on n:

$$H_{0,k} = \Big\{ \bigwedge \Delta : \Delta \subseteq \Lambda_k \Big\},$$
$$H_{n+1,k} = \Big\{ \bigwedge \{ \exists v_{k+1}\, \chi : \chi \in S \} \wedge \forall v_{k+1} \bigvee S \ : \ S \subseteq H_{n,k+1} \Big\}.$$

For a Σ-structure $\mathcal{A}$ and a tuple $\bar{a} \in A^k$, the *Hintikka formula of rank n* of $\bar{a}$ in $\mathcal{A}$ is the formula $\varphi^n_{\bar{a}}$ defined by recursion on n:

$$\varphi^0_{\bar{a}} = \bigwedge \{ \alpha \in \Lambda_k : \mathcal{A} \models \alpha(\bar{a}) \},$$
$$\varphi^{n+1}_{\bar{a}} = \bigwedge \{ \exists v_{k+1}\, \varphi^n_{\bar{a}a} : a \in A \} \wedge \forall v_{k+1} \bigvee \{ \varphi^n_{\bar{a}a} : a \in A \}.$$

The definition and part (1) of the next lemma go together, by induction on n: part (1) guarantees at each stage that the sets over which the conjunctions and disjunctions range are finite. The structure $\mathcal{A}$ is omitted from the notation $\varphi^n_{\bar{a}}$, since it is always clear from the context.

Lemma 18.10. *Let $\mathcal{A}$ and $\mathcal{B}$ be structures in a finite relational signature Σ, let $n, k \geq 0$ be integers, and let $\bar{a} \in A^k$ and $\bar{b} \in B^k$ be given.*

(1) *The set $H_{n,k}$ is a finite set of formulas of quantifier rank at most n whose free variables lie among $v_1, \ldots, v_k$, and it contains $\varphi^n_{\bar{a}}$.*
(2) *The tuple $\bar{a}$ satisfies its Hintikka formula: $\mathcal{A} \models \varphi^n_{\bar{a}}(\bar{a})$.*
(3) *The relation $\mathcal{B} \models \varphi^n_{\bar{a}}(\bar{b})$ holds if and only if Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$.*

Proof. All three parts are proved by induction on n, simultaneously for all k, all structures and all tuples.

(1) The set Λ_k consists of $\top$, the k^2 equations $v_i = v_l$, the k^r formulas $R(v_{i_1}, \ldots, v_{i_r})$ for each symbol $R \in \Sigma$ of arity r, and the negations of all these. It is finite because Σ is finite; this is the only point of the proof of Theorem 18.6 at which the finiteness of the signature is used. The set $H_{0,k}$ therefore has at most $2^{|\Lambda_k|}$ members, all of rank 0, and it contains $\varphi^0_{\bar{a}}$. For the inductive step, suppose that the statement holds for n and every k. Every subset S of the finite set $H_{n,k+1}$ is finite, the formula $\bigwedge\{\exists v_{k+1}\chi : \chi \in S\} \wedge \forall v_{k+1} \bigvee S$ has rank at most $n+1$, and its free variables lie among $v_1, \ldots, v_k$, because v_{k+1} is bound in it. Consequently, the set $H_{n+1,k}$ is finite, with at most $2^{|H_{n,k+1}|}$ members. By the induction hypothesis, the formulas $\varphi^n_{\bar{a}a}$ with $a \in A$ form a subset of $H_{n,k+1}$, and the formula $\varphi^{n+1}_{\bar{a}}$ is the member of $H_{n+1,k}$ that corresponds to this subset.

(2) Every conjunct of $\varphi^0_{\bar{a}}$ holds of $\bar{a}$, by definition. For the inductive step, the induction hypothesis provides $\mathcal{A} \models \varphi^n_{\bar{a}a}(\bar{a}a)$ for each $a \in A$. The element a witnesses the conjunct $\exists v_{k+1} \varphi^n_{\bar{a}a}$ at $\bar{a}$, and the tuple $\bar{a}a$ satisfies the disjunct $\varphi^n_{\bar{a}a}$ of $\bigvee\{\varphi^n_{\bar{a}c} : c \in A\}$; the second conjunct, $\forall v_{k+1} \bigvee\{\varphi^n_{\bar{a}c} : c \in A\}$, therefore holds at $\bar{a}$ as well, which proves that $\mathcal{A} \models \varphi^{n+1}_{\bar{a}}(\bar{a})$.

(3) For every atomic formula $\alpha \in \Lambda_k$, exactly one of α and $\neg\alpha$ is a conjunct of $\varphi^0_{\bar{a}}$. Consequently, the relation $\mathcal{B} \models \varphi^0_{\bar{a}}(\bar{b})$ holds exactly when $\bar{a}$ and $\bar{b}$ satisfy the same atomic formulas with variables among $v_1, \ldots, v_k$. These atomic formulas, other than $\top$, are the equations $v_i = v_l$ and the formulas $R(v_{i_1}, \ldots, v_{i_r})$, which express the conditions of Definition 18.2; thus $\mathcal{B} \models \varphi^0_{\bar{a}}(\bar{b})$ holds exactly when $\bar{a} \mapsto \bar{b}$ is a partial isomorphism, that is, exactly when Player II wins the game $\mathrm{EF}_0(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$, which has no rounds. For the inductive step, the definition of $\varphi^{n+1}_{\bar{a}}$ shows that $\mathcal{B} \models \varphi^{n+1}_{\bar{a}}(\bar{b})$ holds if and only if

(i) for every $a \in A$ there is $b \in B$ with $\mathcal{B} \models \varphi^n_{\bar{a}a}(\bar{b}b)$, and

(ii) for every $b \in B$ there is $a \in A$ with $\mathcal{B} \models \varphi^n_{\bar{a}a}(\bar{b}b)$.

By the induction hypothesis, applied to the tuples $\bar{a}a$ and $\bar{b}b$ of length $k+1$, the relation $\mathcal{B} \models \varphi^n_{\bar{a}a}(\bar{b}b)$ holds exactly when Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{A}, \bar{a}a, \mathcal{B}, \bar{b}b)$. Conditions (i) and (ii) thus become the conditions (forth) and (back) of Lemma 18.8 for the game $\mathrm{EF}_{n+1}(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$, and that lemma completes the induction. ☺

The proof of the theorem. With the three lemmas in hand, the proof of Theorem 18.6 is short.

Proof of Theorem 18.6. We begin with part (1). Suppose first that $(\mathcal{A}, \bar{a}) \equiv_n (\mathcal{B}, \bar{b})$. The Hintikka formula $\varphi^n_{\bar{a}}(v_1, \ldots, v_k)$ has quantifier rank at most n and holds of $\bar{a}$ in $\mathcal{A}$, by parts (1) and (2) of Lemma 18.10; it therefore holds of $\bar{b}$ in $\mathcal{B}$, and part (3) of that lemma provides a winning strategy for Player II in $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$.

For the converse, we show by induction on the formula φ the following claim, which does not use the finiteness of the signature: for every list $\bar{x}$ of distinct variables containing the free variables of φ, every integer $j \geq \mathrm{qr}(\varphi)$, and all structures $\mathcal{A}$ and $\mathcal{B}$ and tuples $\bar{a}$ and $\bar{b}$ of the length of $\bar{x}$ such that Player II has a winning strategy in $\mathrm{EF}_j(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$, we have

$$\mathcal{A} \models \varphi(\bar{a}) \iff \mathcal{B} \models \varphi(\bar{b}).$$

If φ is atomic, the claim holds because the map $\bar{a} \mapsto \bar{b}$ is a partial isomorphism (see Lemma 18.7(1)), and the atomic formulas with variables in $\bar{x}$ express the conditions of Definition 18.2. The cases $\neg\varphi$ and $\varphi \wedge \psi$ follow at once from the induction hypothesis, applied with the same list, the same integer j and the same tuples.

The quantifier is the heart of the matter. Suppose that $\varphi = \exists z\, \psi$, and let $\bar{x}'$, $\bar{a}'$ and $\bar{b}'$ be obtained from $\bar{x}$, $\bar{a}$ and $\bar{b}$ by deleting the entry of z, if z occurs in $\bar{x}$. Since $j \geq \mathrm{qr}(\psi) + 1$, the game has at least one round. The tuples $\bar{a}'$ and $\bar{b}'$ are reindexings of $\bar{a}$ and $\bar{b}$ along one and the same map, and by Lemma 18.7(2) Player II has a winning strategy in $\mathrm{EF}_j(\mathcal{A}, \bar{a}', \mathcal{B}, \bar{b}')$. Suppose that $\mathcal{A} \models \varphi(\bar{a})$, and let $a \in A$ be an element with $\mathcal{A} \models \psi(\bar{a}', a)$. The condition (forth) of Lemma 18.8 provides $b \in B$ such that Player II has a winning strategy in $\mathrm{EF}_{j-1}(\mathcal{A}, \bar{a}'a, \mathcal{B}, \bar{b}'b)$. The induction hypothesis for ψ, applied to the list $\bar{x}'z$ and to $j - 1 \geq \mathrm{qr}(\psi)$, gives $\mathcal{B} \models \psi(\bar{b}', b)$, whence $\mathcal{B} \models \varphi(\bar{b})$. The implication from $\mathcal{B}$ to $\mathcal{A}$ is proved in the same way, with (back) in place of (forth), and the claim is established.

By the claim, applied to the formulas $\varphi(\bar{x})$ of quantifier rank at most n, a winning strategy of Player II in $\mathrm{EF}_n(\mathcal{A}, \bar{a}, \mathcal{B}, \bar{b})$ forces $(\mathcal{A}, \bar{a}) \equiv_n (\mathcal{B}, \bar{b})$, and part (1) is proved. Part (2) is the case $k = 0$ of part (1). Part (3) follows from part (2), because two structures are elementarily equivalent exactly when $\mathcal{A} \equiv_n \mathcal{B}$ for every n. ☺

A statement about all first-order sentences is thus replaced by a combinatorial task: finding a strategy. The proof also yields a description of the rank-n type of a tuple, the set of formulas of quantifier rank at most n that hold of it, and a count of the formulas of rank at most n.

Corollary 18.11. *Let Σ be a finite relational signature, and let $n, k \geq 0$ be integers.*

(1) *If $\mathcal{A}$ and $\mathcal{B}$ are Σ-structures, $\bar{a} \in A^k$ and $\bar{b} \in B^k$, then the relation $(\mathcal{A}, \bar{a}) \equiv_n (\mathcal{B}, \bar{b})$ holds if and only if $\mathcal{B} \models \varphi^n_{\bar{a}}(\bar{b})$.*

(2) *Every formula $\varphi(v_1, \ldots, v_k)$ of quantifier rank at most n is logically equivalent to a disjunction $\bigvee S$ with $S \subseteq H_{n,k}$. Consequently, up to logical equivalence there are at most $2^{|H_{n,k}|}$ formulas of quantifier rank at most n with free variables among $v_1, \ldots, v_k$.*

Proof. Part (1) combines Theorem 18.6(1) with Lemma 18.10(3). For part (2), the idea is to collect the Hintikka formulas of all the tuples that satisfy φ. Let S be the set of formulas

$\varphi^n_{\bar a}$, where $\mathcal{A}$ ranges over all Σ-structures and $\bar a$ over the tuples in A^k with $\mathcal{A} \models \varphi(\bar a)$; it is a subset of the finite set $H_{n,k}$. If $\mathcal{A} \models \varphi(\bar a)$, then the Hintikka formula $\varphi^n_{\bar a}$ is a member of S that holds of $\bar a$ (see Lemma 18.10(2)), and the tuple $\bar a$ satisfies $\bigvee S$. Conversely, suppose that $\mathcal{B} \models \bigvee S(\bar b)$. By the definition of S, there are a structure $\mathcal{A}$ and a tuple $\bar a$ with $\mathcal{A} \models \varphi(\bar a)$ and $\mathcal{B} \models \varphi^n_{\bar a}(\bar b)$, and part (1) gives $(\mathcal{A}, \bar a) \equiv_n (\mathcal{B}, \bar b)$; since φ has rank at most n, it holds of $\bar b$ in $\mathcal{B}$. The formula φ is thus logically equivalent to $\bigvee S$, and the bound follows from the number of subsets of $H_{n,k}$. ☺

Remark 18.12. The finiteness of the signature is used only in the half of the proof of Theorem 18.6 that derives a winning strategy from $\equiv_n$, through the finiteness of the sets Λ_k in Lemma 18.10(1); the other half, which derives $\equiv_n$ from a winning strategy, holds in every relational signature. The first implication, from $\equiv_n$ to a winning strategy, fails for infinite signatures: two structures may satisfy the same sentences, and yet Player I may win in a single round (see Exercise 8 at the end of this section). Function and constant symbols can be absorbed into relational signatures. A function symbol f of arity r is replaced by a relation symbol R_f of arity $r+1$, interpreted as the graph $\{(\bar a, a) : f(\bar a) = a\}$, and a constant symbol by a unary relation symbol. Every formula of the original signature translates into a formula of the new one that holds in the relational version of a structure exactly when the original formula holds in the structure, and conversely; two structures are therefore elementarily equivalent exactly when their relational versions are, and Theorem 18.6(3), applied to the relational versions, characterizes elementary equivalence in every finite signature. The quantifier rank, on the other hand, is not preserved: a nested term must be unnested by quantifiers, as in the translation of $f(f(x)) = y$ into $\exists z\,(R_f(x,z) \wedge R_f(z,y))$.

Dense linear orders. In the first application the strategy of Player II is the one suggested by the back-and-forth proof of Cantor's theorem: she answers every move with an element that occupies the same place, relative to the elements already chosen, as the move of Player I. Density and the absence of endpoints guarantee that such an element exists. For linear orders, the map $\bar a \mapsto \bar b$ is a partial isomorphism exactly when $a_i < a_l \iff b_i < b_l$ and $a_i = a_l \iff b_i = b_l$ for all indices i and l, that is, exactly when the two tuples have the same order type.

Proposition 18.13. *Let $\mathcal{A}$ and $\mathcal{B}$ be dense linear orders without endpoints, and let $k \geq 0$ be an integer. If $\bar a \in A^k$ and $\bar b \in B^k$ are tuples for which the map $\bar a \mapsto \bar b$ is a partial isomorphism, then Player II has a winning strategy in $\mathrm{EF}_n(\mathcal{A}, \bar a, \mathcal{B}, \bar b)$ for every $n \geq 0$. In particular, the orders $\mathcal{A}$ and $\mathcal{B}$ are elementarily equivalent, and so are $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$.*

Proof. We argue by induction on n, for all tuples at once. For $n = 0$ the hypothesis is the winning condition. Suppose that $n \geq 1$ and that the statement holds for $n-1$. By Lemma 18.8 it suffices to verify the conditions (forth) and (back), and the symmetry of the hypotheses in $\mathcal{A}$ and $\mathcal{B}$ reduces the task to (forth). Let $a \in A$ be given. If $a = a_i$ for some index i, we take $b = b_i$: every condition of Definition 18.2 that involves the new entries of $\bar a a$ and $\bar b b$ repeats a condition for the entries with index i. Otherwise each index i lies in exactly one of the sets $P^- = \{i : a_i < a\}$ and $P^+ = \{i : a_i > a\}$. For $i \in P^-$ and $l \in P^+$ we have $a_i < a_l$, hence $b_i < b_l$, and there is therefore an element $b \in B$ with

$$b_i < b < b_l \qquad \text{for all } i \in P^- \text{ and } l \in P^+.$$

Indeed, if both sets are non-empty, density provides b between the largest of the elements b_i with $i \in P^-$ and the smallest of the elements b_l with $l \in P^+$; if only one of them is non-empty, the absence of endpoints provides b below or above the corresponding elements; and if both are empty, which happens only when $k = 0$, any element of B will do. For every index i, the

element b then satisfies $a_i < a \iff b_i < b$ and $a_i > a \iff b_i > b$, while neither $a_i = a$ nor $b_i = b$ holds. Together with the hypothesis on $\bar{a}$ and $\bar{b}$, these relations show that $\bar{a}a \mapsto \bar{b}b$ is a partial isomorphism, and the induction hypothesis yields a winning strategy for Player II in $\mathrm{EF}_{n-1}(\mathcal{A}, \bar{a}a, \mathcal{B}, \bar{b}b)$. This is (forth), and the induction is complete.

The empty tuples satisfy the hypothesis, and Player II therefore wins $\mathrm{EF}_n(\mathcal{A}, \mathcal{B})$ for every n. By Theorem 18.6(3), the two orders are elementarily equivalent. The orders $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$ are dense and have no endpoints, which completes the proof. ☺

The orders $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$ are not isomorphic, since only one of them is countable; elementary equivalence is a strictly weaker relation than isomorphism, and the game measures it exactly.

Finite linear orders. For $m \geq 1$, let L_m denote the linear order $(\{1, \ldots, m\}, <)$. A strategy for Player II in $\mathrm{EF}_n(L_m, L_{m'})$ must keep track of distances. She imagines two extra elements in each order, 0 just below the least element and $m + 1$ just above the greatest, and she watches the gaps between consecutive chosen elements. The intuition is that two elements at distance at least 2^j look infinitely far apart to a formula that has only j quantifiers left to inspect them, since each quantifier can at best halve the distance. The following definition makes this invariant precise.

Definition 18.14. Let $j \geq 0$ be an integer. For integers d and d', we write $d \approx_j d'$ if $d = d'$ or $d, d' \geq 2^j$. Let $m, m' \geq 1$ and $k \geq 0$ be integers, and let $\bar{a} \in \{1, \ldots, m\}^k$ and $\bar{b} \in \{1, \ldots, m'\}^k$ be given. We add the *imaginary elements* $a_- = 0$, $a_+ = m + 1$, $b_- = 0$ and $b_+ = m' + 1$, and we write $K = \{-, 1, \ldots, k, +\}$. Two indices $p, q \in K$ are *consecutive* for $\bar{a}$ if $a_p < a_q$ and no index $s \in K$ satisfies $a_p < a_s < a_q$. The tuples $\bar{a}$ and $\bar{b}$ are *j-similar* if the following two conditions hold:

(1) for all $p, q \in K$, we have $a_p < a_q \iff b_p < b_q$ and $a_p = a_q \iff b_p = b_q$;
(2) for all indices $p, q \in K$ that are consecutive for $\bar{a}$, the gaps satisfy $a_q - a_p \approx_j b_q - b_p$.

By condition (1), the indices p and q are consecutive for $\bar{a}$ exactly when they are consecutive for $\bar{b}$, and the definition is therefore symmetric in the two orders. For $j \geq 1$, the relation $d \approx_j d'$ implies $d \approx_{j-1} d'$. The heart of the strategy is an arithmetic lemma: when Player I cuts a gap of one order into two pieces, Player II can cut the corresponding gap of the other order in such a way that each pair of corresponding pieces stays similar, one level down.

Lemma 18.15. *Let $j \geq 1$ be an integer, and let d, d' and e be positive integers with $d \approx_j d'$ and $e < d$. Then there is an integer e' with $0 < e' < d'$ such that*

$$e \approx_{j-1} e' \qquad \text{and} \qquad d - e \approx_{j-1} d' - e'.$$

Proof. The idea is to copy whichever piece of the first gap is short, measuring it from the corresponding end of the second gap, and to cut in the middle region when both pieces are long. If $d = d'$, the integer $e' = e$ has the required properties. Otherwise d and d' are both at least 2^j, and we distinguish three cases.

Suppose first that $e < 2^{j-1}$; we then take $e' = e$. The integer e' satisfies $0 < e' < 2^{j-1} < d'$, and the remaining pieces satisfy $d - e > 2^j - 2^{j-1} = 2^{j-1}$ and, likewise, $d' - e' > 2^{j-1}$; the left-hand pieces are thus equal, and the right-hand pieces are both at least 2^{j-1}.

Suppose next that $e \geq 2^{j-1}$ and $d - e < 2^{j-1}$; here the right-hand piece is short, and we copy it, taking $e' = d' - (d - e)$. The right-hand pieces are then equal, $d' - e' = d - e > 0$, whence $e' < d'$. Moreover, since $d - e < 2^{j-1}$ and $d' \geq 2^j$, the inequalities $e' > d' - 2^{j-1} \geq 2^{j-1} > 0$ hold, and so does $e \geq 2^{j-1}$, by assumption; the relation $e \approx_{j-1} e'$ follows.

Suppose finally that both pieces are long, that is, $e \geq 2^{j-1}$ and $d - e \geq 2^{j-1}$; we then cut the second gap at $e' = 2^{j-1}$. The integer e' satisfies $0 < e' < 2^j \leq d'$, and $e \approx_{j-1} e'$

because both are at least 2^{j-1}. The right-hand pieces satisfy $d-e \geq 2^{j-1}$ by assumption and $d'-e' \geq 2^j - 2^{j-1} = 2^{j-1}$, whence $d-e \approx_{j-1} d'-e'$. The three cases exhaust the possibilities, and the lemma is proved. ☺

Proposition 18.16. *Let $m, m' \geq 1$ be integers.*

(1) *If $j, k \geq 0$ are integers and the tuples $\bar{a} \in \{1, \ldots, m\}^k$ and $\bar{b} \in \{1, \ldots, m'\}^k$ are j-similar, then Player II has a winning strategy in $\mathrm{EF}_j(L_m, \bar{a}, L_{m'}, \bar{b})$.*

(2) *For every integer $n \geq 0$ with $m, m' \geq 2^n - 1$, the relation $L_m \equiv_n L_{m'}$ holds.*

Proof. We prove part (1) by induction on j, for all tuples at once. For $j = 0$, condition (1) of Definition 18.14, restricted to the indices $1, \ldots, k$, says that $\bar{a} \mapsto \bar{b}$ is a partial isomorphism, which is the winning condition of the game without rounds.

Suppose that $j \geq 1$ and that part (1) holds for $j-1$. By Lemma 18.8 and the induction hypothesis, it suffices to answer every move of Player I with a move that makes the extended tuples $(j-1)$-similar. The definition of similarity is symmetric in the two orders, and we may therefore assume that Player I plays an element $a \in \{1, \ldots, m\}$. Two cases arise.

If $a = a_i$ for some $i \in \{1, \ldots, k\}$, Player II answers $b = b_i$. The extended tuples $\bar{a}a$ and $\bar{b}b$ take the same values as $\bar{a}$ and $\bar{b}$, and the new index behaves exactly as the index i: it satisfies the same comparisons, and the consecutive pairs that contain it have the same gaps as those that contain i. The extended tuples are therefore $(j-1)$-similar, because $d \approx_j d'$ implies $d \approx_{j-1} d'$.

Otherwise the element a differs from every a_p with $p \in K$, and it lies strictly between $a_- = 0$ and $a_+ = m+1$. Let $p \in K$ be an index for which a_p is the largest value below a, and let $q \in K$ be an index for which a_q is the smallest value above a. The indices p and q are consecutive for $\bar{a}$: a value a_s strictly between a_p and a_q would lie below a, above a or at a, and each of these possibilities is excluded. We write $d = a_q - a_p$, $d' = b_q - b_p$ and $e = a - a_p$; these integers satisfy $0 < e < d$ and, by condition (2), $d \approx_j d'$. Let e' be the integer provided by Lemma 18.15, and let Player II answer $b = b_p + e'$. Since $0 \leq b_p < b < b_q \leq m' + 1$, this answer is an element of $\{1, \ldots, m'\}$.

We check that $\bar{a}a$ and $\bar{b}b$ are $(j-1)$-similar. By the choice of p and q, every index $s \in K$ satisfies $a_s \leq a_p$ or $a_s \geq a_q$, and condition (1) transfers this dichotomy to $b_s \leq b_p$ or $b_s \geq b_q$. Consequently, for every $s \in K$,

$$a_s < a \iff a_s \leq a_p \iff b_s \leq b_p \iff b_s < b,$$

and the same argument gives $a_s > a \iff b_s > b$, while neither $a_s = a$ nor $b_s = b$ holds; this is condition (1) for the extended tuples. The consecutive pairs of the extended tuples are those of $\bar{a}$, except the pairs with values a_p and a_q, together with the new pairs that join the new index to the indices with value a_p or a_q. The old pairs satisfy condition (2) for $j-1$ because they satisfy it for j. The new pairs have gaps e and $d-e$ in L_m, and e' and $d'-e'$ in $L_{m'}$, and they satisfy condition (2) for $j-1$ by the choice of e'. This completes the induction, and part (1) is proved.

For part (2), we apply part (1) to the empty tuples, with $j = n$. Their only consecutive pair is $(-,+)$, with gaps $m+1$ and $m'+1$, both at least 2^n because $m, m' \geq 2^n - 1$; hence the relation $m+1 \approx_n m'+1$ holds, and the empty tuples are n-similar. Player II thus has a winning strategy in $\mathrm{EF}_n(L_m, L_{m'})$, and Theorem 18.6(2) gives $L_m \equiv_n L_{m'}$. ☺

Figure 24 shows a play of $\mathrm{EF}_2(L_6, L_5)$ in which Player II follows this strategy. The bound $2^n - 1$ cannot be lowered: if $m \neq m'$ and one of them is less than $2^n - 1$, then Player I wins $\mathrm{EF}_n(L_m, L_{m'})$ (see Exercises 2 and 3 at the end of this section).

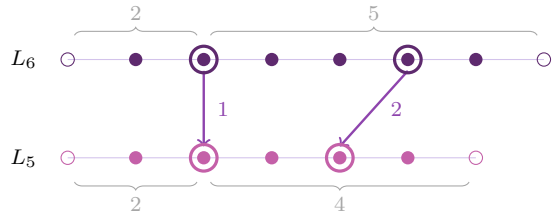


FIGURE 24. A play of $\mathrm{EF}_2(L_6, L_5)$ (see Proposition 18.16), with the imaginary elements below and above each order drawn as open circles. Before the play, the single gap has length 7 in L_6 and 6 in L_5, both at least $2^2 = 4$. In round 1 Player I chooses 2 in L_6, and Player II answers 2, the answer constructed in the proof of Lemma 18.15. The gaps are then 2 and 5 in L_6 and 2 and 4 in L_5: equal, or both at least $2^{2-1} = 2$, as 1-similarity requires. In round 2 Player I chooses an element in the long gap of L_6, and Player II answers in the corresponding gap of L_5; in the last round any element of that gap will do, since any two positive integers d and d' satisfy $d \approx_0 d'$. The chosen elements are in the same order in both structures, and Player II wins.

Parity and connectivity. The second application follows at once from Proposition 18.16.

Corollary 18.17. *There is no first-order sentence θ in the signature $\{<\}$ such that every finite linear order satisfies θ if and only if it has an even number of elements.*

Proof. Suppose, toward a contradiction, that θ is such a sentence, and let n be its quantifier rank. The integers 2^n and $2^n + 1$ are at least $2^n - 1$, and Proposition 18.16(2) gives $L_{2^n} \equiv_n L_{2^n+1}$. The two orders therefore agree on θ, although one of them has an even number of elements and the other an odd number. This contradiction proves the corollary. ☺

The third application rests on a familiar device: every first-order property of a graph defined in a linear order by a first-order formula translates, sentence by sentence, into a first-order property of the order. Let $\delta(x, y)$ be a formula in the signature $\{<\}$ whose free variables lie among the distinct variables x and y. For a $\{<\}$-structure $\mathcal{A}$, we write $\mathcal{A}^\delta$ for the $\{E\}$-structure with universe A and

$$E^{\mathcal{A}^\delta} = \big\{(a, b) \in A^2 : \mathcal{A} \models \delta(a, b)\big\}.$$

Lemma 18.18. *Let $\delta(x, y)$ be a formula in the signature $\{<\}$. For every sentence θ in the signature $\{E\}$ there is a sentence θ^* in the signature $\{<\}$ such that every $\{<\}$-structure $\mathcal{A}$ satisfies θ^* if and only if $\mathcal{A}^\delta$ satisfies θ.*

Proof. The sentence θ^* is obtained from θ by replacing each atomic formula $E(u, w)$ with a copy of δ in which x and y are renamed u and w. To avoid clashes of variables, we first replace the bound variables of δ by variables that do not occur in θ; for variables u and w of θ we then write $\delta(u, w)$ for the formula obtained from δ by substituting u for x and w for y simultaneously. By the substitution lemma of first-order logic, for all $a, b \in A$ we have $\mathcal{A} \models \delta(u, w)$ with u and w interpreted as a and b if and only if $\mathcal{A} \models \delta(a, b)$. For every subformula φ of θ, let φ^* be the result of the replacement; the free variables of φ^* lie among those of φ. We show by induction on φ that, for every list $\bar{x}$ of distinct variables containing the free variables of φ and every tuple $\bar{a}$ of elements of A of the same length,

$$\mathcal{A} \models \varphi^*(\bar{a}) \iff \mathcal{A}^\delta \models \varphi(\bar{a}).$$

For $\varphi = E(u, w)$, the substitution lemma and the definition of $E^{\mathcal{A}^\delta}$ give the equivalence. The other atomic formulas, $\top$ and the equations, are not changed by the replacement, and their truth does not depend on the relations of the structure. The steps for $\neg$ and $\wedge$ are immediate. For $\varphi = \exists z\, \psi$ we have $\varphi^* = \exists z\, \psi^*$, and Tarski's clause, applied in the two structures with the same universe A, reduces the equivalence for φ to the equivalences for ψ at the tuples $\bar{a}'a$ with $a \in A$, which hold by the induction hypothesis. The case of the sentence θ itself, with the empty list, is the lemma. ☺

The translation may raise the quantifier rank, since each atomic formula $E(u, w)$ is replaced by a copy of δ; no harm is done, because below only the rank of the translated sentence θ^* matters.

For connectivity we need a graph, defined from a linear order, that is connected exactly when the order has an odd number of elements. For $m \geq 3$, let G_m denote the graph with vertex set $\{1, \ldots, m\}$ in which i is joined to $i + 2$ for $1 \leq i \leq m - 2$, the vertex m is joined to 2, and the vertex $m - 1$ is joined to 1 (see Figure 25).

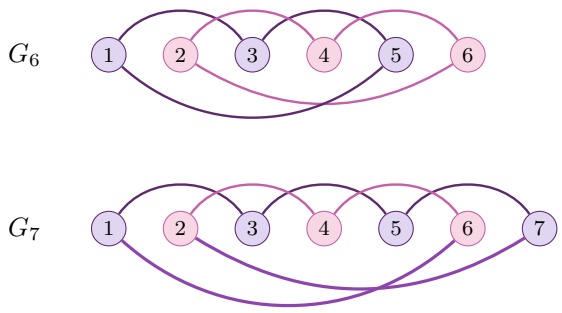


FIGURE 25. The graphs G_6 and G_7 of Lemma 18.19, with the odd vertices in lavender and the even vertices in pink. The arcs above the vertices join i to $i + 2$, and the arcs below are the extra edges, which join m to 2 and $m - 1$ to 1. In G_6 the extra edges close the odd and the even vertices into two separate triangles; in G_7 they join an odd vertex to an even one (orchid), and the graph is the single cycle $1, 3, 5, 7, 2, 4, 6$.

Lemma 18.19. *Let $m \geq 3$ be an integer.*

(1) *The graph G_m is connected if and only if m is odd.*
(2) *There is a formula $\delta(x, y)$ in the signature $\{<\}$, not depending on m, such that $L_m^\delta = G_m$ for every $m \geq 3$.*

Proof. (1) The edges joining i to $i + 2$ connect consecutive odd vertices and consecutive even vertices; they form a path through the odd vertices $1, 3, 5, \ldots$ and a path through the even vertices $2, 4, 6, \ldots$, and both paths are non-empty because $m \geq 3$. If m is odd, the edge joining the odd vertex m to the even vertex 2 connects the two paths, and the graph is connected. If m is even, every edge of G_m joins two vertices of the same parity: this is clear for the edges joining i to $i + 2$, and the extra edges join the even vertices m and 2 and the odd vertices $m - 1$ and 1. No path then leads from the vertex 1 to the vertex 2, and the graph G_m is not connected.

(2) The idea is that the successor, the first two elements and the last two elements of a finite linear order are first-order definable from the order. We use the formulas

$$\begin{aligned} \mathrm{succ}(x,y) &= x<y \wedge \neg\exists z\,(x<z\wedge z<y), & \mathrm{plus}_2(x,y) &= \exists z\,\big(\mathrm{succ}(x,z)\wedge\mathrm{succ}(z,y)\big),\\ \mathrm{first}(x) &= \neg\exists z\,(z<x), & \mathrm{second}(x) &= \exists z\,\big(\mathrm{first}(z)\wedge\mathrm{succ}(z,x)\big),\\ \mathrm{last}(x) &= \neg\exists z\,(x<z), & \mathrm{penult}(x) &= \exists z\,\big(\mathrm{last}(z)\wedge\mathrm{succ}(x,z)\big), \end{aligned}$$

with their bound variables renamed, where they are nested, into variables different from each other and from x and y. In L_m, the formulas $\mathrm{succ}(i,i')$ and $\mathrm{plus}_2(i,i')$ hold exactly when $i'=i+1$ and when $i'=i+2$, respectively; the formulas first, second, last and penult define the elements 1, 2, m and $m-1$. The formula

$$\varepsilon(x,y) = \mathrm{plus}_2(x,y)\vee\big(\mathrm{last}(x)\wedge\mathrm{second}(y)\big)\vee\big(\mathrm{penult}(x)\wedge\mathrm{first}(y)\big)$$

therefore holds of (i,i') in L_m exactly when $i'=i+2$, or $i=m$ and $i'=2$, or $i=m-1$ and $i'=1$. The formula $\delta(x,y)=\varepsilon(x,y)\vee\varepsilon(y,x)$ holds exactly when one of these conditions holds in one of the two directions, which is the definition of an edge of G_m; thus $L_m^{\delta}=G_m$. ☺

Corollary 18.20. *There is no first-order sentence θ in the signature $\{E\}$ such that every finite graph satisfies θ if and only if it is connected.*

Proof. Suppose, toward a contradiction, that θ is such a sentence. Let δ be the formula of Lemma 18.19(2), and let θ^* be the sentence provided by Lemma 18.18. For every $m\geq 3$, the two lemmas provide the equivalences

$$L_m\models\theta^* \iff G_m\models\theta \iff G_m \text{ is connected} \iff m \text{ is odd}.$$

Let n be the quantifier rank of θ^*. The integer $m=2^n+2$ is at least 3, and Proposition 18.16(2) gives $L_m\equiv_n L_{m+1}$, since m and $m+1$ are at least 2^n-1. The orders L_m and L_{m+1} therefore agree on θ^*, although exactly one of the integers m and $m+1$ is odd. This contradiction completes the proof. ☺

The two corollaries are statements about all sentences, and neither mentions a game; in both, the game did the work, by reducing an infinite family of sentences to a single strategy, the halving of gaps in Lemma 18.15.

18.5. **Further remarks (optional).**

Back-and-forth systems. Fraïssé's original formulation contains no players. A *back-and-forth system* of length n between two structures $\mathcal{A}$ and $\mathcal{B}$ is a sequence $I_0,\dots,I_n$ of sets of finite partial isomorphisms from $\mathcal{A}$ to $\mathcal{B}$ (finite maps p from a subset of A to B such that $\bar a\mapsto p(\bar a)$ is a partial isomorphism for an enumeration $\bar a$ of the domain of p), with I_n nonempty, such that, for $1\leq j\leq n$ and every $p\in I_j$, every $a\in A$ has a partner $b\in B$ with $p\cup\{(a,b)\}\in I_{j-1}$, and every $b\in B$ has a partner $a\in A$ with the same property. The strategies of Player II and the back-and-forth systems carry the same information, one in dynamic and the other in static form. If Player II wins $\mathrm{EF}_n(\mathcal{A},\mathcal{B})$, the sets I_j of finite partial isomorphisms p for which she wins $\mathrm{EF}_j(\mathcal{A},\bar a,\mathcal{B},p(\bar a))$, where $\bar a$ enumerates the domain of p, form such a system: the choice of the enumeration does not matter, by Lemma 18.7(2), and the conditions are those of Lemma 18.8. Conversely, a system provides a winning strategy for Player II, with the help of Lemma 18.7(2) and of Lemma 18.8, applied n times. Theorem 18.6(2) is thus Fraïssé's theorem: the relation $\mathcal{A}\equiv_n\mathcal{B}$ holds exactly when there is a back-and-forth system of length n between the two structures. For dense linear orders without endpoints, the set of all finite partial isomorphisms works for every j at once, as the proof of Proposition 18.13 shows. When both orders are countable, Player II can keep the matched pairs a partial

isomorphism for ω rounds; if Player I enumerates the two universes, one element of each in turn, the union of the matched pairs is an isomorphism, and Cantor's theorem appears as the infinite form of the game.

Ultrapowers. Ultrafilters give a second answer to the problem, algebraic where Theorem 18.6 is combinatorial: two structures are elementarily equivalent if and only if they have isomorphic ultrapowers. Keisler proved this theorem in 1961 under the generalized continuum hypothesis (GCH) [Kei61], and Shelah proved it in ZFC in 1971 [She71]. One direction follows from Łoś's theorem, by which every structure embeds elementarily, through the diagonal map, in each of its ultrapowers. For the other direction, elementarily equivalent finite structures are isomorphic, and for infinite ones we follow the proof under GCH as it is usually given today, with Keisler's good ultrafilters. Let λ be an infinite cardinal at least as large as the cardinalities of the structures and of the signature, and let $\mathcal{D}$ be a countably incomplete λ^+-good ultrafilter on λ. Such ultrafilters exist under GCH by Keisler's theorem of 1964, and in ZFC by Kunen's theorem of 1972 [Kun72], whose proof manufactures them from an independent family of 2^λ subsets of λ. The ultrapowers $\mathcal{A}^\lambda/\mathcal{D}$ and $\mathcal{B}^\lambda/\mathcal{D}$ of elementarily equivalent infinite structures $\mathcal{A}$ and $\mathcal{B}$ are then λ^+-saturated and have cardinality 2^λ, which equals λ^+ under GCH. Now the game comes back, played for λ^+ rounds instead of n. In the Ehrenfeucht–Fraïssé game of length λ^+ between the two ultrapowers, Player II answers each move by realizing, in the other structure, the type of the new element over the fewer than λ^+ elements already chosen; saturation guarantees that she can, and if Player I enumerates both structures, moving in each in turn, the completed play is an isomorphism. Shelah removed GCH by building the ultrafilter and the isomorphism together, in a single transfinite construction of length 2^λ.

Exercises.

(1) Show that $L_m \equiv_1 L_{m'}$ for all $m, m' \geq 1$, by describing a winning strategy for Player II in $\mathrm{EF}_1(L_m, L_{m'})$. Show that Player I wins $\mathrm{EF}_2(L_1, L_2)$, and find a sentence of quantifier rank 2 true in L_2 and false in L_1.

(2) Show that Player I wins $\mathrm{EF}_2(L_2, L_3)$ and that Player II wins $\mathrm{EF}_2(L_3, L_4)$, describing the strategies directly. Thus the orders L_3 and L_4 satisfy the same sentences of quantifier rank 2, while L_2 and L_3 do not, and for $n = 2$ the bound $2^n - 1$ of Proposition 18.16(2) cannot be lowered.

(3) $(*)$ Let $n \geq 1$ be an integer, and let $m \neq m'$ be positive integers with $\min\{m, m'\} < 2^n - 1$. Show that Player I wins $\mathrm{EF}_n(L_m, L_{m'})$, and conclude that $L_m \equiv_n L_{m'}$ holds if and only if $m = m'$ or $m, m' \geq 2^n - 1$. (Hint: show by induction on j that Player I wins in j rounds from any pair of tuples that violates condition (1) of Definition 18.14, or that has a consecutive pair of indices whose gaps have lengths $d \neq d'$ with $\min\{d, d'\} < 2^j$; if $d < d'$, he plays in the longer gap, at distance $\min\{2^{j-1}, \lfloor d'/2 \rfloor\}$ from its left end.)

(4) Show that $(\mathbb{Z}, <) \equiv_2 (\mathbb{Q}, <)$, but that Player I wins $\mathrm{EF}_3((\mathbb{Z}, <), (\mathbb{Q}, <))$. Find a sentence of quantifier rank 3 that distinguishes the two orders.

(5) Draw the graphs G_5 and G_8 of Lemma 18.19. Show that for odd $m \geq 3$ the graph G_m is a single cycle through all its vertices, and that for even $m \geq 6$ it consists of two disjoint cycles, one through the odd and one through the even vertices. Compute the quantifier rank of the formula δ in the proof of Lemma 18.19(2).

(6) Show that there is no first-order sentence in the signature $\{E\}$ that holds in a finite graph if and only if the graph is bipartite. (Hint: argue as in the proof of Corollary 18.20, with the cycle obtained from L_m by joining each $i < m$ to $i+1$ and m to 1.)

(7) $(*)$ Let $\mathbb{Z}+\mathbb{Z}$ denote the order obtained by placing one copy of $(\mathbb{Z},<)$ after another. Show that $(\mathbb{Z},<) \equiv (\mathbb{Z}+\mathbb{Z},<)$. (Hint: adapt the invariant used for finite linear orders, regarding two elements in different copies as being infinitely far apart.)

(8) $(*)$ Let $\Sigma = \{P_0, P_1, P_2, \dots\}$ be an infinite signature of unary relation symbols. Let $\mathcal{A}$ be the Σ-structure with universe $\mathbb{N}$ in which P_i is interpreted as $\{k \in \mathbb{N} : k > i\}$, and let $\mathcal{B}$ be obtained from $\mathcal{A}$ by adding a new element ∞ that lies in every P_i. Show that $\mathcal{A} \equiv \mathcal{B}$, but that Player I wins $\mathrm{EF}_1(\mathcal{A}, \mathcal{B})$. (Hint: for every N, the reducts of $\mathcal{A}$ and $\mathcal{B}$ to the signature $\{P_0, \dots, P_N\}$ are isomorphic.) Thus Theorem 18.6(2) fails for infinite signatures, as stated in Remark 18.12.

Part V. The Scottish Book, von Neumann, Ulam, Texas, Toronto, and the triumph of games, logic, topology and combinatorics

This epilogue is about the people behind the mathematics of these notes: where they worked, what happened to them, and how their problems traveled. Many of them were very young. When the Scottish Book was started, in the summer of 1935, Stanisław Ulam was twenty-six, Stanisław Mazur was thirty, Mark Kac was twenty, and Stefan Banach, who led the discussions at the café table, was forty-three. Figure 26 shows the journey of the Scottish Book and of the people who carried it, and the timeline in Figure 27 is worth keeping at hand throughout.

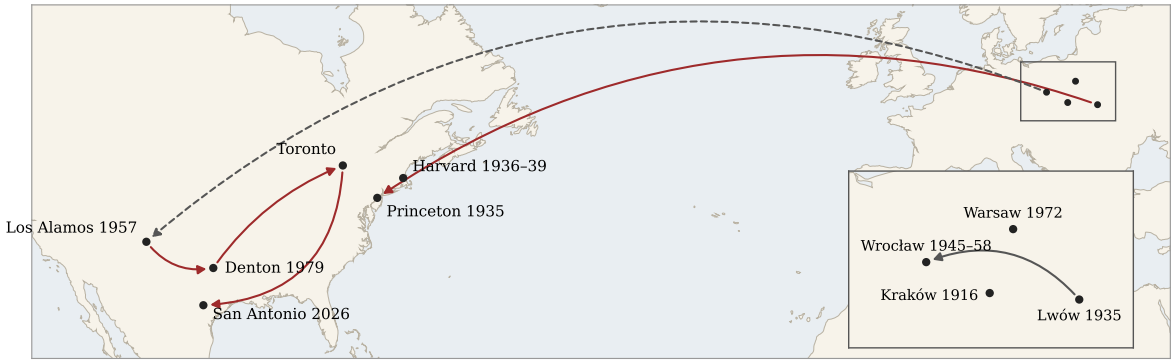


FIGURE 26. The journey of the Scottish Book. Red: Ulam's crossing in 1935, and the road from Los Alamos to Denton, Toronto and San Antonio. Dashed: the typewritten copy that Steinhaus sent from Wrocław to Ulam in 1956. The inset shows Poland: Kraków, where Steinhaus met Banach in 1916; Lwów, home of the Scottish Café; Wrocław, home of the New Scottish Book after the war; and Warsaw, where Enflo received his goose in 1972.

19. The marble tables of the Scottish Café

Between the two world wars, the city of Lwów (then in southeastern Poland, now Lviv, in Ukraine) was home to one of the liveliest mathematical communities of the century. Led by Stefan Banach and Hugo Steinhaus, with Stanisław Mazur, Juliusz Schauder, Stanisław

1916	Steinhaus meets Banach and Nikodym in the Planty park, Kraków.
1920	Banach's doctorate in Lwów, without a university degree.
1930	Ulam on measures defined on all subsets of a set (measurable cardinals).
1932	Banach's *Théorie des opérations linéaires*.
1935	17 July: Problem 1, by Banach. Mazur's Problem 43; 4 August: Banach's answer. End of the year: Ulam goes to Princeton.
1936	6 November: Mazur's Problem 153, the goose problem.
1939	20 August: Ulam leaves Poland for the last time. September: war.
1941	31 May: last problem, number 193, by Steinhaus. July: the Lwów professors are murdered.
1945	31 August: Banach dies in Lwów. Steinhaus moves to Wrocław.
1946	2 July: first entry in the New Scottish Book, by Steinhaus.
1946	Ulam's idea of the Monte Carlo method.
1957	Ulam's English translation, mimeographed at Los Alamos; Oxtoby publishes the first proof of Banach's solution of Problem 43.
1962	Mycielski and Steinhaus propose the Axiom of Determinacy.
1972	Mazur gives Enflo a live goose in Warsaw.
1975	Martin proves Borel determinacy.
1979	The Scottish Book Conference in Denton, organized by Mauldin.
1981	*The Scottish Book*, edited by Mauldin (second edition 2015).
1994–98	Distortion problem (Odell–Schlumprecht); Gowers's dichotomy; Fields Medal for Gowers.
2026	The UTSA–Toronto seminar.

FIGURE 27. A timeline of the Scottish Book.

Ulam, Mark Kac and many others, the Lwów School did much to create modern functional analysis and, through Mazur's Problem 43, gave topology its first infinite game.

The school began with a chance meeting. One evening in 1916, walking in the Planty, the park around the old town of Kraków, Steinhaus overheard the words "Lebesgue integral" and stopped to talk to the two young men who had said them. They were Stefan Banach and Otto Nikodym. Banach, then twenty-four, had studied engineering in Lwów without finishing his degree, and was teaching himself mathematics. Steinhaus mentioned a problem on which he had been working; a few days later Banach came back with the solution, and their first joint paper followed. In 1920, with Steinhaus's support, Banach was allowed to submit a doctoral thesis in Lwów without a university degree; the thesis, published in 1922, contains what we now call Banach spaces. Steinhaus, whose own contributions to mathematics were many, later said that Banach was his greatest scientific discovery. In 1935, fifteen years after the thesis, Banach opened the Scottish Book with Problem 1, which asks when a metric space can be remetrized so as to become complete and compact while every sequence that converged in the old metric still converges in the new one; within weeks he had also answered Mazur's Problem 43.

The school's real seminar room was the *Kawiarnia Szkocka*, the Scottish Café, on Plac Akademicki (see Figure 28). The mathematicians spent hours at its tables over coffee and, as the stories go, wrote their formulas directly on the marble tabletops, which were wiped clean when the café closed. To save at least the problems, Banach's wife, Łucja Banachowa, is said to have provided a large notebook, which was kept by the headwaiter and brought to the table on request. Thus was born the *Scottish Book* (*Księga Szkocka*); its first problem, by Banach, is dated 17 July 1935. The sessions could be long: Ulam recalled one with Mazur and Banach that lasted seventeen hours, interrupted only for meals [Ula76]. He also recorded

Banach's view of what a mathematician does: "Good mathematicians see analogies between theorems or theories, the very best ones see analogies between analogies." Problems came with prizes: a cup of coffee, a small beer, five beers, a bottle of wine, 100 grams of caviar, a dinner at the Hotel George, a fondue in Geneva, a bottle of whisky "of measure > 0" (von Neumann, 1937), a kilo of bacon (Saks, 1940), and a live goose (see Section 25).



FIGURE 28. Plac Akademicki in Lwów, the square of the Scottish Café, on a hand-colored postcard of 1909 (Kraków: Wydawnictwo Salonu Malarzy Polskich). National Library of Poland (Polona), public domain.

20. Problem 43

Mazur's Problem 43, the subject of these notes, was written in the summer of 1935, in the first weeks of the book. In the Introduction we saw what it asks and how Banach answered it on 4 August (see Figure 1), and in Section 6 we followed Oxtoby's proof of 1957, which turned the answer into a theorem. Right after Problem 43 in the book, Ulam proposed a variant in which the players choose binary digits. This "Ulam game" (see Section 13) later inspired Steinhaus and Jan Mycielski to propose, in 1962, a new axiom for set theory, the *Axiom of Determinacy* [MS62]. A question about the size of sets of real numbers, asked over coffee, had become a question in the foundations of mathematics.

21. Von Neumann's invitation and the bridge to America

John von Neumann met Ulam in Warsaw in 1935 and invited him to spend a few months at the Institute for Advanced Study in Princeton. Ulam, then twenty-six, sailed for America in December 1935.

Ulam was the most prolific author of the Scottish Book: forty problems are his alone, and about twenty more are joint with others, among them Banach, Mazur and Schreier. His problems have had a long life. Problem 19 asks whether a solid of uniform density that floats in water in equilibrium in every position must be a ball. In 2022 Dmitry Ryabogin showed that the answer is no: there is a strictly convex body of revolution in $\mathbb{R}^3$, of density $1/2$ and not a ball, that floats in equilibrium in every position [Rya22]. Ulam's earlier work, of 1930, on measures defined on all subsets of a set led to the measurable cardinals (see Section 28 below).

At a tea in von Neumann's house, Ulam met G. D. Birkhoff, who asked whether he would like to join the Society of Fellows at Harvard. From 1936 to 1939 Ulam divided his time between Harvard and Lwów, returning each summer to Poland and to the tables of the Scottish Café. On 20 August 1939 he left Poland for the last time, sailing from Gdynia with his seventeen-year-old brother Adam. Twelve days later Germany invaded Poland; on 17 September the Soviet Union followed. The brothers' father, Józef, who had seen them off at the quay, was killed in the Holocaust, together with their sister Stefania and much of the family.

22. War, and the survival of the book

The war destroyed the Lwów School. From September 1939 the city was under Soviet occupation, and the café meetings went on: Saks's kilo of bacon dates from February 1940. German troops entered Lwów on 30 June 1941. On the night of 3–4 July, in a planned action against the Polish intelligentsia, they arrested professors of the city's universities together with members of their families, and shot them on the Wuleckie Hills: twenty-five professors and twenty-five of their relatives died. Among them were the mathematicians Antoni Łomnicki and Włodzimierz Stożek, together with two of Stożek's sons. Kazimierz Bartel, geometer and former prime minister of Poland, was arrested on 2 July and killed later that month.

The killing did not stop there. Stanisław Saks was shot by the Gestapo in Warsaw in 1942. Herman Auerbach died the same year, deported from the Lwów ghetto to the Bełżec death camp. Juliusz Schauder, of the Schauder fixed point theorem, was murdered in 1943. Mark Kac had left for the United States in 1938; his parents and his brother were killed in 1942. Steinhaus survived by hiding in the countryside under a false name.

Banach survived because of Rudolf Weigl, a biologist who had developed a vaccine against typhus. The vaccine was grown in lice, and the lice had to be fed on human blood: "louse feeders" let cages of lice feed on their legs. The German army needed the vaccine, and the papers of Weigl's institute protected those who worked there; Weigl used that protection to employ mathematicians, scientists, members of the underground and Jews. Banach fed lice there, and so did several other members of the school. Weigl also smuggled vaccine into the Warsaw and Lwów ghettos. In 2003 Yad Vashem recognized him as Righteous Among the Nations. Banach lived to see the end of the war; he died of lung cancer in Lwów on 31 August 1945.

The book survived too. Its last entry, Problem 193, had been written by Steinhaus on 31 May 1941, a month before the Germans arrived. Ulam recalled that Mazur had once proposed, in case the city was bombed, to put the manuscripts and the Scottish Book in a case and to bury the case near a goal post of a football field outside the city; whether he did so is not known. What is known is that after the war Banach's son, Stefan Banach Jr., brought the book to Wrocław, where Steinhaus had moved.

23. Ulam after Lwów

The mathematics of the café went on in Ulam's life (see Figure 29) in unexpected ways. In 1943, after asking von Neumann for war work, he received an invitation, signed by Hans Bethe, to an unnamed project near Santa Fe; in February 1944 he arrived at Los Alamos and joined the Manhattan Project. In 1946, while recovering from encephalitis and brain surgery, Ulam played a great deal of solitaire. He wondered what the chance of winning a game was, and saw that it was easier to estimate that chance by playing many random games than by counting. With von Neumann he turned this idea into the *Monte Carlo method*, named by their colleague Nicholas Metropolis after Monte Carlo, where an uncle of Ulam's liked to gamble [MU49]; it is now among the most widely used methods in science. With Edward Teller he conceived the Teller–Ulam design of the hydrogen bomb. In the early 1950s, with Enrico Fermi and John Pasta, he simulated on the MANIAC computer a chain of oscillators that, unexpectedly, did not reach thermal equilibrium; their report appeared in 1955 [FPU55]. The programs were written by Mary Tsingou, whose role was acknowledged only decades later, when the problem became known as the Fermi–Pasta–Ulam–Tsingou problem [Dau08].

24. The 1957 Los Alamos mimeograph

Steinhaus had a typewritten copy of the book made and in 1956 sent it to Ulam, who was by then at the Los Alamos Scientific Laboratory in New Mexico. Ulam translated it into English and, in his words, "had some three hundred mimeographed copies of the translation made. I had to pay for this myself—Los Alamos is a government laboratory, and one cannot use taxpayers' money for such frivolous purposes." The copies circulated among mathematicians around the world, and for a generation they were how many mathematicians came to know the problems.

25. Problem 153: a live goose

The best-known prize of the Scottish Book was paid thirty-six years after it was offered. Mazur's Problem 153 of 6 November 1936 asked a question about approximating continuous functions of two variables and offered a live goose. The problem turned out to be equivalent to the approximation problem formulated by Grothendieck in 1955, the question whether, on every Banach space, every compact operator is a norm limit of operators of finite rank. A negative answer would also settle Banach's basis problem, since a space with a Schauder basis has the approximation property. In 1972 the young Swedish mathematician Per Enflo constructed a separable Banach space without the approximation property, and therefore without a Schauder basis [Enf73]. Mazur, then in Warsaw, kept his word: that same year, in a ceremony at the Stefan Banach Center, he presented the twenty-eight-year-old Enflo with a live goose. According to several accounts, the ceremony was broadcast on Polish television.

Enflo is also a concert pianist. In 1956, at the age of eleven, he won the Swedish competition for young pianists, and he won it again in 1961; among his teachers was Géza Anda. He has gone on performing ever since.

26. The road to Denton: Dan Mauldin and North Texas

In the 1970s the story moved to Texas. R. Daniel Mauldin, a measure theorist and descriptive set theorist at North Texas State University (now the University of North Texas) in

Denton, took on the task of finding out what had become of the 193 problems. In May 1979 he organized the *Scottish Book Conference* in Denton.

The list of speakers reads like a family reunion. Ulam, Mark Kac, Antoni Zygmund, Paul Erdős and Andrzej Granas all lectured, Ulam on the anecdotal history of the book, Kac on his personal history of it, and Zygmund on Steinhaus and the development of Polish mathematics. R. D. Anderson and D. A. Martin spoke about work that went back to the Scottish Book. Martin had proved Borel determinacy in 1975: every infinite game of perfect information in which the players choose natural numbers, and whose payoff set is Borel, is determined (see Section 15). John Oxtoby, who had published the first proof of Banach's solution of Problem 43, toasted a "condensation of Poles" at the conference; R. H. Bing, a pillar of Texas topology, also offered a toast.

From the conference came the book edited by Mauldin, *The Scottish Book: Mathematics from the Scottish Café* (Birkhäuser, 1981) [Mau81]. It contains lectures from the conference, Ulam's "anecdotal history" of the book among them, and the problems, with commentaries on what became of them. A second edition appeared in 2015 [Mau15].

27. The New Scottish Book

The tradition did not die in Lwów. In Wrocław, where much of the Lwów university was re-established after the war, Steinhaus started the *New Scottish Book* (*Nowa Księga Szkocka*). Its first entry, by Steinhaus, is dated 2 July 1946; problems were entered in it regularly until 1979, and occasionally until 1987. The second edition of Mauldin's book includes a short history of the New Scottish Book and a selection of its problems.

28. Set-theoretic topology and Ramsey spaces at Toronto

The questions of the Scottish Book on measure, category and infinite games also run through set-theoretic topology, a field in which the University of Toronto has long been a leading center.

In 1930 Ulam proved that a countably additive probability measure that is defined on all subsets of a set and vanishes on points can exist only if the set is very large: at least as large as a weakly inaccessible cardinal, and, if the measure takes only the values 0 and 1, at least as large as a strongly inaccessible one [Ula30]. In particular, such a measure on all subsets of the real line can exist only if the continuum is at least as large as a weakly inaccessible cardinal; in fact, the least cardinal that carries such a measure is then weakly inaccessible and no larger than the continuum. Along the way Ulam introduced what are now called Ulam matrices. A set carries a two-valued measure of this kind exactly when it is at least as large as a measurable cardinal, and measurable cardinals, with the larger cardinals that followed, became a standard tool for consistency proofs in topology. At Toronto, Franklin D. Tall and his collaborators applied such set-theoretic methods to classical problems of point-set topology, most notably the normal Moore space problem. Infinite games of the Banach–Mazur type, and their relatives such as the Menger game and the point-open game of Part III, are now everyday tools in the field.

Stevo Todorcevic, at Toronto, developed the theory of Ramsey spaces, which unifies the Galvin–Prikry and Ellentuck theorems, Gowers's FIN_k theorem and much else in a single axiomatic framework [Tod10]. The circle closes in the present: Tall is one of the authors, with Eduardo Dueñez and José Iovino of UT San Antonio and Tonatiuh Matos-Wiederhold and Luciano Salvetti of Toronto, of recent work on the complexity of deep computations

through the topology of function spaces [DIMW+26]; that work is the subject of the seminar for which these notes were written.

29. The triumph of games: combinatorics, logic and the real-valued setting

Texas also became a center of the Banach space theory that Banach's 1932 monograph and the Scottish Book had launched. Edward Odell at the University of Texas at Austin and Thomas Schlumprecht at Texas A&M University were at the heart of it.

In 1991 Schlumprecht constructed the first arbitrarily distortable Banach space [Sch91], and in 1994 Odell and Schlumprecht used his ideas to solve the distortion problem: the Hilbert space ℓ_2 is itself arbitrarily distortable [OS94]. Schlumprecht's space was also the starting point of W. T. Gowers and B. Maurey, who in 1992 (published in 1993) answered the old unconditional basic sequence problem in the negative. They constructed a "hereditarily indecomposable" space, one in which no infinite-dimensional closed subspace is the direct sum of two infinite-dimensional closed subspaces; in particular, no infinite-dimensional subspace has an unconditional basis [GM93].

The last step needed a different kind of mathematics. Banach's homogeneous space problem, from his 1932 monograph, asks whether ℓ_2 is the only infinite-dimensional Banach space isomorphic to all of its infinite-dimensional closed subspaces. To answer it, Gowers turned to infinite combinatorics and logic.

The bridge was infinite-dimensional Ramsey theory, the subject of Section 11: the theorems of Galvin and Prikry, of Silver and of Ellentuck, all proved by logicians, say that definable colorings of the infinite subsets of ω have infinite homogeneous sets. Gowers saw that this machinery could be imported into Banach space theory if it were translated into a continuous, real-valued setting, with block sequences of vectors in place of infinite sets of integers, and with a winning strategy in a two-player game in place of a homogeneous set. His Ramsey theorem for this game yields his dichotomy theorem. The dichotomy, combined with the Gowers–Maurey theorem that a hereditarily indecomposable space is isomorphic to none of its proper subspaces and with a theorem of Komorowski and Tomczak-Jaegermann, implies that every infinite-dimensional homogeneous Banach space is isomorphic to ℓ_2 [Gow96] (see Section 12).

When Gowers received the Fields Medal in 1998, the cycle was complete. Infinite games, born at a café table in Lwów in 1935, had passed through the foundations of mathematics (determinacy) and through the hands of logicians and combinatorialists (infinite Ramsey theory); by way of Texas, Paris and Cambridge, they had come back to analysis to settle a problem posed by Banach himself. The notebook that survived a war, and the people who survived it, or did not, and the colleagues who carried their problems across an ocean, are part of that triumph.

30. A Scottish book of our own

The Scottish Book was not a journal. It was a notebook on a café table, open to anyone who sat down, where a problem could be written before it was understood and a prize could be a beer or a goose. That spirit is easy to revive. We propose a Scottish Book of our own for the participants of this seminar, in San Antonio and in Toronto: a shared notebook of the problems that come up in our discussions, each signed and dated, with a prize chosen by its author, and with room for the answers, partial results and counterexamples that follow. Students are especially welcome: Ulam was twenty-six and Kac twenty when the original book was opened, and some of its best problems are short questions that anyone can state.

Further reading

Ulam's autobiography [Ula76] and Kac's [Kac85] are the best first-hand accounts of the Lwów School and of what came after. Kałuża's biography of Banach [Kał96], written with a reporter's eye, tells the story of his life and of his city. Mauldin's book [Mau81, Mau15] contains the problems themselves, with commentaries on what became of them.



Figure 29. Stanisław Ulam in 1962. Photograph by Alfred Eisenstaedt for *LIFE*, from a series of portraits of mathematicians that Eisenstaedt made for the magazine in 1962–63, which also includes Kurt Gödel and Claude Shannon.

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